UNIT 1-

WHAT IS CLIMATE CHANGE ?

The science is clear. Climate change is real. Climate change is happening now. Climate change requires immediate and ambitious action to prevent the worst effects it can have on people and wildlife all over the world.

We know that the planet has warmed by an average of nearly 1°C in the past century. If we are to prevent the worst effects of climate change, there is global agreement that temperature rises need to be kept well below 2°C from the pre-industrial era, with an ambition to keep it below 1.5°C. Currently, however, assessments suggest that we are currently on course for temperature rises of up to as much as 4°C or higher.

We have recently seen a number of unwanted developments due to climate change and global warming:

- 16 of the 17 warmest years on record have occurred since 2001, with 2016 being the warmest yet.
- The current levels of atmospheric concentrations of greenhouse gases are unprecedented in the last 800,000 years.
- And recently, scientists have declared a new geological time period: the Anthropocene, in which human activity is said to be the dominant influence on the environment, climate, and ecology of the earth.

As the planet continues to warm, climate patterns change. Extreme and unpredictable weather will become more common across the world as climate patterns change, with some places being hotter, some places being wetter, and some places being drier. These changes can have (and are already having) drastic impacts on all life on Earth.

A rise of just 2°c would mean:

- Severe storms and floods in many countries, particularly impacting coastal areas, with droughts also affecting many parts of the world
- Seas become more acidic, coral and krill die, food chains are destroyed
- Little or no Arctic sea ice in summer – which not only means less habitat for polar bears, but also means the global climate warms faster, as there is less polar ice to reflect sunlight

Beyond 2°c:

- Rainforests dying
- Unthinkable loss of ancient ice sheets of Greenland and Antarctica, causing dramatic sea level rises
- Mass displacement of people and widespread species loss and extinction

That's why we must act now.

Some people might try to tell you that global warming is natural, or that the Earth is actually cooling. Or they might suggest there's nothing we can do. But here's what the science tells us about climate change:

Yes, the Earth's climate has always changed, with temperatures rising and falling over thousands of years. But now, it's happening at a far faster rate than ever before, giving people and wildlife very little time to react and adapt.

There is overwhelming evidence (97% scientific consensus) that global warming is mostly man-made – largely down to burning fossil fuels and deforestation on a mass scale. This is not a natural process, no matter how much climate change deniers may claim it is.

The good news? We **can** do something about it. But we have to do it together, and we have to do it now.

Interesting Climate Change Facts

1. We Are Certain We Caused It

The UN's Intergovernmental Panel on Climate Change (IPCC) kicked off its 2021 report with the following statement: "It is unequivocal that human influence has warmed the atmosphere, ocean and land."

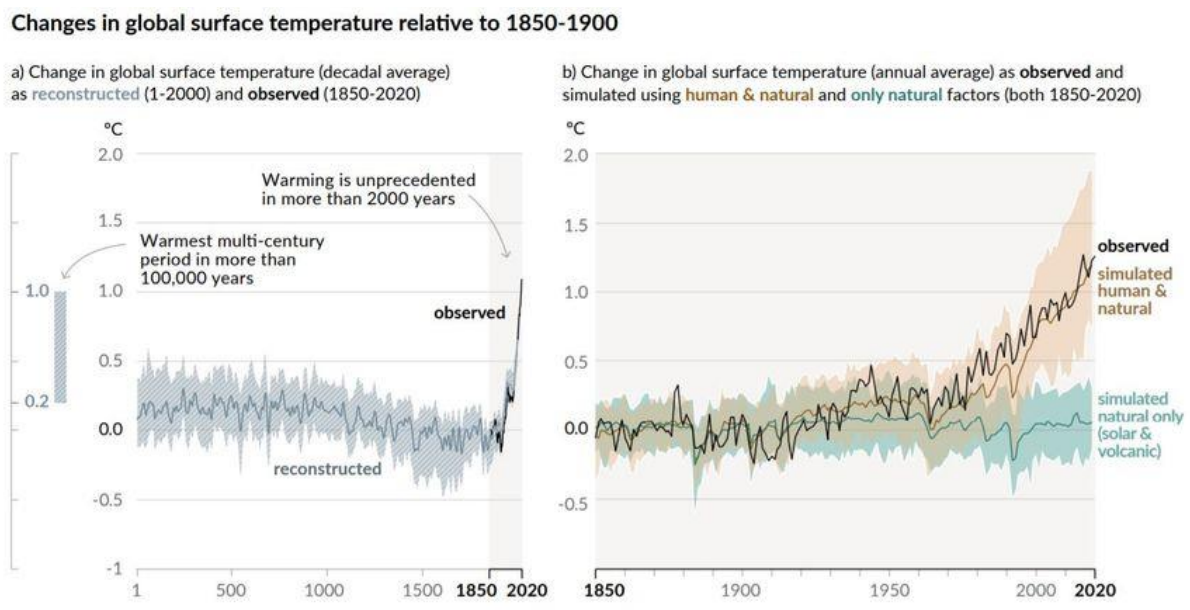
How are we so certain?

It took a while, but climate modelling is now refined enough to predict how things would go without human influence, within a margin of error. What we are observing today, however, is beyond that margin of error, therefore proving that we have driven the change.

2. The Last Decade Was the Hottest in 125,000 Years

Most straightforward of our climate change facts: according to the [IPCC's sixth assessment report] on the state of our climate, the past decade is likely to have been the hottest period in the last 125,000 years. For about 100,000 years, we have been oscillating between glacial (ice ages) and warmer interglacial periods like the one we currently live in.

Yet, this is also the warmest multi-century period we have had in this timespan.



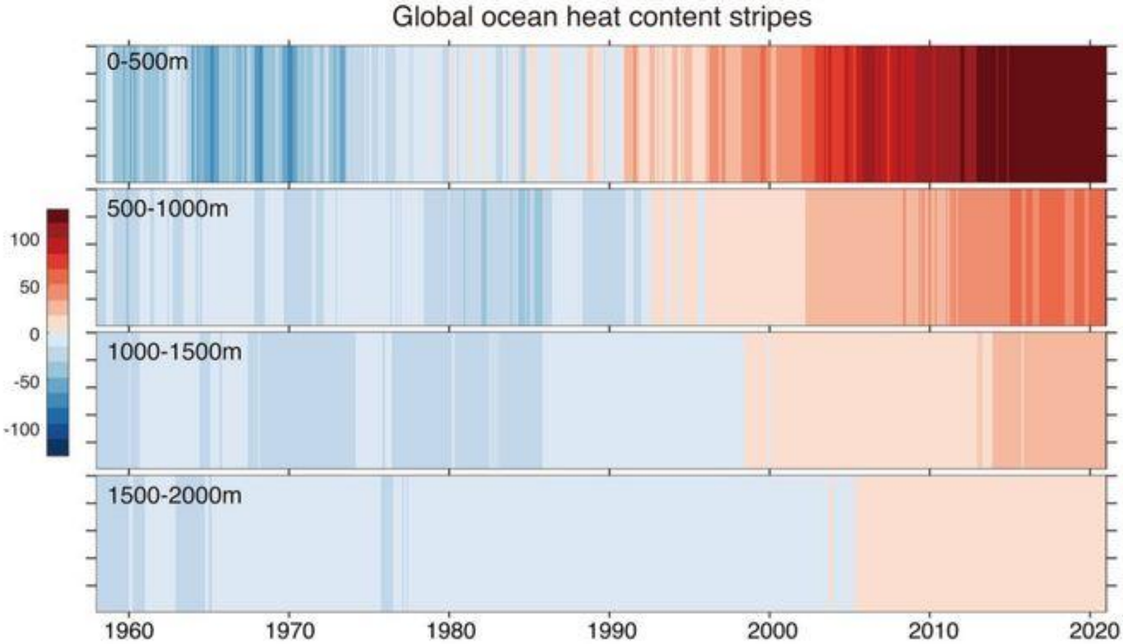
Source: IPCC, 2021.

IPCC - INTERGOVERNMENTAL PANEL ON CLIMATE CHANGE

The vertical bar on the left shows the estimated temperature (very likely range) during the warmest multi-century period in the last 100,000 years, which occurred around 6,500 years ago during our current era called the Holocene. Only around 125,000 years ago, a time prior to the last ice age, might have had higher temperature than the ones we are currently experiencing. Each of these past warm periods were caused by slow (multi-millennial) orbital variations that are not in play today.

3. The Ocean Absorbs Most of the Heat We Produce

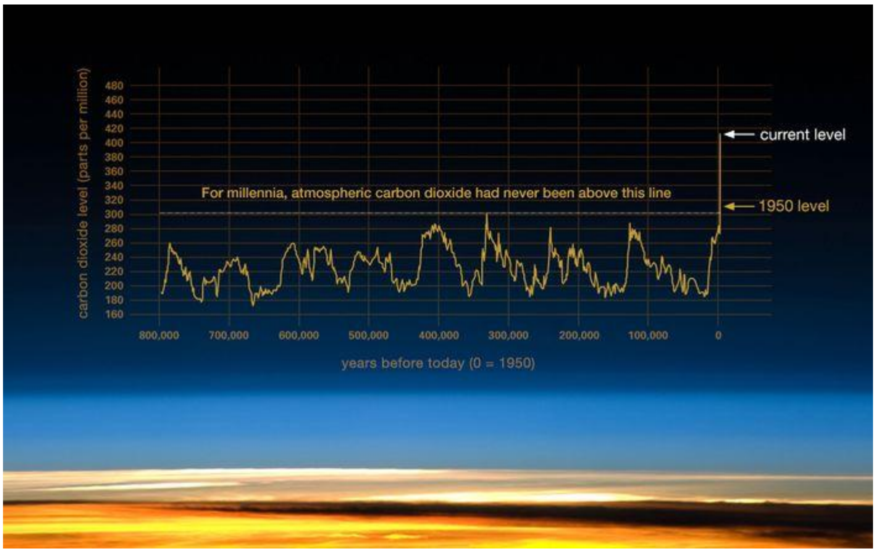
A 2019 study found that oceans had sucked up **90% of the heat** gained by the planet between 1971 and 2010. Another found that it absorbed 20 sextillion joules of heat in 2020 – equivalent to two Hiroshima bombs per second.



The ocean has tremendous volume and heat-storage capacity, which is why some organisms are used to temperatures being quite stable. Of these, coral reefs are particularly sensitive to temperature levels, reason for which many are now dying off.

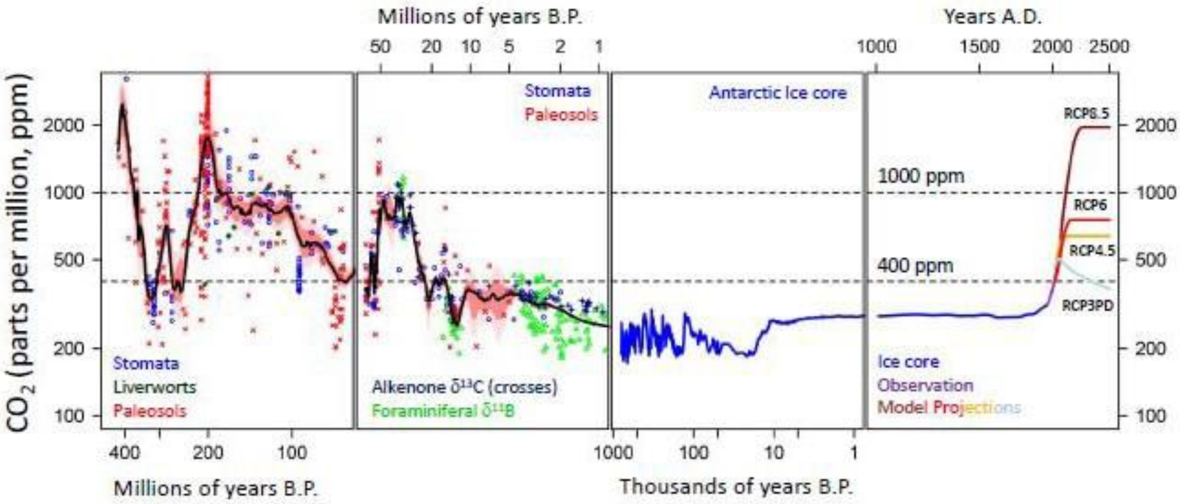
4. CO2 Is At Its Highest in 2 Million Years

Pre-industrial CO2 levels were around 280 parts per million (ppm). Today, we stand close to 420 ppm.



Source: NASA.

The most distant period in time for which we have estimated CO2 levels is around the Ordovician period, 500 million years ago.



Source: Descent into the Icehouse.

Once again, the ocean comes to our rescue by absorbing about one-third of the carbon in the atmosphere. Before the industrial revolution, it was actually a source of carbon, and not a sink, but the massive amount of CO2 now in the atmosphere has forced it to start absorbing the gas.

5. We Are Losing 1.2 Trillion Tons of Ice Each Year

This item on our list of climate change facts can be hard to comprehend because we are dealing with volume beyond our comprehension.

Since the mid-1990s, we've lost around 28 trillion tons of ice, with today's melt rate standing at 1.2 trillion tons a year. To help you put that into perspective, the combined weight of all human-made things is 1.1 trillion tons. That's about the same weight as all living things on earth.

6. Air pollution Is Both Good and Bad

It was recently found that air pollution kills more than 9 million people per year. Developing hotspots in south Asia and Africa will be dealing with poor air quality for years to come, but there is a silver lining.

Polluting particles, such as PM10 or PM2.5, which cause adverse health effects similar to those of cigarettes, actually reflect the sun's heat rather than trap it. We've pumped enough greenhouse gases into the atmosphere to warm it by 1.5C already, but fine particles have kept it around 1.1C so far.

Some have proposed intentionally disseminating particles into the atmosphere to help reflect more sunlight, but potential unforeseen consequences have prevented us from doing so.

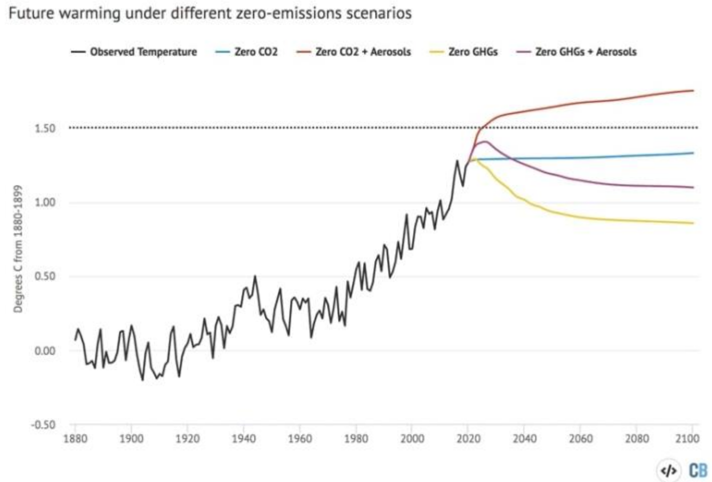
7. Attribution Is Now Possible (Extreme Weather)

We can now attribute natural disasters to human-driven climate change with certainty. This hasn't always been the case, as a lack of data and refined techniques for detecting attribution made it hard to tell how much we had to do with each extreme weather events.

We can now say with precision how much likelier we made things like the North American summer 2021 heatwave, which the World Weather Attribution says was "virtually impossible" without climate change as well as the Indian heatwave, which experts believe it was made 30 times more likely because of climate change.

8. Global Warming Is (Partially) Reversible

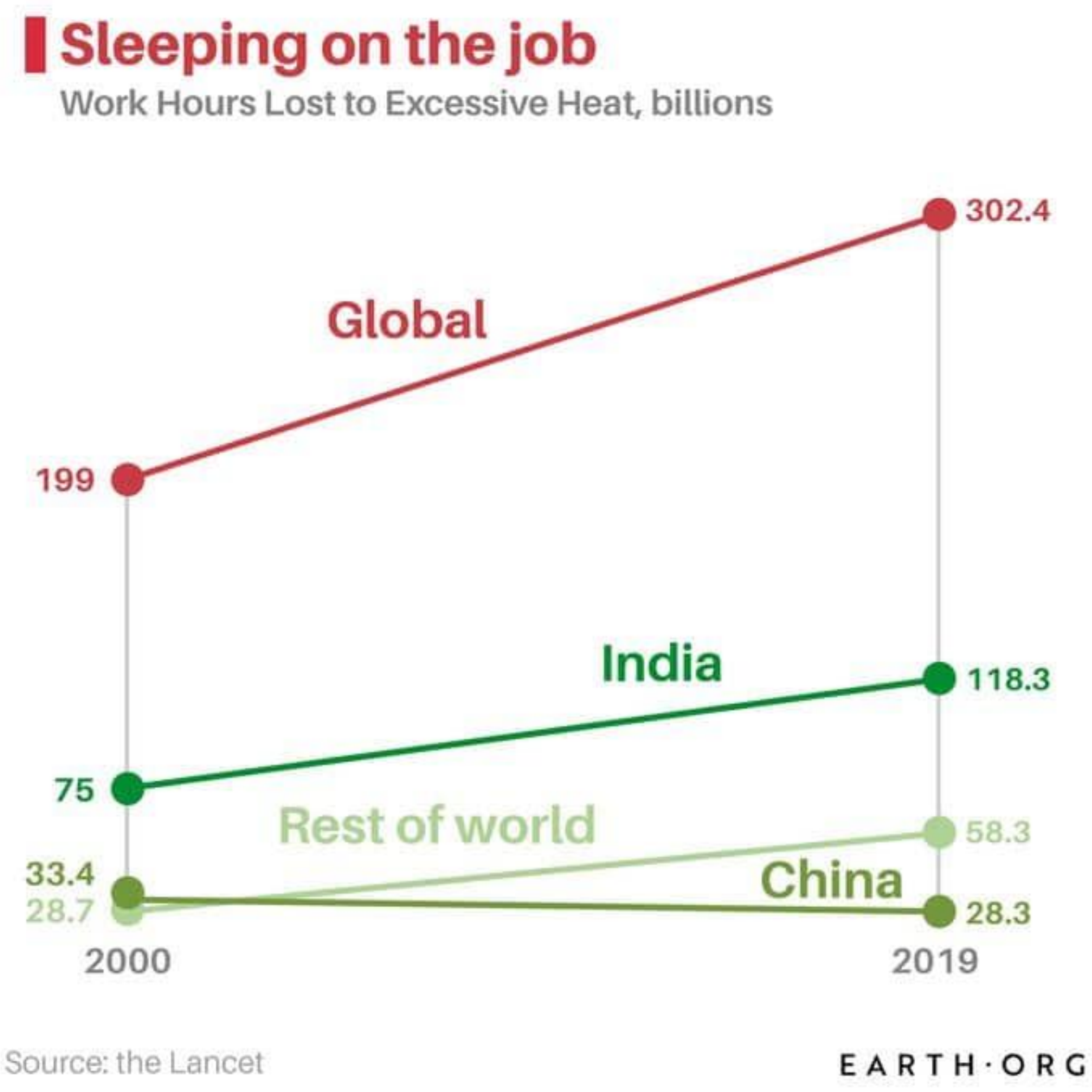
If global net emissions were entirely ceased, the warming we've caused would gradually reverse but other climate-induced changes would continue for decades if not centuries. For example, sea level rise would probably take millennia to reverse its course.



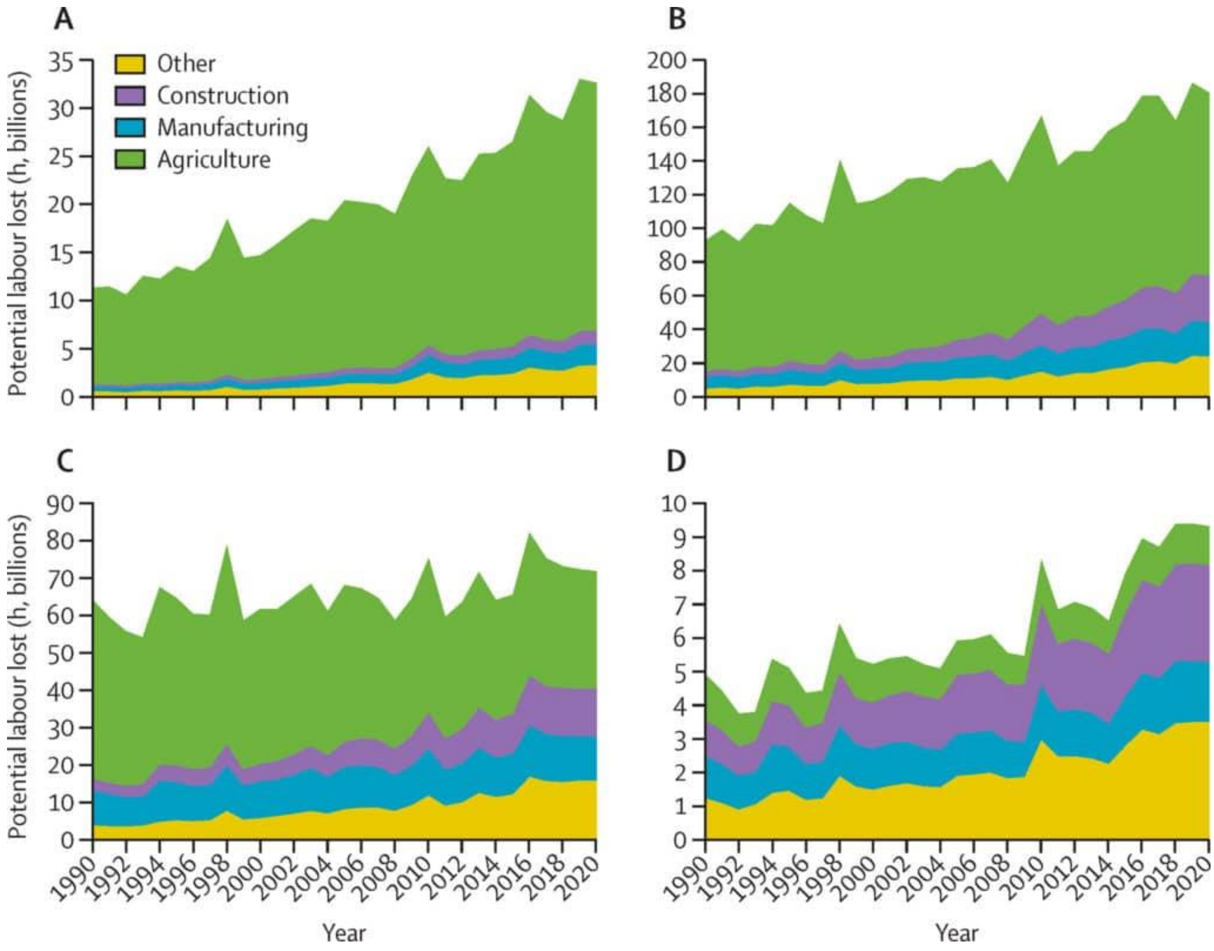
Source: Carbon Brief.

9. We Lost 302.4 Billion Work Hours to Excessive Heat In 2019

If you have ever been in humid South East Asia on a hot August day, you will know that working outdoors with shade is barely feasible, and without, simply dangerous. A report from The Lancet found that the number of work hours lost to heat across the globe increased from 199 billion in 2000 to 295 billion in 2020. That is equivalent to 88 work hours per employed person.



Of course, daytime outdoor labor is most exposed, which often targets lower-income areas and professions, especially agriculture.

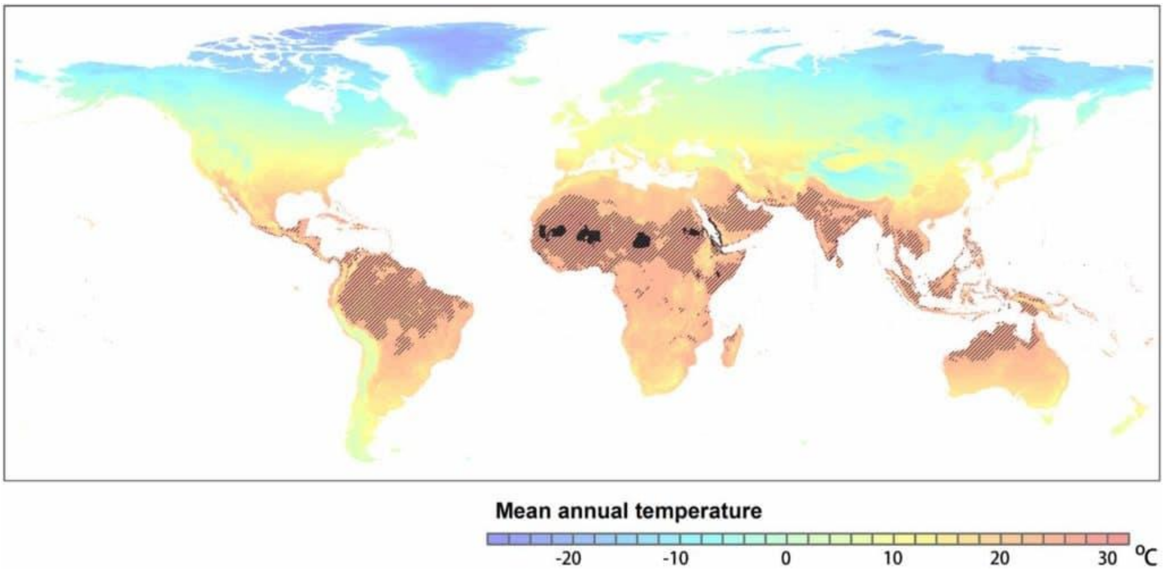


Potential labour lost due to heat-related factors in each sector (1990–2000) Low HDI (A), medium HDI (B), high HDI (C), and very high HDI (D) groups (2019 HDI country group). HDI=human development index. Source: The Lancet

10. It Could Become Too Hot to Live in Many Places By the End of the Century

This may the most catastrophic of our climate change facts. As of now, only 0.8% of the planet's land surface has mean annual temperatures above 29°C, mostly in the Sahara desert and Saudi Arabia (solid black in the map below).

A study by Xu et al. (2020) called "*Future of the Human Niche*" found that by 2070, under a high emissions scenario, these unbearable temperatures could expand to affect up to 3 billion people (black hashes).



Source: Xu et al. 2020, *Future of the Human Niche*.

Lost work hours are only one example of the impacts of extreme heat, especially when sustained over long periods. Others include agriculture yield losses, the spread of disease-carrying mosquitoes, and the increased need for air conditioning with its accompanying energy consumption.

11. The Cost of Inaction is Higher Than the Opposite

On the current path, climate change could end up costing us 11 to 14% of the global GDP by **mid-century**. Regression into a high emissions scenario would mean an 18% loss, while staying below 2°C would reduce the damage to only 4%.

It has been proposed that ending climate change would take between $300 billion and $50 trillion over the next two decades. Even if $50 trillion is the price tag, that comes down to $2.5 trillion a year, or just over 3% of the global GDP.

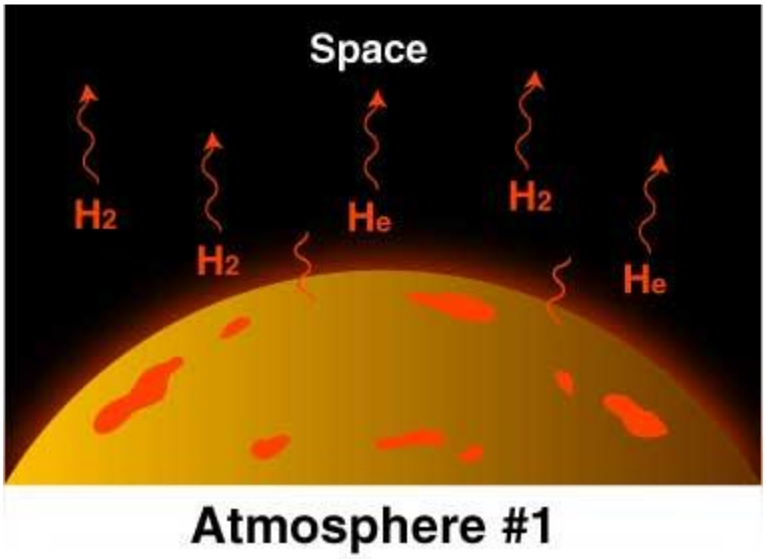
Origin of Atmosphere

- **Earth is believed to have formed about 5 billion years ago**.
- In the first 500 million years, a **dense atmosphere emerged from the vapor and gases** that were expelled **during degassing of the planet's interior.**
    - **During the cooling of the earth, gases and water vapour were released from the interior solid earth**. This started the evolution of the present atmosphere. **This process is called degassing.**

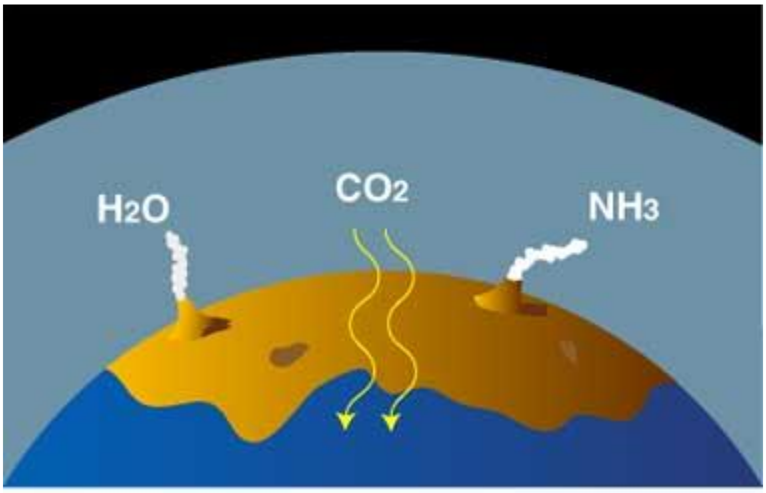
- These gases may have consisted of **hydrogen (H_2), water vapor, methane (CH_4) , and carbon oxides.**
- **Prior to 3.5 billion years ago** the atmosphere probably consisted of **carbon dioxide (CO_2), carbon monoxide (CO), water (H_2O), nitrogen (N_2), and hydrogen.**
- The **hydrosphere was formed 4 billion years ago** from the **condensation of water vapor**, resulting in oceans of water in which sedimentation occurred.
- The most important feature of the ancient environment was the **absence of free oxygen**. Evidence of such an **anaerobic reducing atmosphere** is hidden in early rock formations that contain many elements, such as iron and uranium, in their reduced states. Elements in this state are not found in the rocks of mid-Precambrian and younger ages, less than 3 billion years old.
- **One billion years ago**, early **aquatic organisms** called blue-green algae began **using energy from the Sun to split molecules of H_2O and CO_2** and recombine them into organic compounds and molecular oxygen (O_2). **This solar energy conversion process is known as photosynthesis**. Some of the photosynthetically created oxygen combined with organic carbon to recreate CO_2 molecules. The remaining oxygen accumulated in the atmosphere, touching off a massive ecological disaster with respect to early existing anaerobic organisms. As oxygen in the atmosphere increased, CO_2 decreased.
- **High in the atmosphere, some oxygen (O_2) molecules absorbed energy from the Sun's ultraviolet (UV) rays and split to form single oxygen atoms.** These atoms combining with remaining oxygen (O_2) to form ozone (O_3) molecules, which are very effective at absorbing UV rays. The thin layer of ozone that surrounds Earth acts as a shield, protecting the planet from irradiation by UV light.
- The amount of ozone required to shield Earth from biologically lethal UV radiation, wavelengths from 200 to 300 nanometers (nm), is believed to have been in existence **600 million years** ago. **At this time, the oxygen level was approximately 10% of its present atmospheric concentration.**
- **Prior to this period, life was restricted to the ocean.** The **presence of ozone enabled organisms to develop and live on the land. Ozone played a significant role in the evolution of life on Earth and allows life as we presently know it to exist.**

Phases of Atmosphere Formation

- ***Just formed Earth***: Like Earth, the **hydrogen (H2) and helium (He) was very warm**. These molecules of gas **moved so fast they escaped Earth's gravity and eventually all drifted off into space.**
    - **Earth's original atmosphere was probably just hydrogen and helium** because **these were the main gases in the dusty, gassy disk around the Sun from which the planets formed**. The **Earth and its atmosphere were very hot.** Molecules of hydrogen and helium **move really fast, especially when warm**. Actually, they moved so fast they eventually all escaped Earth's gravity and drifted off into space.

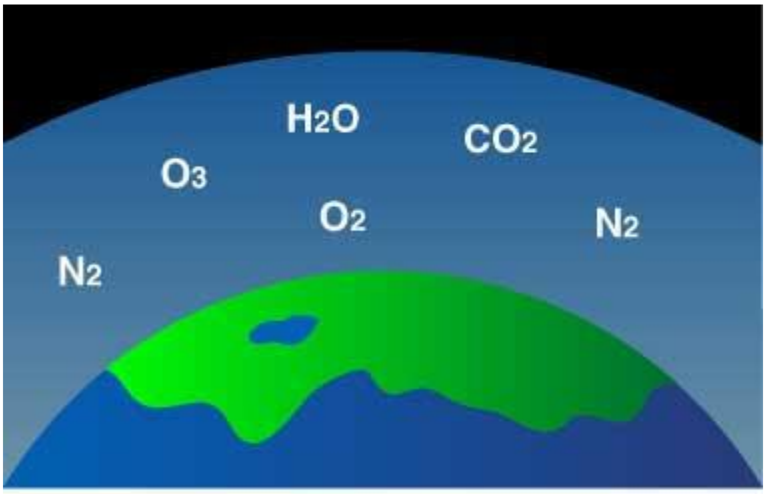


- ***Young Earth*:** Volcanoes released gases H2O (water) as steam, carbon dioxide (CO2), and ammonia (NH3). Carbon dioxide dissolved in seawater. Simple bacteria thrived on sunlight and CO2. By-product is oxygen (O2).
    - **Earth's "second atmosphere" came from Earth itself.** There were **lots of volcanoes**, many more than today because Earth's crust was still forming. The volcanoes released
        - **steam (H2O, with two hydrogen atoms and one oxygen atom),**
        - **carbon dioxide (CO2, with one carbon atoms and two oxygen atoms),**
        - **ammonia (NH3, with one nitrogen atom and three hydrogen atoms).**



Atmosphere #2

- ***Current Earth:*** Plants and animals thrive in balance. Plants take in carbon dioxide (CO2) and give off oxygen (O2). Animals take in oxygen (O2) and give off CO2. Burning stuff also gives off CO2.
    - **Much of the CO2 dissolved into the oceans**. Eventually, a simple form of **bacteria developed** that could live on energy from the Sun and carbon dioxide in the water, producing oxygen as a waste product. Thus, **oxygen began to build up in the atmosphere, while the carbon dioxide levels continued to drop**. Meanwhile, the **ammonia molecules in the atmosphere were broken apart by sunlight, leaving nitrogen and hydrogen**. The **hydrogen, being the lightest element, rose to the top of the atmosphere and much of it eventually drifted off into space.**



Atmosphere #3

Now we have Earth's "third atmosphere," the one we all know and love—an atmosphere containing enough oxygen for animals, including ourselves, to evolve.

So, **Plants and some bacteria use carbon dioxide and give off oxygen**, and **animals use oxygen and give off carbon-dioxide**—***how convenient!*** The atmosphere upon which life depends was created by life itself.

Do you know

Formation of the layered structure in the lithosphere:

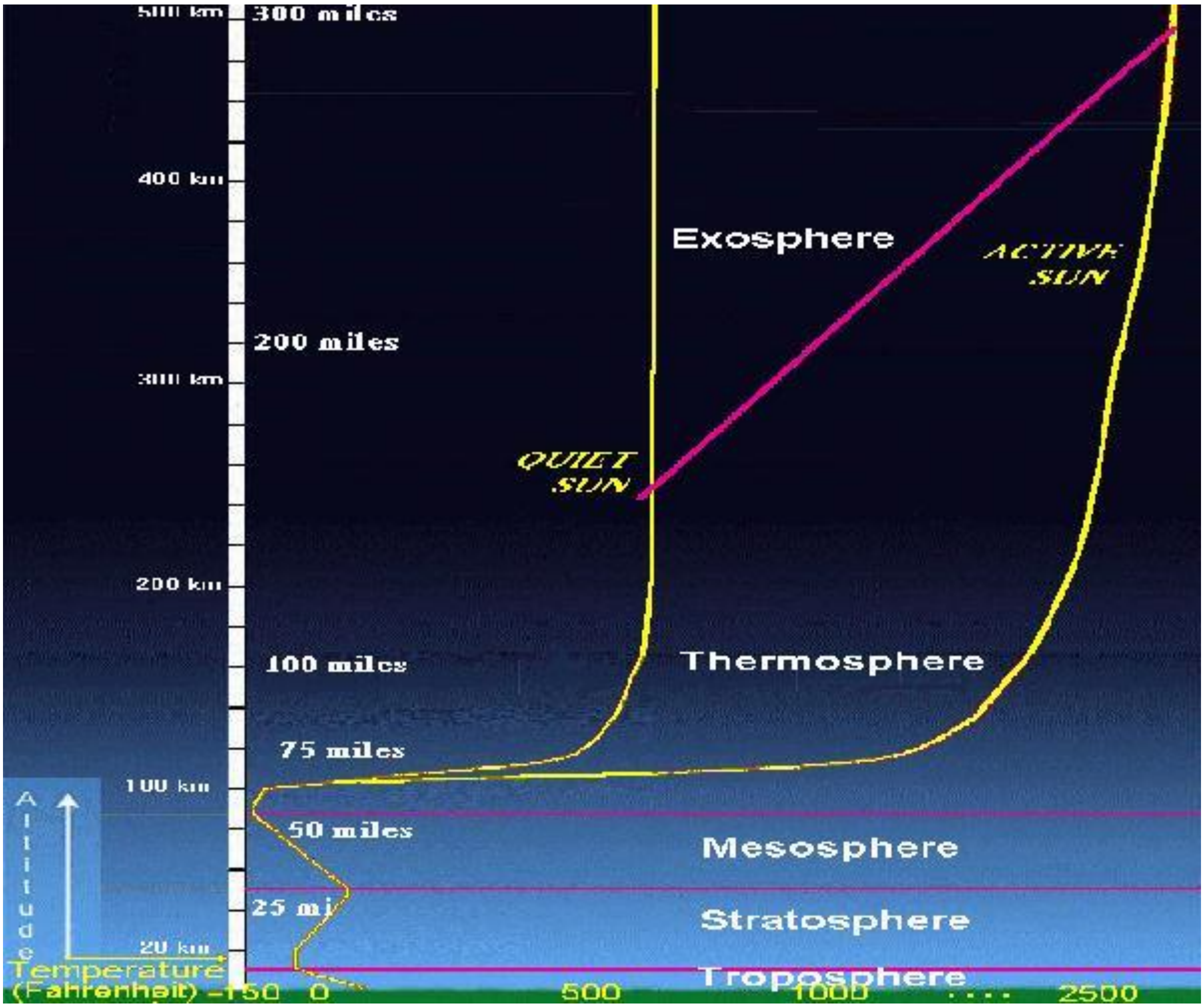
- The earth was mostly in a volatile state during its primordial stage.
- Due to the gradual increase in density the temperature inside has increased. As a result of the material inside started getting separated depending on their densities. **This process is called differentiation.**
- This allowed heavier materials (like iron) to sink towards the centre of the earth and the lighter ones to move towards the surface.
- Due to this earth got divided into layers like the crust (outermost), mantle, outer core, and inner core (innermost).
- **From the crust to the core, the density of the material increases.**

Atmospheric Structure

The gaseous area surrounding the planet is divided into several concentric strata or layers. About 99% of the total atmospheric mass is concentrated in the first 20 miles (32 km) above Earth's surface on the discovery of atmospheric structure.

THERMAL STRUCTURE

Atmospheric layers are characterized by variations in temperature resulting primarily from the absorption of solar radiation; visible light at the surface, near ultraviolet radiation in the middle atmosphere, and far ultraviolet radiation in the upper atmosphere.



Troposphere

The troposphere is the atmospheric layer closest to the planet and contains the largest percentage (around 80%) of the mass of the total atmosphere. Temperature and water vapor content in the troposphere decrease rapidly with altitude. Water vapor plays a major role in regulating air temperature because it absorbs solar energy and thermal radiation from the planet's surface. The troposphere contains 99 % of the water vapor in the atmosphere. Water vapor concentrations vary with latitude. They are greatest above the tropics, where they may be as high as 3 %, and decrease toward the polar regions.

All *weather* phenomena occur within the troposphere, although turbulence may extend into the lower portion of the stratosphere. Troposphere means "region of mixing" and is so named because of vigorous convective air currents within the layer.

The upper boundary of the layer, known as the *tropopause*, ranges in height from 5 miles (8 km) near the poles up to 11 miles (18 km) above the equator. Its height also varies with the seasons; highest in the summer and lowest in the winter.

Stratosphere

The stratosphere is the second major strata of air in the atmosphere. It extends above the tropopause to an altitude of about 30 miles (50 km) above the planet's surface. The air temperature in the stratosphere remains relatively constant up to an altitude of 15 miles (25 km). Then it increases gradually to up to the *stratopause*. Because the air temperature in the stratosphere increases with altitude, it does not cause convection and has a stabilizing effect on atmospheric conditions in the region. Ozone plays the major role in regulating the thermal regime of the stratosphere, as water vapor content within the layer is very low. Temperature increases with ozone concentration. Solar energy is converted to kinetic energy when ozone molecules absorb ultraviolet radiation, resulting in heating of the stratosphere.

The *ozone layer* is centered at an altitude between 10-15 miles (15-25 km). Approximately 90 % of the ozone in the atmosphere resides in the stratosphere. Ozone concentration in the this region is about 10 parts per million by volume (ppmv) as compared to approximately 0.04 ppmv in the troposphere. Ozone absorbs the bulk of solar ultraviolet radiation in wavelengths from 290 nm - 320 nm (UV-B radiation). These wavelengths are harmful to life because they can be absorbed by the nucleic acid in cells. Increased penetration of ultraviolet radiation to the planet's surface would damage plant life and have harmful environmental consequences. Appreciably large amounts of solar ultraviolet radiation would result in a host of biological effects, such as a dramatic increase in cancers.

A popular pastime of late seems to be sending things up on balloons into the stratosphere and posting the video on YouTube
A beer can goes to 90,000 feet altitude (17 miles, 27 km) and lands in the "drink"
A toy robot, a Lego Space Shuttle, a Thomas the Train toy, and even an armchair
A human goes to 128,000 feet (24+ miles, 39 km) and jumps breaking the sound barrier as he falls
A balloon is for fine tourists, but try a rocket to get there in a hurry!

Mesosphere

The mesosphere a layer extending from approximately 30 to 50 miles (50 to 85 km) above the surface, is characterized by decreasing temperatures. The coldest temperatures in Earth's atmosphere occur at the top of this layer, the *mesopause*, especially in the summer near the pole. The mesosphere has sometimes jocularly been referred to as the "ignorosphere" because it had been probably the least studied of the atmospheric layers. The stratosphere and mesosphere together are sometimes referred to as the *middle atmosphere*.

Thermosphere

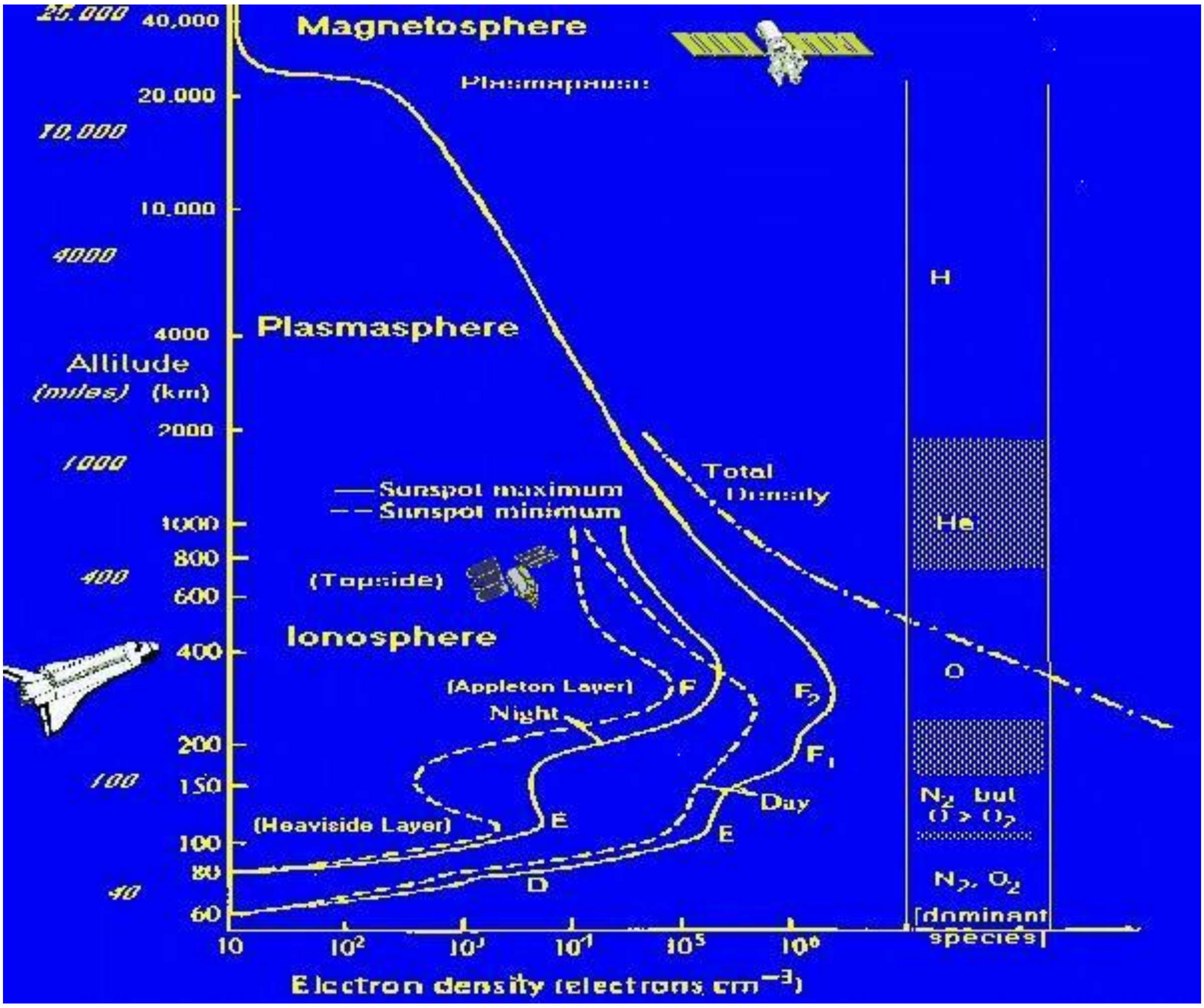
The thermosphere is located above the mesosphere. The temperature in the thermosphere generally increases with altitude reaching 600 to 3000 F (600-2000 K) depending on solar activity. This increase in temperature is due to the absorption of intense solar radiation by the limited amount of remaining molecular oxygen. At this extreme altitude gas molecules are widely separated. Above 60 miles (100 km) from Earth's surface the chemical composition of air becomes strongly dependent on altitude and the atmosphere becomes enriched with lighter gases (atomic oxygen, helium and hydrogen). Also at 60 miles (100 km) altitude, Earth's atmosphere becomes too thin to support aircraft and vehicles need to travel at orbital velocities to stay aloft. This demarcation between aeronautics and astronautics is known as the Karman Line. Above about 100 miles (160 km) altitude the major atmospheric component becomes atomic oxygen. At very high altitudes, the residual gases begin to stratify according to molecular mass, because of gravitational separation.

Exosphere

The exosphere is the most distant atmospheric region from Earth's surface. In the exosphere, an upward travelling molecule can escape to space (if it is moving fast enough) or be pulled back to Earth by gravity (if it isn't) with little probability of colliding with another molecule. The altitude of its lower boundary, known as the *thermopause* or *exobase*, ranges from about 150 to 300 miles (250-500 km) depending on solar activity. The upper boundary can be defined theoretically by the altitude (about 120,000 miles, half the distance to the Moon) at which the influence of solar radiation pressure on atomic hydrogen velocities exceeds that of the Earth's gravitational pull. The exosphere observable from space as the geocorona is seen to extend to at least 60,000 miles from the surface of the Earth. The exosphere is a transitional zone between Earth's atmosphere and interplanetary space.

MAGNETO-ELECTRONIC STRUCTURE

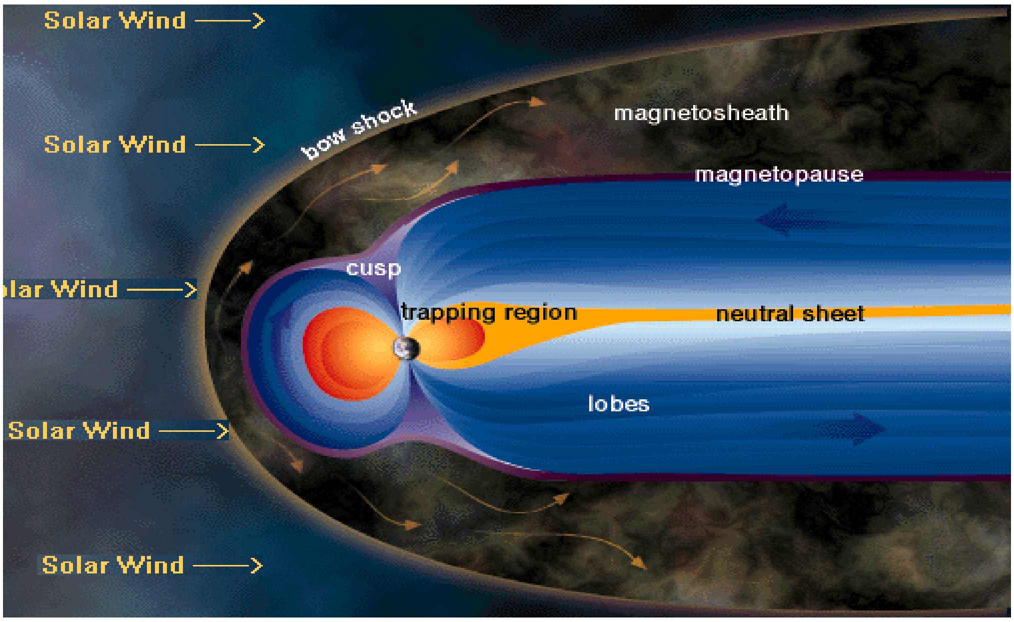
The upper atmosphere is also divided into regions based on the behavior and number of free electrons and other charged particles.



Ionosphere

The ionosphere is defined by atmospheric effects on radiowave propagation as a result of the presence and variation in concentration of free electrons in the atmosphere.

Magnetosphere



Outside the plasmapause, magnetic field lines are unable to corotate because they are influenced strongly by electric fields of solar wind origin. The magnetosphere is a cavity (also not spherical) in which the Earth's magnetic field is constrained by the solar wind and interplanetary magnetic field (IMF). The outer boundary of the magnetosphere is called the *magnetopause*. The magnetosphere is shaped like an elongated teardrop (like a Christmas Tree ornament) with the tail pointing away from the Sun. The magnetopause is typically located at about 10 Earth radii or some 35,000 miles (about 56,000 km) above the Earth's surface on the day side and stretches into a long tail, the *magnetotail*, a few million miles long (about 1000 Earth radii), well past the orbit of the Moon (at around 60 Earth radii), on the night side of the Earth. However, the Moon itself is usually not within the magnetosphere except for a couple of days around the Full Moon.

Beyond the magnetopause are the *magnetosheath* and *bow shock* which are regions in the solar wind disturbed by the presence of Earth and its magnetic field.

Composition of the atmosphere

- **The atmosphere is made up of different gases, water vapour and dust particles.**
- **The composition of the atmosphere is not static and it changes according to the time and place.**

Gases of the atmosphere

Constituent	Percent by Volume	Concentration in Parts Per Million (PPM)
Nitrogen (N_2)	78.084	780,840.0
Oxygen (O_2)	20.946	209,460.0
Argon (Ar)	0.934	9,340.0
Carbon dioxide (CO_2)	0.036	360.0
Neon (Ne)	0.00182	18.2
Helium (He)	0.000524	5.24
Krypton (Kr)	0.000114	1.14
Hydrogen (H_2)	0.00005	0.5

- **The atmosphere is a mixture of different types of gases.**
- **Nitrogen and oxygen are the two main gases in the atmosphere and 99 percentage of the atmosphere is made up of these two gases.**
- **Other gases like argon, carbon dioxide, neon, helium, hydrogen, etc. form the remaining part of the atmosphere.**
- **The portion of the gases changes in the higher layers of the atmosphere in such a way that oxygen will be almost negligible quantity at the heights of 120 km.**
- **Similarly, carbon dioxide (and water vapour) is found only up to 90 km from the surface of the earth.**

CARBON DIOXIDE:

- **Carbon dioxide is meteorologically a very important gas.**
- **It is transparent to the incoming solar radiation (insolation) but opaque to the outgoing terrestrial radiation.**
- **It absorbs a part of terrestrial radiation and reflects back some part of it towards the earth's surface.**
- **Carbon dioxide is largely responsible for the greenhouse effect.**
- **When the volume of other gases remains constant in the atmosphere, the volume of the carbon dioxide has been rising in the past few decades mainly because of the burning of fossil**

fuels. This rising volume of carbon dioxide is the main reason for global warming.

OZONE GAS:

- **Ozone is another important component of the atmosphere found mainly between 10 and 50 km above the earth's surface.**
- **It acts as a filter and absorbs the ultra-violet rays radiating from the sun and prevents them from reaching the surface of the earth.**
- **The amount of ozone gas in the atmosphere is very little and is limited to the ozone layer found in the stratosphere.**

Water Vapour

- **Gases form of water present in the atmosphere is called water vapour.**
- **It is the source of all kinds of precipitation.**
- **The amount of water vapour decreases with altitude. It also decreases from the equator (or from the low latitudes) towards the poles (or towards the high latitudes).**
- **Its maximum amount in the atmosphere could be up to 4% which is found in the warm and wet regions.**
- **Water vapour reaches in the atmosphere through evaporation and transpiration. Evaporation takes place in the oceans, seas, rivers, ponds and lakes while transpiration takes place from the plants, trees and living beings.**
- **Water vapour absorbs part of the incoming solar radiation (insolation) from the sun and preserves the earth's radiated heat. It thus acts like a blanket allowing the earth neither to become too cold nor too hot.**
- **Water vapour also contributes to the stability and instability in the air.**

Dust Particles

- **Dust particles are generally found in the lower layers of the atmosphere.**
- **These particles are found in the form of sand, smoke-soot, oceanic salt, ash, pollen, etc.**

- **Higher concentration of dust particles is found in subtropical and temperate regions due to dry winds in comparison to equatorial and polar regions.**
- **These dust particles help in the condensation of water vapour. During the condensation, water vapour gets condensed in the form of droplets around these dust particles and thus clouds are formed.**

What is Weather?

Weather describes the condition of the atmosphere over a short period of time e.g. from day to day or week to week, while climate describes average conditions over a longer period of time.

As an air mass warms, it becomes lighter and rises higher into the atmosphere. As an air mass cools, it becomes heavier and sinks. Pressure differences between masses of air generate winds, which tend to blow from high-pressure areas to areas of low pressure. Fast-moving, upper atmosphere winds known as jet streams help move weather systems around the world.

Large weather systems called cyclones rotate counterclockwise in the Northern Hemisphere (clockwise in the Southern Hemisphere); they are also called "lows," because their centers are low-pressure areas. Clouds and precipitation are usually associated with these systems. Anticyclones, or "highs," rotate in the opposite direction and are high-pressure areas – usually bringing clearer skies and more settled weather.

The boundary between two air masses is called a weather front. Here, wind, temperature, and humidity change abruptly, producing atmospheric instability. When things get "out of balance" in the atmosphere, storms develop, bringing rain or snow and sometimes thunder and lightning too.

The weather you experience is influenced by many factors, including your location's latitude, elevation, and proximity to water bodies. Even the degree of urban development, which creates "heat islands," and the amount of snow cover, which chills an overlying air mass, play important roles.

The next time you watch a weather report on television, think about the many factors that influence our weather, some thousands of miles away, that help make the weather what it is.

What is Climate?

Weather describes the condition of the atmosphere over a short period of time e.g. from day to day or week to week, while climate describes average conditions over a longer period of time.

The climate of an area or country is known through the average weather over a long period of time. If an area has more dry days throughout the year than wet days, it would be described as a dry climate; a place which has more cold days than hot days would make it known to have a cold climate.

FACTORS THAT INFLUENCE CLIMATE & WEATHER

FACTORS THAT INFLUENCE WEATHER

There are many factor that influence weather, many of which we cannot see.

The Water Cycle

As the sun warms the surface of the Earth, water rises in the form of water vapor from lakes, rivers, oceans, plants, the ground, and other sources. This process is called evaporation.

Water vapor provides the moisture that forms clouds; it eventually returns to Earth in the form of precipitation, and the cycle continues.

Air Masses

When air hovers for a while over a surface area with uniform humidity and temperature, it takes on the characteristics of the area below. For example, an air mass over the tropical Atlantic Ocean would become warm and humid; an air mass over the winter snow and ice of northern Canada would become cold and dry. These massive volumes of air often cover thousands of miles and reach to the stratosphere. Overtime, mid-latitude cyclonic storms and global wind patterns move them to locations far from their source regions.

What happens when 2 air masses meet? The cold air pushes the hot air upwards, when going up the temperature drops and the air can't hold as much water when it is warm, so the cold water molecules condense and form clouds.

Jet Stream

A jet stream is the name given to the area of air above where two air masses of different temperature converge e.g. a cold front meeting a warm front. The greater the temperature difference between the air masses, the greater the air pressure difference, and the faster the wind blows in the jet stream. This river of air has wind speeds which often exceed 100 mph, and sometimes over 200 mph. Jet streams more commonly form in the winter, when there is a greater difference between the temperature of the cold continental air masses and warm oceanic air masses.

FACTORS THAT INFLUENCE CLIMATE

There are lots of factors that influence our climate

Elevation or Altitude effect climate

Normally, climatic conditions become colder as altitude increases. "Life zones" on a high mountain reflect the changes, plants at the base are the same as those in surrounding countryside, but no trees at all can grow above the timberline. Snow crowns the highest elevations.

Prevailing global wind patterns

There are 3 major wind patterns found in the Northern Hemisphere and also 3 in the Southern Hemisphere. These are average conditions and do not essentially reveal conditions on a particular day. As seasons change, the wind patterns shift north or south. So does the intertropical convergence zone, which moves back and forth across the Equator. Sailors called this zone the doldrums because its winds are normally weak.

Latitude and angles of the suns rays. As the Earth circles the sun, the tilt of its axis causes changes in the angle of which sun's rays contact the earth and hence changes the daylight hours at different latitudes. Polar regions experience the greatest variation, with long periods of limited or no sunlight in winter and up to 24 hours of daylight in the summer.

Topography

The Topography of an area can greatly influence our climate. Mountain ranges are natural barriers to air movement. In California, winds off the Pacific ocean carry moisture-laden air toward the coast. The Coastal Range allows for some condensation and light precipitation. Inland, the taller Sierra Nevada range rings more significant precipitation in the air. On the western slopes of the Sierra Nevada, sinking air warms from compression, clouds evaporate, and dry conditions prevail.

Effects of Geography

The position of a town, city or place and its distance from mountains and substantial areas of water help determine its prevailing wind patterns and what types of air masses affect it. Coastal areas may enjoy refreshing breezes in summer, when cooler ocean air moves ashore. Places south and east of the Great Lakes can expect "lake effect" snow in winter, when cold air travels over relatively warmer waters.

In spring and summer, people in Tornado Alley in the central United States watch for thunderstorms, these storms are caused where three types of air masses frequently converge: cold and dry from the north, warm and dry from the southwest, and warm and moist from the Gulf of Mexico – these colliding air masses often generate tornado storms.

Surface of the Earth

Just look at any globe or a world map showing land cover, and you will see another important factor which has a influence on climate: the surface of the Earth. The amount of sunlight that is absorbed or reflected by the surface determines how much atmospheric heating occurs. Darker areas, such as heavily vegetated regions, tend to be good absorbers; lighter areas, such as snow and ice-covered regions, tend to be good reflectors. The ocean absorbs and loses heat more slowly than land. Its waters gradually release heat into the atmosphere, which then distributes heat around the globe.

Climate change over time

Cold and warm periods punctuate Earth's long history. Some were fairly short; others spanned hundreds of thousands of years. In some cold periods, glaciers grew and spread over large regions. In subsequent warm periods, the ice retreated. Each period profoundly affected plant and animal life. The most recent cool period, often called the "Little Ice Age," ended in western Europe around 1850.

Since the turn of the 20th century, temperatures have been rising steadily throughout the world. But it is not yet clear how much of this global warming is due to natural causes and how much derives from human activities, such as the burning of fossil fuels and the clearing of forests.

CLIMATE CLASSIFICATION

climate classification, the formalization of systems that recognize, clarify, and simplify climatic similarities and differences between geographic areas in order to enhance the scientific understanding of climates. Such classification schemes rely on efforts that sort and group vast amounts of environmental data to uncover patterns between interacting climatic processes. All such classifications are limited since no two areas are subject to the same physical or biological forces in exactly the same way. The creation of an individual climate scheme follows either a genetic or an empirical approach.

Koeppen's Climate Classification

Koeppen's Climate Classification

- Koeppen's Classification of climate is the most commonly used classification of climate.
- This climate classification scheme was developed by Wladimir Peter Koeppen in 1884.
- He recognized a close relationship between the distribution of vegetation and climate.
- The categories are based on the data of annual and monthly averages of temperature and precipitation.

- He selected specific values of temperature and precipitation and related them to the distribution of vegetation and used these values for classifying the climates.
- The Koeppen climate classification system recognizes five major climatic types and each type is designated by a capital letter- A, B, C, D, E, and H.
- The seasons of dryness are indicated by the small letters: f, m, w, and s.
  - f -no dry season
  - m – Monsoon climate
  - w- Winter dry season
  - s – Summer dry season
- The small letters a, b, c, and d refer to the degree of severity of temperature.

List of climatic groups and their characteristics according to Koeppen

Group	**Characteristics**
A- Tropical	The average temperature of the coldest month is 18° C or higher
B- Dry Climates	Potential evaporation exceeds precipitation
C- Warm Temperate	The average temperature of the coldest month of the (Mid-latitude) climates years is higher than minus 3°C but below 18°C
D- Cold Snow forest	The average temperature of the coldest month is minus 3° C or below
E- Cold Climates	Cold Climates Average temperature for all months is below 10° C

H- Highlands	Cold due to elevation

Climatic Types According to Koeppen

Group	Type	Letter Code	Characteristics
A-Tropical Humid Climate	Tropical Wet Tropical Monsoon Tropical wet and dry	Af Am Aw	No dry season Monsoonal, Short dry season Winter dry season
B-Dry Climate	Subtropical Steppe Subtropical Desert Mid-latitude Steppe Mid-latitude Desert	BSh BWh BSk BWk	Low-latitude semi-arid or dry Low-latitude arid or dry Mid-latitude semi-arid or dry Mid-latitude arid or dry
C-Warm temperate Climates	Humid subtropical Mediterranean Marine west coast	Cfa Cs Cfb	No dry season Dry hot summer No dry season, warm and cool summer
D- Cold Snow-forest Climates	Humid Continental Subarctic	Df Dw	No dry season, severe winter Winter dry and very severe
E-Cold climates	Tundra Polar ice cap	ET EF	No true summer Perennial ice
H-highland	Highland	H	Highland with snow cover

Human Impact on the Atmosphere

Objectives:

1. Discuss how human activities can affect the atmosphere.

2. Compare and contrast acid rain, smog, and global warming.

Key Terms:

As an industrialized nation we are constantly forced to weigh the effects of manufacturing versus the benefits of the products. Pollution is a term given to a harmful (gas or particulate) substance that is released into the environment. Although some pollution occurs naturally (volcanoes) most pollutants are produced as a result of industry.

There are several different classifications:

1. irritant - causing inflammation and discomfort
2. teratogen - causing birth defects
3. carcinogen - causing cancers

Proliferation of of pollutants depletes our ozone layer, contributes to increased global warming, increases the amount of acid rain, and puts people at risk for health problems. In 1970 the Clean Air Act sought to address 6 of the worst known pollutants. The EPA (Environmental Protection Agency) monitors these pollutants and sets acceptable levels for their release.

1. CO - Carbon monoxide
2. NO2 - Nitrogen dioxide
3. SO2 - Sulfur dioxide
4. Particulate pollutant
5. Pb - lead
6. O3 - ozone

(www.tanplusforhealth.com/Images/ Ess%20Air/Pollutants.gif)

Acid Rain

Acid rain is created as CO2, NO2 and SO2 mix with water in the atmosphere to create weak acids that return to the Earth's surface as rain. This rain accumulates in the water table and in the fresh water systems where it becomes concentrated killing wildlife, vegetation, and bacteria.

Smog

Smog is a term used to describe a collection of pollution that is condensed in a particular area.

- generally a brownish color
- mostly formed from the burning of fossil fuels
- rich in NO2, SO2 and CO2
- creates surface ozone
- creates a temperature inversion (top layer warmer than the lower layer) which aids in the trapping of more smog at increasingly lower levels

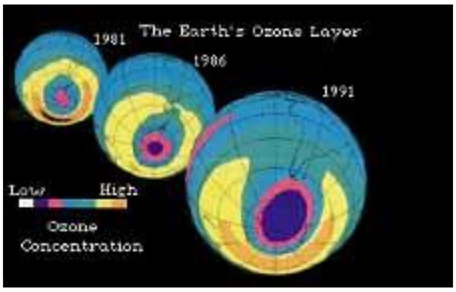
(www.epa.gov/airnow/health/ images/smog.jpg)

Ozone Concentration (ppm) (8-hour average, unless noted)	Air Quality Index Values	Air Quality Descriptor

0.0 to 0.064	0 to 50	Good
0.065 to 0.084	51 to 100	Moderate
0.085 to 0.104	101 to 150	Unhealthy for Sensitive Groups
0.105 to 0.124	151 to 200	Unhealthy
0.125 (8-hr.) to 0.404 (1-hr.)	201 to 300	Very Unhealthy

Ozone Depletion

Ozone depletion is perhaps the most overlooked environmental hazard. Although the banning of CFCs in the 1970s helped slow the thinning of this layer, the depletion continues today. Effects of thinning are:



✖increased skin cancers
✖premature aging
✖plant damage
✖reduction in the amount of phytoplankton in the oceans

(www.epa.gov/grtlakes/seahome/ thumbs/ozone.jpg)

Global Warming

Although global warming is vital to the survival of the planet, its rate of increase threatens life. Most scientists will conclude that global warming is a natural process that has cycles of increase and decrease but there are undeniable contributions that we are making that have significantly sped up the process of warming.

✖Burning of fossil fuels - releases CO_2
✖Global deforestation - increases CO_2 levels

<u>These activities have led to the the following consequences:</u>

✖Rising sea levels
✖Increased frequency of damaging storms - hurricanes and tornados
✖Increased frequency of severe weather - heat waves and droughts

✖Relocation of global crop growing areas

Origin and Evolution of the Atmosphere

Prebiotic atmosphere

The atmosphere of the Earth (and also of Venus and Mars) is generally believed to have its origin in relatively volatile compounds that were incorporated into the solids from which these planets accreted. Such compounds could include nitrides (a source of N_2), water (which can be taken up by silica, for example), carbides, and hydrogen compounds of nitrogen and carbon. Many of these compounds (and also some noble gases) can form clathrate complexes with water and some minerals which are fairly stable at low temperatures. The high temperatures developed during the later stages of accretion as well as subsequent heating produced by decay of radioactive elements presumably released these gases the surface. Even at the present time, large amounts of CO_2 , water vapor, N_2, HCl, SO_2 and H_2S are emitted from volcanos. The more reactive of these gases would be selectively removed from the atmosphere by reaction with surface rocks or dissolution in the ocean, leaving an atmosphere enriched in its present major components with the exception of oxygen which is discussed in the next section.

Any hydrogen present would tend to escape into space, causing the atmosphere to gradually became less reducing. However there is now some doubt that hydrogen and other volatiles (mainly the inert gases) were present in the newly accreted planets in anything like their cosmic abundance. The main evidence against this is the observation that gases such as helium, neon and argon, which are among the ten most abundant elements in the universe, are depleted on the earth by factors of 10^{-7} to 10^{-11} . This implies that there was a selective removal of these volatiles prior to or during the planetary accretion process.

The overall oxidation state of the earth's mantle is not consistent with what one would expect from equilibration with highly reduced volatiles, and there is no evidence to suggest that the composition of the mantle has not remained the same. If this is correct, then the primitive atmosphere may well have had about the same composition as the gases emitted by volcanos at the present time. These consist mainly of water and CO_2, together with small amounts of N_2, H_2, H_2S, SO_2, CO, CH_4, NH_3, HCl and HF. If water vapor was a major component of outgassing of the accreted earth, it must have condensed quite rapidly into rain. Any significant concentration of water vapor in the atmosphere would have led to a runaway greenhouse effect, resulting in temperatures as high as 400°C.

Origin of Atmospheric Oxygen

Free oxygen is never more than a trace component of most planetary atmospheres. Thermodynamically, oxygen is much happier when combined with other elements as oxides; the pressure of O_2 in equilibrium with basaltic magmas is only about 10^{-7} atm. Photochemical decomposition of gaseous oxides in the upper atmosphere is the major source of O_2 on most planets. On Venus, for example, CO_2 is broken down into CO and O_2 . On the earth, the major inorganic source of O_2 is the photolysis of water vapor; most of the resulting hydrogen escapes into space, allowing the O2 concentration to build up. An estimated 2 10^{11} g of O_2 per year is generated in this way. Integrated over the earth's history, this amounts to less than 3% of the present oxygen abundance. The partial pressure of O_2 in the prebiotic atmosphere is estimated to be no more than 10^{-3} atm, and may have been several orders of magnitude less. The major source of atmospheric oxygen on the earth is photosynthesis carried out by green plants and certain bacteria:

H2O+CO2⟶(CH2O)x+O2(3.2.1)(3.2.1)H2O+CO2⟶(CH2O)x+O2

A historical view of the buildup of atmospheric oxygen concentration since the beginning of the sedimentary record (3.7 10^9 ybp) can be worked out by making use of the fact that the carbon in the product of the above reaction has a slightly lower C^{13} content than does carbon of inorganic origin. Isotopic analysis of carbon-containing sediments thus provides a measure of the amounts of photosynthetic O_2 produced at various times in the past.

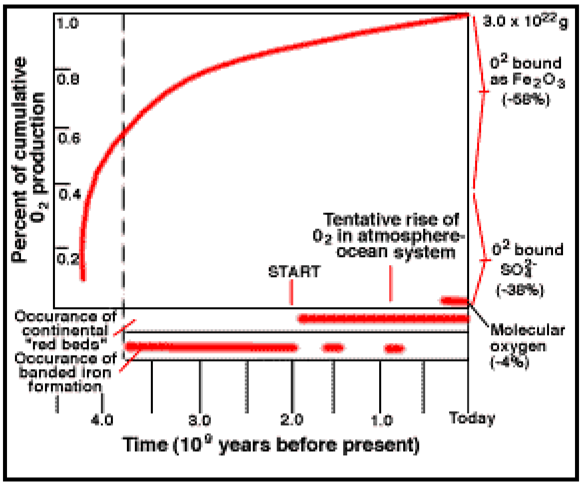


Figure 3.2.13.2.1: ***Evolution of atmospheric oxygen content.*** *Note carefully that the curve plots cumulative O_2 production, but that until a few hundred million years ago, most of this was taken up by Fe(II) compounds in the crust and by reduced sulfur; only after this massive "oxygen sink" became filled did free O_2 begin to accumulate in the atmosphere.*

Carbon Dioxide

Carbon dioxide has probably always been present in the atmosphere in the relatively small absolute amounts now observed (around 54 x 10^{15} mol = 54

Pmol). The reaction of CO_2 with silicate-containing rocks to form precambrian limestones suggest a possible moderating influence on its atmospheric concentration throughout geological time.

CaSiO3+CO2→CaCO3+SiO2(3.2.2)(3.2.2)CaSiO3+CO2→CaCO3+SiO2

About ten percent of the atmospheric CO2��2 is taken up each year by photosynthesis. Of this, all except 0.05 percent is returned by *respiration*, almost entirely due to microorganisms. The remainder leaks into the slow part of the geochemical cycle, mostly as buried carbonate sediments.

Since the advent of the industrial revolution in 1860, the amount of CO2��2 in the atmosphere has been increasing. Isotopic analysis shows that most of this has been due to fossil-fuel combustion; in recent years, the mass of carbon released to the atmosphere by this means has been more than ten times the estimated rate of natural removal into sediments. The large-scale destruction of tropical forests, which has accelerated greatly in recent years, is believed to exacerbate this effect by removing a temporary sink for CO2��2.

The oceans have a large absorptive capacity for CO2��2 owing to its reaction with carbonate:

CO2+CO2–3⇌2HCO3–(3.2.3)(3.2.3)CO2+CO32–⇌2HCO3–

There is about 60 times as much inorganic carbon in the oceans as in the atmosphere. However, efficient transfer takes place only into the topmost (100 m) wind-mixed layer of the ocean, which contains only about one atmosphere equivalent of CO2��2. Further uptake is limited by the very slow transport of water into the deep ocean, which takes around 1000 years. For this reason, the buffering effect of the oceans on atmospheric CO2��2 is not very effective; only about ten percent of the added CO2��2 is taken up by the oceans.

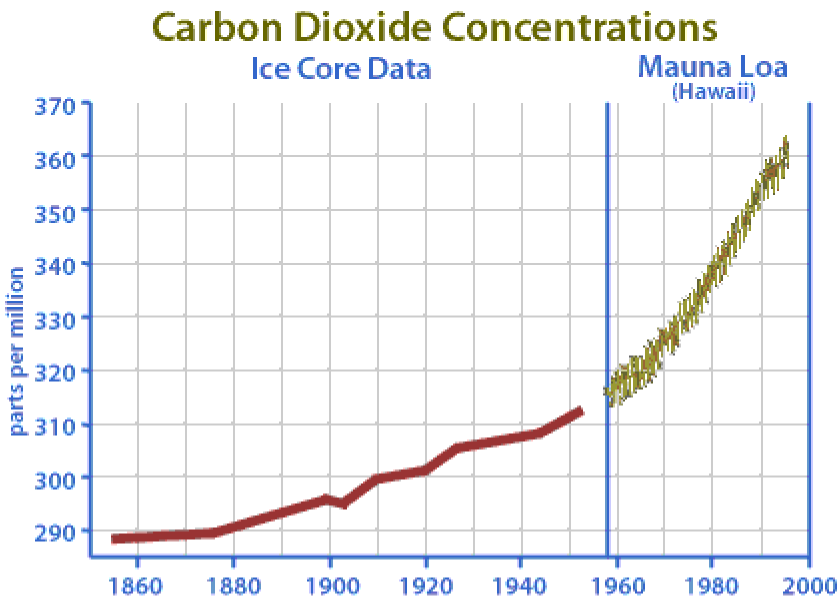


Figure. This set of plots, taken from an [ETE page] with data from NASA, shows the dramatic increase in atmospheric CO_2since the beginning of the industrial age. The "squiggles" on the Mauna Loa data reflect seasonal variations; more $CO_2$is taken up during the longer-daylight periods of the summer.

If we examine the vertical structure the atmosphere in different places we will find it varies in height, being lowest at the poles and highest at the equator. The varying height is due to the spatial variation in heating of the Earth's surface and thus the atmosphere above. This fact makes it difficult to define exact heights for the layers of the atmosphere. The solution is to subdivide the atmosphere not on the basis of fixed heights, but on temperature change.
Figure 3.2.13.2.1 illustrates the way in which the atmosphere is divided using temperature change as the primary criterion.

Troposphere and Tropopause

The ***troposphere*** is the layer closest to the Earth's surface. The graph of temperature change indicates that air temperature decreases with an increase in altitude through this layer. Air temperature normally decreases with height above the surface because the primary source of heating for the air is the Earth.

The rate of change in temperature with altitude is called the ***environmental lapse rate of temperature (ELR)*** The ELR varies from day-to-day at a place, and from place to place on any given day. The ***normal lapse rate of temperature*** is the average value of the ELR, 0.65^o C /100 meters. That is, at any particular place and on any given day the actual ELR may be larger or smaller, but on average has a value of 0.65^o C /100 m. So if I went outside today it could be 0.62^o C /100 m and then tomorrow it might be 0.68^o C /100 m. The ELR also varies from place to place on a given day. That is, at Chicago, Illinois it might be 0.65^o C /100 m and on the same day it could be 0.62^o C /100 m over London, England.

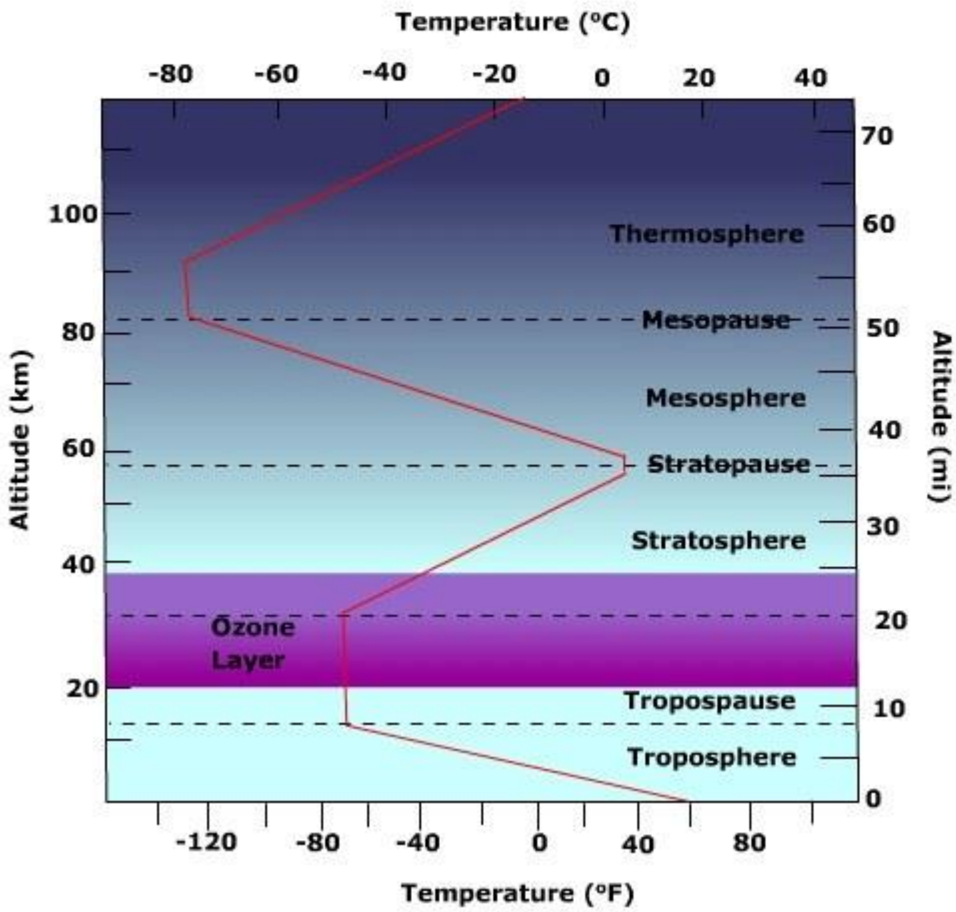


Figure 3.2.13.2.1: Structure of the Atmosphere

Under the right conditions, the air temperature may actually increase with an increase in altitude above the Earth. When this occurs we are experiencing an ***inverted lapse rate of temperature***, or simply an ***inversion***. Shallow surface inversions are typical over the snow covered surfaces of subarctic and polar regions, and sometimes occur when high pressure cells inhabit a region.

The ***tropopause*** lies above the troposphere. Here the temperature tends to stay the same with increasing height. The tropopause acts as a "lid" on the troposphere preventing air from rising upwards into the stratosphere.

Stratosphere

Above the tropopause lies the stratosphere. Note in Figure 3.2.13.2.1 that the temperature of the air does not change with an increase in elevation. If a layer of air exhibits no change in temperature with an increase in elevation we typically refer to it as an *isothermal layer* i.e. layer of equal temperature. Through most of the stratosphere the air temperature increases with an increase in elevation creating a temperature inversion. The inverted lapse rate of temperature is due to the presence of stratospheric ozone which is a good absorber of ultra-violet radiation emitted by the Sun. As energy penetrates downward, less and less is available for lower layers and hence the temperature decreases toward the bottom of the stratosphere. The downward reduction of heat transfer due to solar energy absorption from above is offset by the heat given off by the Earth creating the isothermal layer at the bottom of the stratosphere. At the top of the stratosphere lies the stratopause. Like the tropopause, the stratopause is an isothermal layer that separates the stratosphere from the mesosphere.

Mesosphere and beyond

It is the properties of the previously discussed layers that affects most of what we study in physical geography. Processes acting in layers above the stratopause have relatively little impact on our elemental study of Earth near-surface processes. In the **mesosphere** air temperatures begin to decrease with increasing altitude. 99.9 percent of the gases that comprise the atmosphere lie below this level. The air of the mesosphere is thus extremely thin and air pressure very small. With very few molecules like ozone capable of absorbing solar radiation, especially near the top of the layer, the air temperature decreases with height. The mesopause separates the mesosphere from the thermosphere above.

Figure 3.2.13.2.1 shows air temperature increasing with increasing altitude in the **thermosphere**. Here, energetic oxygen molecules absorb incoming solar radiation raising the layer's temperature. Because solar activity determines the temperature of the layer, temperature at the top of the layer is warmer than that near the bottom of the thermosphere. Even though the temperatures are quite high in the thermosphere, the heat content of the layer is very low due the low density of air at this level.

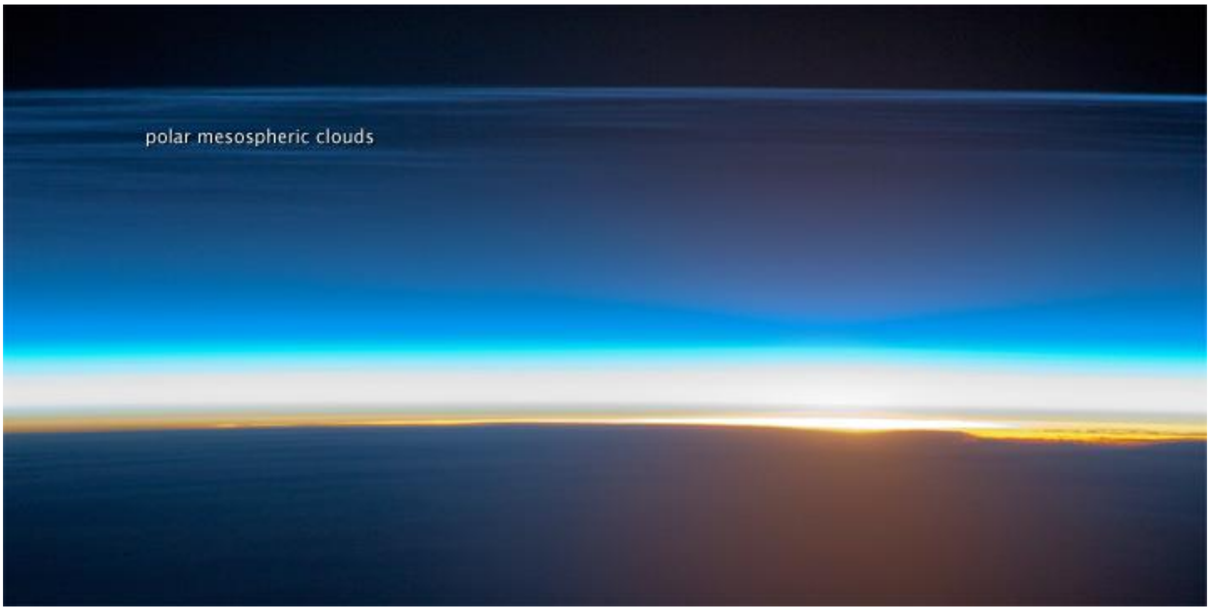


Figure 3.2.23.2.2: Polar Mesospheric clouds. (Courtesy NASA EOS, Source)

Polar mesospheric clouds are also known as noctilucent or "night-shining" clouds as they are usually seen at twilight. These clouds form at high latitudes and high altitudes (76 - 85 kilometers) near the boundary of the mesosphere and thermosphere.

Functional layers

Two layers, the **ionosphere** and ozonosphere are identified when using function as the criterion for subdivision. The ionosphere is not really a layer of the atmosphere, but an electrified field of ions and free electrons. The ionosphere absorbs cosmic rays, gamma rays, X-rays, and shorter wavelengths of ultraviolet radiation. The spectacular display of auroral lights is generally found in this region.

The **ozonosphere**, also called the "ozone layer", is the concentrated layer of ozone found in the stratosphere. Ozone (O_3) absorbs ultraviolet light between 0.1 - 0.3ตm. The ozone layer absorbs 97 - 99% of the Sun's ultraviolet light that can be harmful to life on earth. Though relatively constant through millions of years, seasonal fluctuations of ozone especially over the Arctic and Antarctic are common. The ozone layer is thinner at the equator and thicker at the poles. Ozone concentrations are highest in the spring and generally lowest during the autumn

UNIT 2

Global Warming and the Greenhouse Effect

 OBJECTIVES :Explain the impact of the greenhouse effect on planet Earth Describe greenhouse gases and their effects Explain how human activities have contributed to global warming Describe the effects of global warming on people and the land5. Give examples of what people can do about the amount of greenhouse gases in the atmosphere.

 Global Warming What is it?
Earth has warmed by about 1 º F over the past 100 years. But why? And How?Scientists are not exactly sure .The earth could be getting warm on it's own.

 However … Many of the world's leading climate scientists think that things people do are helping to make the Earth warmer . Scientists are sure about the greenhouse effect. They know that greenhouse gases make the earth warmer by trapping energy in the atmosphere.

 What is the Greenhouse effect?
The greenhouse effect is the rise in temperature that the Earth experiences because certain gases in the atmosphere trap heat from the Sun's rays.

 Have you seen a greenhouse?
Most greenhouses look like small glasshouses.Green houses are used to grow plants, especially in the winter.

 How do greenhouses work?
Greenhouses work by trapping heat from the sun. The glass panels of the greenhouse let in light but keep heat from escaping.

How do greenhouses work? This causes the greenhouse to heat up much like the inside of a car parked in sunlight, and keeps the plants warm enough to live in the winter.

The Greenhouse Effect The Earth's atmosphere is all around us. It is the air we breathe.Greenhouse gases in the atmosphere behave much like the glass panes in a greenhouse.

The Greenhouse Effect Sunshine enters the Earth's atmosphere passing through the blanket of greenhouse gases. As it reaches the Earth's surface, land, water, and biosphere absorb the sunlight's energy! Once absorbed this energy is sent back into the atmosphere.

How do greenhouses work? Some of the energy passes back into space. Much of it remains trapped in the atmosphere by the greenhouse gases, causing our world to heat up.

The greenhouse effect is important. Without the greenhouse effect, the Earth would not be warm enough for humans to live. But if the greenhouse effect becomes stronger, it could make the Earth warmer than usual. Even a little warming causes problems for plants and animals.

Greenhouse Effect Without these gases, heat would escape back into space and Earth's average temperature would be about 60 º F colder. Because of how they warm our world, these gases are referred toas greenhouse gases.

What are these gases? The greenhouse gases are: Water Vapour Carbon dioxide Nitrous Oxide Methane CFCs

Water Vapour There is more water in the atmosphere than carbon dioxide so most of the greenhouse heating of the Earth's surface is due to water vapour. The water vapour content in the atmosphere is constant which means it hasn’t changed.

- Water Vapour Water vapour is the biggest contributor to the “natural greenhouse effect ”Human activities have little impact on the level of water vapour.

- Carbon Dioxide Our atmosphere contains many natural gases other than ozone. One of these natural gases is carbon dioxide. Our atmosphere needs a certain amount of this gas. It is carbon dioxide that help to keep the Earth warm.

- Carbon Dioxide This gas holds in just enough heat from the sun to keep animals and plants alive.If it held in more heat than it does the climate on Earth would grow too hot for some kinds of life.If it held in less heat, Earth’s climate would be too cold.

- Carbon Dioxide Carbon Dioxide is probably the most important of the greenhouse gases and is currently responsible for 60 % of the ‘enhanced greenhouse effect ’Enhanced Human activities, not natural. Global carbon dioxide emissions

- Carbon Dioxide For the past 100 years, the amount of carbon dioxide in our atmosphere seems to have been increasing. Why is this happening? What is it doing to the Earth’s atmosphere?

- **Where do all the carbon dioxide gases come from?** Human respiration. Industrialization Burning of fossil fuel to generate electricity Burning of forest (lesser trees)CO2 is now 1/3 more than before Industrial Revolution

- Carbon Dioxide Burning fossil fuels release the carbon dioxide stored millions of years ago. We use fossil fuels to run vehicles (petrol, diesel, and kerosene), heat homes, businesses, and power factories.

- Nitrous Oxide Nitrous oxide makes up an extremely small amount of the atmosphere – It is less than one-thousandth as abundant as carbon dioxide. However it is 200 to 300 times more effective in trapping heat than carbon dioxide.

- Nitrous Oxide Nitrous Oxide has one of the longest atmosphere lifetimes of the greenhouse gases, lasting for up to 150 years. Since the Industrial Revolution, the level of nitrous oxide in the atmosphere has increased by 16%.

- **Nitrous Oxide The impact of human activities** Burning fossil fuels and wood Widespread use of fertilizers Sewage treatment plants

- **Where do all nitrous oxide gases come from?** Vehicle exhaust Nitrogen based fertilisers

- Methane The importance of methane in the greenhouse effect is it’s warming effect.It occurs in lower concentrations than carbon dioxide but it produces 21 times as much warming as carbon dioxide.

- **Methane Methane accounts for 20%of the ‘enhanced greenhouse effect’.** It remains in the atmosphere for years. (Less than other greenhouse gases)

- **Methane Human Activities** An increase in livestock farming and rice growing has led to an increase in atmospheric methane. Other sources are the extraction of fossil fuels, landfill sites and the burning of biomass . Methane concentration in the atmosphere has more than doubled during the last 200 yr. Some of this methane is produced by rice fields

□ **Where do all the methane gases come from?** Produced by bacteria living in swampy areas. Wet rice cultivation Waste in landfills Rearing of livestock When cows belch (burp)Each molecule can trap 20 times as much heat as a CO_2 molecule.

□ **Where do all the CFCs come from?** CFCs (Chlorofluorocarbons)Aerosol sprays Making foam packaging Coolants in fridge and air cons Cleaning solvents Each CFC molecule can trap as much heat as CO_2 molecule. Can remain in the atmosphere for a long time (up to years)

□ Global Warming The average global temperature has increased by almost 1º F over the past century. Scientists expect the average global temperature to increase an additional 2º to 6º F over the next hundred years.

□ Global Warming At the peak of the last ice age (18, 000 years ago) the temperature was only 7 º colder than it is today, and glaciers covered much of North America.

□ Global Warming Even a small increase in temperature over a long time can change the climate. When the climate changes, there may be big changes in the things that people depend on.

□ Global Warming These things include the level of the oceans and the places where we plant crops. They also include the air we breathe and the water we drink.

□ Global Warming It is important to understand that scientists don’t know for sure what global warming will bring . Some changes may be good . Eg. If you live in a very cool climate , warmer temperatures might be welcome.

☐ Global Warming Days and nights would be more comfortable and people in the area may be able to grow different and better crops than they could before.

☐ **Global Warming Changes in some places will not be good at all.** Human Health Ecological Systems (Plants and animals) Sea Level Rise Crops and Food Supply

☐ Human Health Heat stress and other heat related health problems are caused directly by very warm temperatures and high humidity . Heat stress – A variety of problems associated with very warm temperatures and high humidity eg. Heat exhaustion and heat stroke.

☐ **Ecological Systems Plants and animals** Climate change may alter the world's habitats .All living things are included in and rely on these places. Most past climate changes occurred slowly, allowing plants and animals to adapt to the new environment or move someplace else .Plants and animals may not be able to react quickly enough to survive if future climate changes occur as rapidly as scientists predict.

☐ Sea Level Rise Global Warming may make the sea level become higher. Why? Warmer weather makes glaciers melt . Melting glaciers add more water to the ocean. Warmer weather also makes water expand . When water expands in the ocean, it takes up more space and the level of the sea rises.

☐ Rising Sea Levels When earth's temperature rises, sea level is likely to rise too: Higher temperature ◊ sea water to expand in volume Ice caps at poles to melt

☐ Sea Level Rise Sea level may rise between several inches and as much as 3 feet during the next century. Coastal flooding could cause saltwater to flow into areas where salt is harmful, threatening plants and animals in those areas.

Oceanfront property would be affected by flooding. Coastal flooding may also reduce the quality of drinking water in coastal areas.

 Crops and Food Supply Global warming may make the Earth warmer in cold places. People living in these areas may have the chance to grow crops in new areas. But global warming might bring droughts to other places where we grow crops.

 What Might Happen? This warming trend is expected to bring droughts and flooding of low lying coastal areas as the polar ice caps melt and raise sea level. The expected negative impact of the greenhouse effect on human life has been assessed by some scientists to be second only to global nuclear war.

 Climatic Change Global warming will lead to an increase in the evaporation of water ◊ more water vapour. With more water vapour, more rain fall is expected. But it is not evenly distributed: Dry areas ◊ severe drought condition, water shortage and heat waves occurs Wet areas ◊ floods and avalanches (landslides)

 Climatic Change Other problems may arise: Destroy food crop ◊ rice, wheat and corn Affect animals ◊ need to migrate Encourage growth of weed and pests ◊ may lead to diseases like dengue fever, cholera which are deadly.

 What can we do about it ? There are many little things that we can do to make a difference to reduce the amount of greenhouse gases that we put into the atmosphere. Many greenhouse gases come from things we do every day. Driving a car or using electricity is not wrong. We just have to be smart Eg. Try carpooling

☐ **Ways you can help make our planet better.** Read – Learning about the environment is very important. Save Electricity – Whenever we use electricity, we help put greenhouse gases into the air. Turn off lights, the television and the computer.

☐ **Ways you can help make our planet better.** Bike, Bus and Walk- You can saves energy by sometimes taking the bus, riding a bike or walking. Talk to Your Family and Friends – about global warming. Let them know what you've learned.

☐ **Ways you can help make our planet better.** Recycle – When you recycle, you send less trash to the landfill and you help save natural resources like trees and elements such as aluminum. Recycle cans, bottles, plastic bags and newspapers.

☐ **Ways you can help make our planet better.** When You Buy, Buy Cool Stuff Buy Products that don't use as much energy Buy recyclable products instead of non-recyclable ones. Solar Energy – can be used to heat homes, buildings, water and to make electricity.

☐ **Ways you can help make our planet better.** Cars – cause pollution and release a lot of greenhouse gases into the air. Some cars are better for the environment – They travel longer on a smaller amount of fuel. They don't pollute as much. Using these cars can help reduce can help reduce the amount of greenhouse gases in the air.

☐ **Song of the Air Play song by clicking the button here.** Hey, you guys, don't gimme smoke and gas. I'm a tough guy, how can I last? Not hard to see damage you've done. Irritate your eyes, blacken your lungs. Let's get smart, find a solution to stop right now all that pollution. Oh man… nasty stuff. All right now, check me out on this one. All around the earth, listen for my sound.

Wherever you go, I'll be hangin' 'round. You can't see me, I can see you. Air's what you want, I do it for you. Play song by clicking the button here.

- **What else can we do? To reduce the emission of greenhouse gases** International efforts: Kyoto treaty (1997) ◊ was started to reduce emission of greenhouse gases by 5% of 1990s levels by 2012. World's major polluters

- **Animation on Global Warming** Read all directions before proceeding. To recap what we've learned in this lesson, click on the button on the bottom right , read the directions and watch the animation on Global Warming. Please be patient while the program loads. When it's completed be sure to take the globe warming quiz by clicking on the green button on the animation screen. Then exit the browser and return to this presentation.

- **Summary / Conclusion Environmental Crisis will affect us: Health** Air pollution ◊ asthma or other respiratory problems Water pollution ◊ poison our food source e.g fish Destruction of forest ◊ lost of possible medical solutions Property Floods ◊ property lost Pollution ◊ destroy streets and beaches Soil erosion ◊ desertification, lost of farm lands

- **Can we not be concerned? Summary / Conclusion** Environmental Crisis will affect us: Economic Costs Lost in terms of monetary values, industry and businesses. Money need to be spent to restore the original Public Health Services need to be provided by the government .Can we not be concerned?

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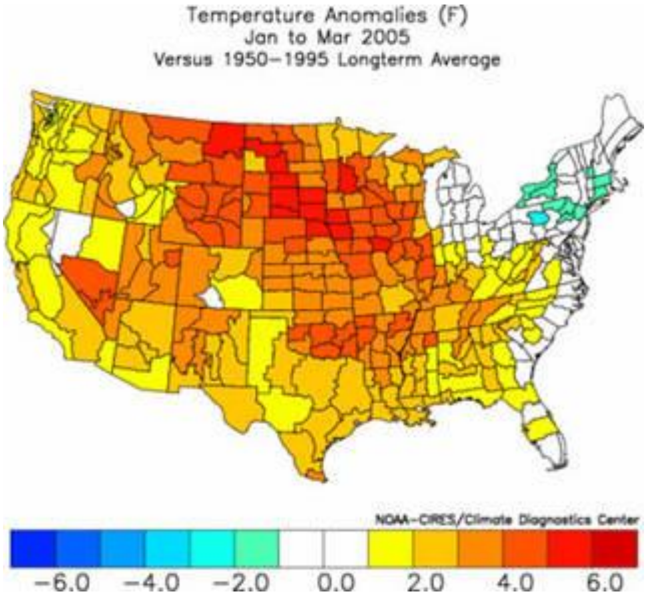
NASA - What's the Difference Between Weather and Climate?



Latest three month average temperature and precipitation anomalies for the United States.

Credits: NOAA

The difference between weather and climate is a measure of time. Weather is what conditions of the atmosphere are over a short period of time, and climate is how the atmosphere "behaves" over relatively long periods of time.

When we talk about climate change, we talk about changes in long-term averages of daily weather. Today, children always hear stories from their parents and grandparents about how snow was always piled up to their waists as they trudged off to school. Children today in most areas of the country haven't experienced those kinds of dreadful snow-packed winters, except for the Northeastern U.S. in January 2005. The change in recent winter snows indicate that the climate has changed since their parents were young.

If summers seem hotter lately, then the recent climate may have changed. In various parts of the world, some people have even noticed that springtime comes earlier now than it did 30 years ago. An earlier springtime is indicative of a possible change in the climate.

In addition to long-term climate change, there are shorter term climate variations. This so-called climate variability can be represented by periodic or intermittent changes related to El Niño, La Niña, volcanic eruptions, or other changes in the Earth system.

What Weather Means

Weather is basically the way the atmosphere is behaving, mainly with respect to its effects upon life and human activities. The difference between weather and climate is that weather consists of the short-term (minutes to months) changes in the atmosphere. Most people think of weather in terms of temperature, humidity, precipitation, cloudiness, brightness, visibility, wind, and atmospheric pressure, as in high and low pressure.

In most places, weather can change from minute-to-minute, hour-to-hour, day-to-day, and season-to-season. Climate, however, is the average of weather over time and space. An easy way to remember the difference is that climate is what you expect, like a very hot summer, and weather is what you get, like a hot day with pop-up thunderstorms.

Things That Make Up Our Weather

There are really a lot of components to weather. Weather includes sunshine, rain, cloud cover, winds, hail, snow, sleet, freezing rain, flooding, blizzards, ice storms, thunderstorms, steady rains from a cold front or warm front, excessive heat, heat waves and more.

In order to help people be prepared to face all of these, the National Oceanic and Atmospheric Administration's (NOAA) National Weather Service (NWS), the lead forecasting outlet for the nation's weather, has over 25 different types of warnings, statements or watches that they issue. Some of the reports NWS issues are: Flash Flood Watches and Warnings, Severe Thunderstorm Watches and Warnings, Blizzard Warnings, Snow Advisories, Winter Storm Watches and Warnings, Dense Fog Advisory, Fire Weather Watch, Tornado Watches and Warnings, Hurricane Watches and Warnings. They also provide Special Weather Statements and Short and Long Term Forecasts.

NWS also issues a lot of notices concerning marine weather for boaters and others who dwell or are staying near shorelines. They include: Coastal Flood Watches and Warnings, Flood Watches and Warnings, High Wind Warnings, Wind Advisories, Gale Warnings, High Surf Advisories, Heavy Freezing Spray Warnings, Small Craft Advisories, Marine Weather Statements, Freezing Fog Advisories, Coastal Flood Watches, Flood Statements, Coastal Flood Statement.

Who is the National Weather Service?

According to their mission statement, "The National Weather Service provides weather, hydrologic, and climate forecasts and warnings for the United States, its territories, adjacent waters and ocean areas, for the protection of life and property and the enhancement of the national economy. NWS data and products form a national information database and infrastructure which can be used by other governmental agencies, the private sector, the public, and the global community."

To do their job, the NWS uses radar on the ground and images from orbiting satellites with a continual eye on Earth. They use reports from a large national network of weather reporting stations, and they launch balloons in the air to measure air temperature, air pressure, wind, and humidity. They put all this data into various computer models to give them weather forecasts. NWS also broadcasts all of their weather reports on special NOAA weather radio, and posts them immediately on their Interactive Weather Information Network website at: http://iwin.nws.noaa.gov/iwin/graphicsversion/bigmain.html.

What Climate Means

In short, climate is the description of the long-term pattern of weather in a particular area.

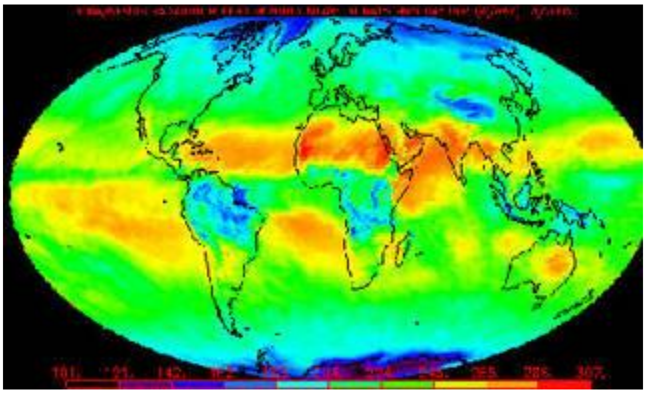
Some scientists define climate as the average weather for a particular region and time period, usually taken over 30-years. It's really an average pattern of weather for a particular region.

When scientists talk about climate, they're looking at averages of precipitation, temperature, humidity, sunshine, wind velocity, phenomena such as fog, frost, and hail storms, and other measures of the weather that occur over a long period in a particular place.

For example, after looking at rain gauge data, lake and reservoir levels, and satellite data, scientists can tell if during a summer, an area was drier than average. If it continues to be drier than normal over the course of many summers, than it would likely indicate a change in the climate.

Why Study Climate?

The reason studying climate and a changing climate is important, is that will affect people around the world. Rising global temperatures are expected to raise sea levels, and change precipitation and other local climate conditions. Changing regional climate could alter forests, crop yields, and water supplies. It could also affect human health, animals, and many types of ecosystems. Deserts may expand into existing rangelands, and features of some of our National Parks and National Forests may be permanently altered.



An example of a Monthly Mean Outgoing Longwave Radiation (OLR) product produced from NOAA polar-orbiter satellite data, which is frequently used to study global climate change.

Credits: NOAA

The National Academy of Sciences, a lead scientific body in the U.S., determined that the Earth's surface temperature has risen by about 1 degree Fahrenheit in the past century, with accelerated warming during the past two decades. There is new and stronger evidence that most of the warming over the last 50 years is attributable to human activities. Yet, there is still some debate about the role of natural cycles and processes.

Human activities have altered the chemical composition of the atmosphere through the buildup of greenhouse gases – primarily carbon dioxide, methane, and nitrous oxide. The heat-trapping property of these gases is undisputed although uncertanties exist about exactly how Earth's climate responds to them. According to the U.S. Climate Change Science Program (http://www.climatescience.gov), factors such as aerosols, land use change and others may play important roles in climate change, but their influence is highly uncertain at the present time.

Who Studies Climate Change?

Modern climate prediction started back in the late 1700s with Thomas Jefferson and continues to be studied around the world today.

At the national level, the U.S. Global Change Research Program coordinates the world's most extensive research effort on climate change. In addition, NASA, NOAA, the U.S. Environmental Protection Agency (EPA) and other federal agencies are actively engaging the private sector, states, and localities in partnerships based on a win-win philosophy and aimed at addressing the challenge of global warming while, at the same time, strengthening the economy. Many university and private scientists also study climate change.

What is the U.S. Global Change Research Program?

The United States Global Change Research Program (USGCRP) was created in 1989 as a high-priority national research program to address key uncertainties about changes in the Earth's global environmental system, both natural and human-induced; to monitor, understand, and predict global change; and to provide a sound scientific basis for national and international decision-making.

Since its inception, the USGCRP has strengthened research on global environmental change and fostered insight into the processes and interactions of the Earth system, including the atmosphere, oceans, land, frozen regions, plants and animals, and human societies. The USGCRP was codified by Congress in the Global Change Research Act of 1990. The basic rationale for establishing the program was that the issues of global change are so complex and wide-ranging that they extend beyond the mission, resources, and expertise of any single agency, requiring instead the integrated efforts of several agencies.

Some Federal Agencies Studying Climate

In the 1980s the National Weather Service established the Climate Prediction Center (CPC), known at the time as the Climate Analysis Center (CAC). The CPC is best known for its United States climate forecasts based on El Niño and La Niña conditions in the tropical Pacific.

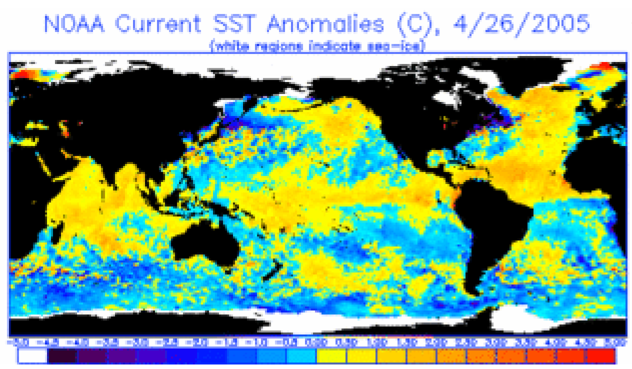


Image Above: The operational SST anomaly charts are useful in assessing ENSO (El Niño - Southern Oscillation) development, monitoring hurricane "wake" cooling, and even major shifts in coastal upwelling.

Credits: NOAA

CPC was established to give short-term climate prediction a home in NOAA. CPC's products are operational predictions or forecasts of how climate may change and includes real-time monitoring of climate. They cover the land, the ocean, and the atmosphere, extending into the upper atmosphere (stratosphere). Climate prediction is very useful in various industries, including agriculture, energy, transportation, water resources, and health.

NASA has been using satellites to study Earth's changing climate. Thanks to satellite and computer model technology, NASA has been able to calculate actual surface temperatures around the world and measure how they've been warming. To accomplish the calculations, the satellites actually measure the Sun's radiation reflected and absorbed by the land and oceans.NASA satellites keep eyes on the ozone hole, El Nino's warm waters in the eastern Pacific, volcanoes, melting ice sheets and glaciers, changes in global wind and pressure systems and much more.

At the global level, countries around the world have expressed a firm commitment to strengthening international responses to the risks of climate change. The U.S. is working to strengthen international action and broaden participation under the support of the United Nations Framework Convention on Climate Change.

Today, scientists around the world continue to try and solve the puzzle of climate change by working with satellites, other tools and computer models that simulate and predict the Earth's conditions.

The Köppen Climate Classification

The Köppen Climate Classification divides the Earth's climate into five main climate groups:

- A (tropical)
- B (dry)
- C (temperate)
- D (continental)
- E (polar)

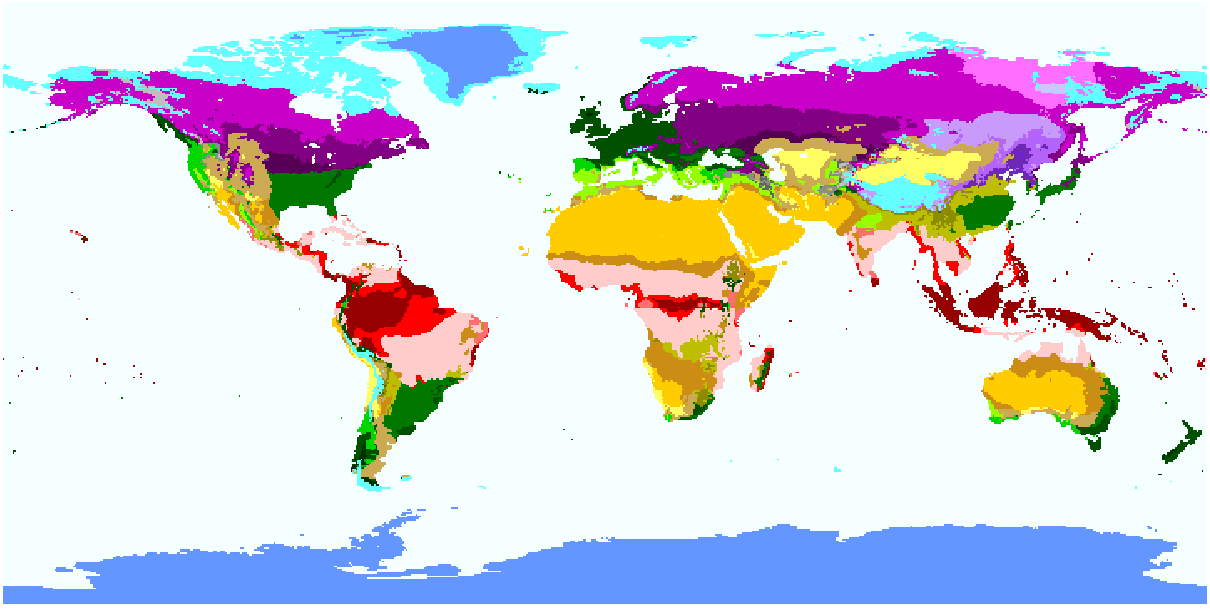
These are subdivided by seasonal precipitation and heat.

It was first published by the Russian-German climatologist Wladimir Köppen in

1884, with several later modifications by Köppen and others, notably Rudolf Geiger, hence the system is sometimes also called the Köppen–Geiger climate classification system.

The map below shows the distribution (as of the 2006) of climatic zones on the land surface of the Earth to an approximate accuracy of 0.5 degrees. Click on any climate classification type for more details.

Ref: Kottek, M., J. Grieser, C. Beck, B. Rudolf, and F. Rubel, 2006: World Map of the Köppen-Geiger climate classification updated. Meteorol. Z., 15, 259-263. doi:10.1127/0941-2948/2006/0130.



Climatic Classifications

Code	**Description**	**Group**	**Precipitation Type**	**Level of Heat**
Af	Tropical rainforest climate	Tropical	Rainforest	
Am	Tropical monsoon climate	Tropical	Monsoon	
As	Tropical dry savanna climate	Tropical	Savanna, Dry	
Aw	Tropical savanna, wet	Tropical	Savanna, Wet	
BSh	Hot semi-arid (steppe) climate	Arid	Steppe	Hot

BSk	Cold semi-arid (steppe) climate	Arid	Steppe	Cold
BWh	Hot deserts climate	Arid	Desert	Hot
BWk	Cold desert climate	Arid	Desert	Cold
Cfa	Humid subtropical climate	Temperate	Without dry season	Hot summer
Cfb	Temperate oceanic climate	Temperate	Without dry season	Warm summer
Cfc	Subpolar oceanic climate	Temperate	Without dry season	Cold summer
Csa	Hot-summer Mediterranean climate	Temperate	Dry summer	Hot summer
Csb	Warm-summer Mediterranean climate	Temperate	Dry summer	Warm summer
Csc	Cool-summer Mediterranean climate	Temperate	Dry summer	Cold summer
Cwa	Monsoon-influenced humid subtropical climate	Temperate	Dry winter	Hot summer
Cwb	Subtropical highland climate or temperate oceanic climate with dry winters	Temperate	Dry winter	Warm summer
Cwc	Cold subtropical highland climate or subpolar oceanic climate with dry winters	Temperate	Dry winter	Cold summer
Dfa	Hot-summer humid continental climate	Cold (continental)	Without dry season	Hot summer
Dfb	Warm-summer humid continental climate	Cold (continental)	Without dry season	Warm summer

Dfc	Subarctic climate	Cold (continental)	Without dry season	Cold summer
Dfd	Extremely cold subarctic climate	Cold (continental)	Without dry season	Very cold winter
Dsa	Hot, dry-summer continental climate	Cold (continental)	Dry summer	Hot summer
Dsb	Warm, dry-summer continental climate	Cold (continental)	Dry summer	Warm summer
Dsc	Dry-summer subarctic climate	Cold (continental)	Dry summer	Cold summer
Dwa	Monsoon-influenced hot-summer humid continental climate	Cold (continental)	Dry winter	Hot summer
Dwb	Monsoon-influenced warm-summer humid continental climate	Cold (continental)	Dry winter	Warm summer
Dwc	Monsoon-influenced subarctic climate	Cold (continental)	Dry winter	Cold summer
Dwd	Monsoon-influenced extremely cold subarctic climate	Cold (continental)	Dry winter	Very cold winter
EF	Ice cap climate	Polar	Ice cap	
ET	Tundra	Polar	Tundra	

What is the Greenhouse Effect?

The greenhouse effect is a natural process that maintains the temperature of the earth. But when the number of greenhouse gases increases in the atmosphere, it results in a phenomenon known as global warming, which is one of the major problems that the world is facing today. We will discuss it in brief in this article below.

The greenhouse effect is a natural phenomenon that warms the surface of the Earth. The atmosphere present on the earthworks like the surface of a glasshouse. When the sun rays reach the surface of the Earth, 30% of the solar energy is reflected back to space. 21% of this energy is absorbed by the atmosphere. The remaining part of solar energy is absorbed by the sea, plants, and other structures present on the surface of the Earth. This absorbed energy is radiated out of the earth in the form of infrared waves that warm the Earth. Some gases present in the atmosphere are called 'greenhouse gases' because of their heat-trapping effect. These gases absorb the heat and radiate it back to the earth's surface. Greenhouse gases are carbon dioxide, methane, and water vapor, etc. These gases help to maintain the greenhouse temperature at around 30℃ warmer than it would otherwise be. Without the greenhouse effect, the earth would be too cold to allow life on earth.

Step1: Sun rays reach the earth's atmosphere. Some of this energy is reflected back into space.

Step2: The remaining part of the sun's energy is absorbed by the sea, plants, land, and other structures present on the Earth.

Step3: Heat radiates from the Earth towards space.

Step4: Some of this heat is trapped by greenhouse gases present in the atmosphere.

Step5: This keeps the earth warm enough to sustain life.Humans have increased the levels of greenhouse gases in the air by excessive use of fossil fuels like coal and petroleum.

This breaks the equilibrium maintained by the greenhouse effect, resulting in an increase in temperature of our earth's atmosphere. Simply, global warming refers to an increase in the earth's temperature. Global warming is a major problem that the world is facing today.Global Warming Greenhouse Gas Formula % on Earth Water vapour H2O35-70%Carbon DioxideCO29-25%MethaneCH44-9%Nitrous Oxide N2O3-7%

We’ve always had some greenhouse gases in the atmosphere. We take oxygen and release carbon dioxide in our respiration process.

Plants take up carbon dioxide to make food by a process called photosynthesis. Once forests are cut down for human uses, they can no longer take in carbon dioxide, and this gas begins building up in the air instead of fuelling the growth of plants.

So, by cutting down trees more carbon dioxide is added into the atmosphere. We've deforested large parts of the planet i.e increased the number of greenhouse gases, and as a result, have changed the temperature of the planet.We’ve burned a lot of fossil fuels.

Greenhouse gases are also released when organic matter (including fossil fuels like oil or gas) is burned. These greenhouse gases, trap heat in Earth’s atmosphere.(image will be uploaded soon)Effects of Global WarmingHuman activities such as the burning of fossil fuel and deforestation increase the level of greenhouse gases that causes long-term heating of the earth's climate system.

This human-produced temperature increase is commonly referred to as global warming.

Scientists use observations from ground, air, and space to study and monitor climate change.

This given graph illustrates the change in global surface temperature.(image will be uploaded soon)

Effects of global warming are given below:

Year by year, records have been broken for the longest heatwaves which affect the sustainability of humans and other forms of life in the earth.

Increased temperatures are melting glaciers and ice caps. This melted ice increases the volume of water in oceans, leading to the rise of sea levels. Extreme weather events like cyclones, drought, and floods are becoming more frequent and more intense as a result of global warming.The oceans have absorbed most of the extra heat and carbon dioxide so far. More than the air, global warming is making the seas both warmer and more acidic. This is a threat to marine ecosystems.Reduced rainfall and increasingly severe droughts may lead to water shortage.Heatwaves may lead to death and illness, especially among the elderly.

High temperatures and humidity could also produce more mosquito-borne diseases.Measures to Reduce Global Warming

As we know, global warming is a major problem that poses a risk and threat to all living systems. At the international level, various conventions, treaties, and agreements have been signed to combat global warming by increased mutual cooperation of various countries. But more than the governments trying to solve the problem, the individual should also learn to take responsibility for their actions and need to take small steps at a personal level to reduce the overall emissions of greenhouse gases in order to make the planet better and healthier.

As individuals, we can do the following:

Use bicycles instead of automobile vehicles for short distances.

Plant more and more trees by organizing social events and plantation drives.

MAJOR GREENHOUSE GASES SOURCES AND SINKS OF GREENHOUSE GASES

Greenhouse gases (GHGs) is considered as one of the biggest reasons behind climate change in Anthropocene. The GHGs (carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), ozone (O_3), chlorofluorocarbons (CFCs) and carbon tetrachloride (CCl_4) play a major role in balancing the radiative budget, by absorbing or emitting some of the infrared rays reflecting from the earth's surface. The increasing levels of most of the GHGs (except CH_4 and some hydrocarbons) in atmosphere is due to increasing rate of anthropogenic activities from past few decades.

The present chapter summarizes the varied sources of GHGs that includes burning of fossil fuels,

growth of urbanization,

waste management,

transport,

agriculture,

habitat fragmentation,

land use land cover change (LULCC),

cultural and economic activities,

weak policies and strategies by the government, etc.

GHGs being an integral part of an ecosystem consist of natural sinks such as terrestrial vegetation,

soil,

geological, biological and stratospheric sinks;

although artificial sinks are also available as well that can act as a sink.

However, it is necessary to prevent or reduce the emission of GHGs to mitigate climate change.

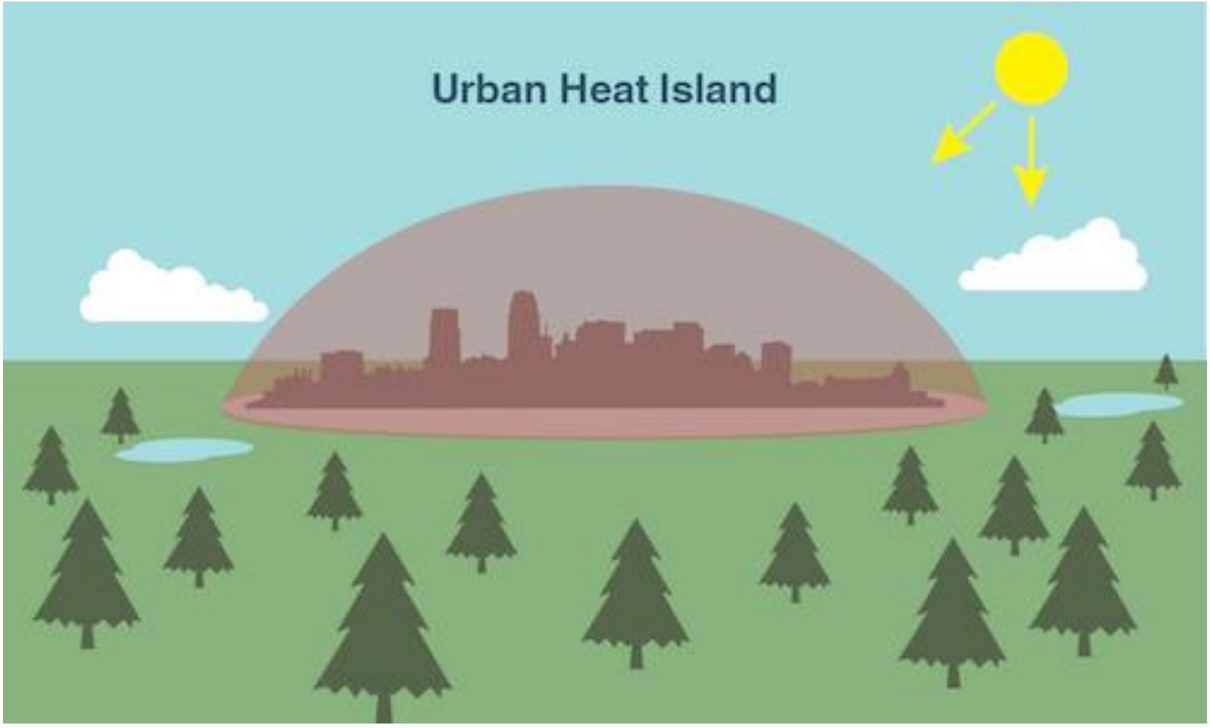
Mitigation refers to the concept of using new technologies, making older equipment energy-efficient, using renewable energy, changing management practices, etc.

Thus, the present chapter explains about the overall view of GHGs including their different sources, sink and impacts.

URBAN HEAT ISLANDS EFFECTS

An urban heat island occurs when a city experiences much warmer temperatures than nearby rural areas. The difference in temperature between urban and less-developed rural areas has to do with how well the surfaces in each environment absorb and hold heat.

An **urban heat island** occurs when a city experiences much warmer temperatures than nearby rural areas.



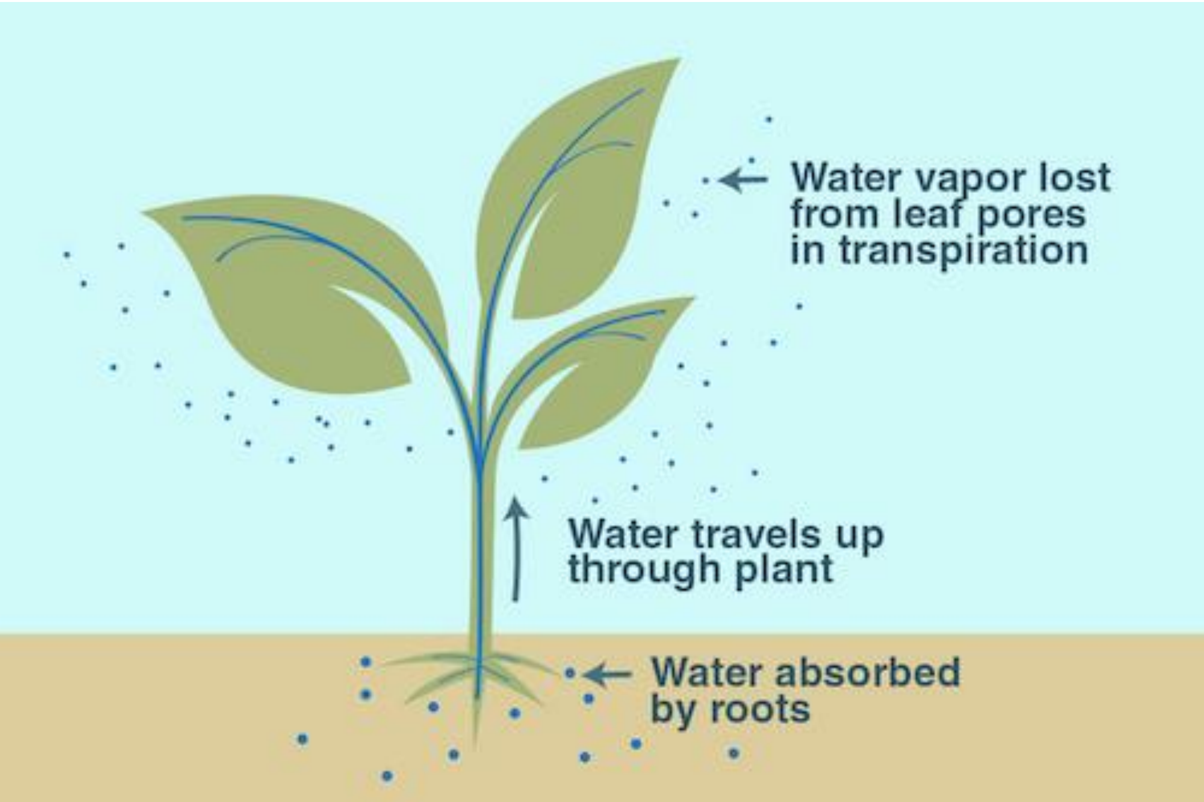
An illustration of an urban heat island. Image credit: NASA/JPL-Caltech

Why does this happen?

An **urban** area is a city. A **rural** area is out in the country. The sun's heat and light reach the city and the country in the same way. The difference in temperature between urban and less-developed rural areas has to do with how well the surfaces in each environment absorb and hold heat.

If you travel to a rural area, you'll probably find that most of the region is covered with plants. Grass, trees and farmland covered with crops, as far as the eye can see.

Plants take up water from the ground through their roots. Then, they store the water in their stems and leaves. The water eventually travels to small holes on the underside of leaves. There, the liquid water turns into water vapor and is released into the air. This process is called **transpiration**. It acts as nature's air conditioner.



An illustration of the process of transpiration. Image credit: NASA/JPL-Caltech

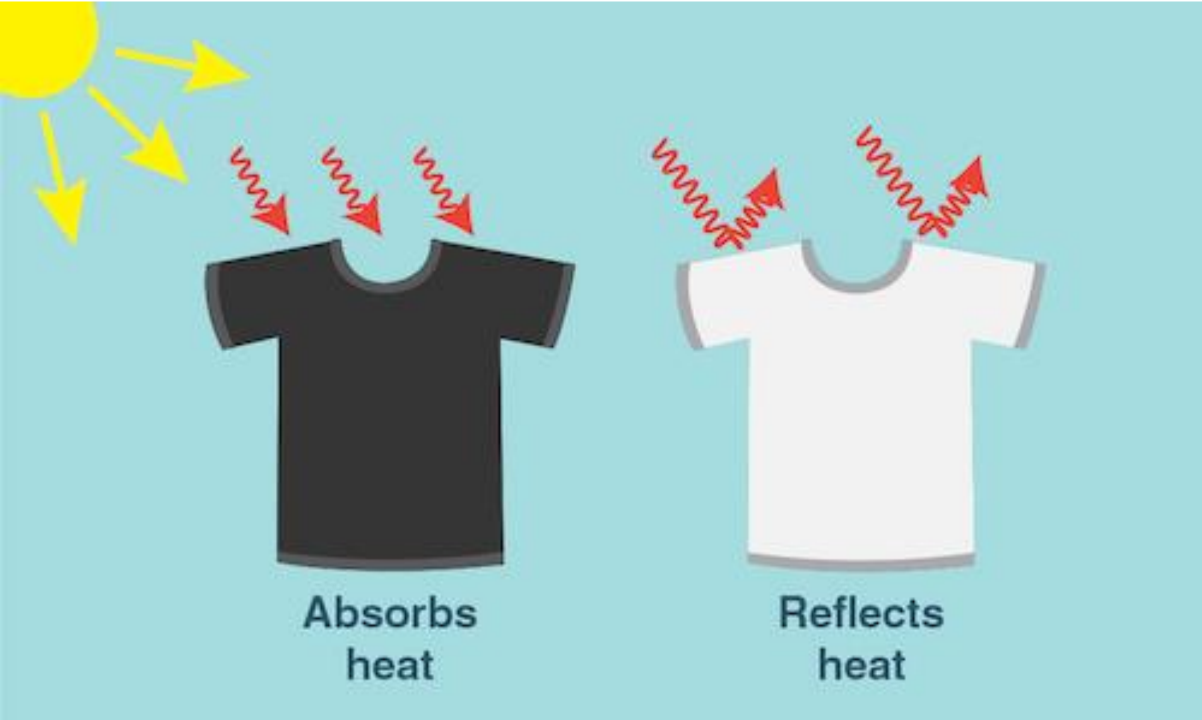
See it for yourself!

You can feel cooling transpiration at work on a hot summer day. On a sunny day, go outside and find a sidewalk that is right next to a patch of grass. Feel both surfaces. The grass should feel cooler on your skin than the pavement—and that’s mostly due to transpiration!

When you visit a big city, you won’t see many plants. Instead, you’ll see sidewalks, streets, parking lots and tall buildings. These structures are usually made up of materials such as cement, asphalt, brick, glass, steel and dark roofs.

What do urban building materials have in common?

First of all, materials such as asphalt, steel, and brick are often very dark colors—like black, brown and grey. A dark object **absorbs** all wavelengths of light energy and converts them into heat, so the object gets warm. In contrast, a white object **reflects** all wavelengths of light. The light is not converted into heat and the temperature of the white object does not increase noticeably. Thus, dark objects—such as building materials—absorb heat from the sun.



Dark surfaces--whether a black t-shirt or an asphalt street--absorb the sun's heat, while lighter colored surfaces reflect heat from the sun. Credit: NASA/JPL-Caltech

To cool down urban heat islands, some cities are ‘lightening’ streets. This is done by covering black asphalt streets, parking lots, and dark roofs with a more reflective gray coating. These changes can drop urban air temperatures dramatically, especially during the heat of summer.

Planting gardens on urban rooftops can also help to cool down the city, too! In fact, a study in Los Angeles, California, calculated that changes like these would be enough to save close to $100 million per year in energy costs!

Urban building materials are another reason that urban areas trap heat. Many modern building materials are **impervious** surfaces. This means that water can't flow through surfaces like a brick or a patch of cement like it would through a plant. Without a cycle of flowing and evaporating water, these surfaces have nothing to cool them down.

Skyscrapers in Chicago. Image credit: Flickr user GiuseppeYahoo Cortese

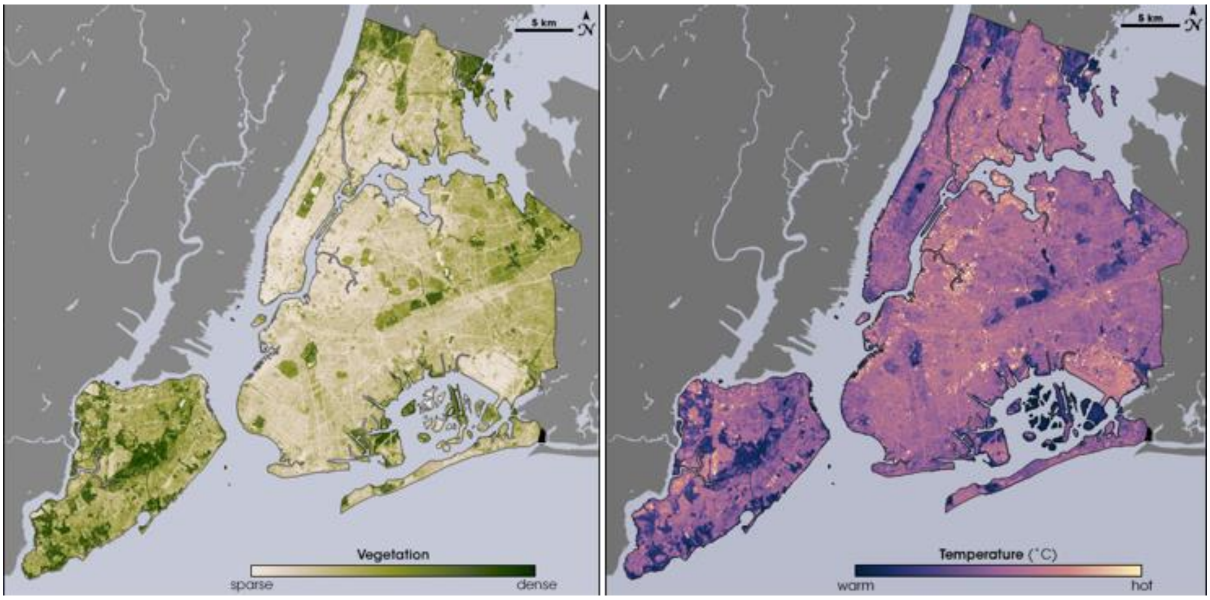
To help cool the heat island, builders can use materials that will allow water to flow through. These building materials—called permeable materials—promote the capture and flow of water, which cools urban regions.

What does it mean?

Urban heat islands are one of the easiest ways to see how human impact can change our planet. After all, sidewalks, parking lots and skyscrapers wouldn't exist if humans weren't there to build them. And although these structures are essential to city living, the heat islands they create can be dangerous for humans.

In the summer, New York City is about 7°F (4°C) hotter than its surrounding areas. That doesn't seem like much, but these higher temperatures can cause people to become dehydrated or suffer from heat exhaustion. The hot temps also require more energy to operate fans and air conditioners. This can lead to power outages and a serious danger to public health.

But, there are things we can do to help cool the cities down. And NASA satellites can help to figure out where these cities are the hottest.



Caption: These images from the NASA/USGS satellite Landsat show the cooling effects of plants on New York City's heat. On the left, areas of the map that are dark green have dense vegetation. Notice how these regions match up with the dark purple regions—those with the coolest temperatures—on the right. Image credit: Maps by Robert Simmon, using data from the Landsat Program.

Earth-observing satellites, such as Landsat and Suomi-NPP, can keep a close eye on the Earth's vegetation and surface temperature. Scientists can use this information to track hotspots in cities across the planet. NASA scientists, with their global satellite views, are working to understand urban heat islands and help urban planners to build more energy efficient, cooler and safer cities.

Ground level ozone

Figure 1. Ozone (O_3) consists of 3 oxygen atoms bonded together.[1] While essential in the upper atmosphere, near the ground ozone causes breathing problems.

Ground level ozone is a highly reactive secondary pollutant. This pollutant forms when primary pollutants, like hydrocarbons and nitrogen oxides, react with sunlight. Ozone irritates people's lungs and is a major component of photochemical smog.[2] Ground level ozone gives photochemical smog its unpleasant odor and can cause lung function complications, especially in young children.

Formation

Ground level ozone is created by a chemical reaction between volatile organic compounds and nitrogen oxides (Figure 1). The sun and high temperatures act as catalysts to this reaction. This makes its production within warm, densely populated, urban cities a common occurrence. However, since ozone takes some time to form, it will not necessarily stay within these cities. The pollutants may blow down-wind before ozone begins to form, therefore small towns and rural populations may also experience high levels of ozone

Although emissions of ozone-producing substances have decreased in recent years in Canada (see data visualization below) many urban areas do not yet meet air quality standards.

The pollutants responsible for the formation of ozone are produced by hydrocarbon combustion in vehicles, factories, and power plants. Within cities, vehicles can make up nearly 90% of these pollutants. Nitrogen oxides and VOCs are produced in the morning by cars traveling to work, and ozone is produced shortly after, typically in the afternoon. Since pollutants may flow down-wind from where they are produced, the ozone can find its way around the globe and be produced in safer quantities. However if there is an inversion layer present over the area of production, then the pollutants can be "trapped" over populated cities and form to unsafe levels in the form of smog.

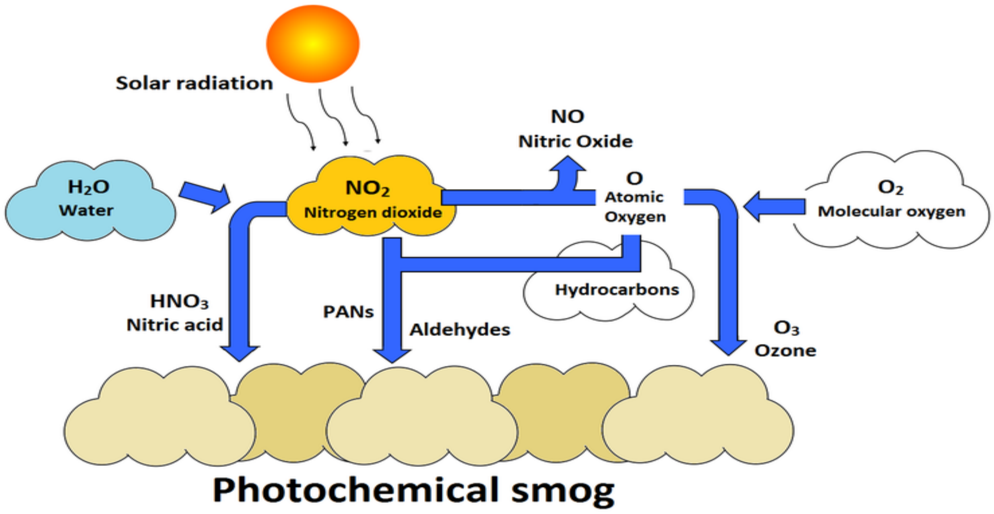


Figure 2. Ozone formation can be seen on the right, in which it is produced from photolyzed nitrogen dioxide.[

Health effects

Due to its reactivity and close proximity to the Earth's surface, ground level ozone can have large effects on people. These effects include breathing problems, coughing, and irritation to the eyes, nose, and throat. People with asthma, bronchitis, emphysema, and heart disease may experience aggravated symptoms when exposed to ground level ozone. It can also reduce the body's immune system, increasing the potential for colds and flu. lung function, in particular, may be affected adversely

Ground level ozone can affect the health of plants and animals. It can also cause damage to rubber, fabrics, and paints

Ozone

Ozone (O_3) is present throughout the atmosphere although there are concentration peaks at two levels, the stratosphere (15 - 50 km) and troposphere (0-15 km), with the largest fraction and concentrations being in the stratospheric O_3 layer (Royal Society, 2008). Stratospheric O_3 is important as it regulates the transmittance of ultraviolet light to the surface of the earth. Hence reductions in stratospheric O_3 in polar regions, particularly the Antarctic "ozone hole", are of concern regarding the health effects of exposure to increased levels of UV-B.

In contrast, O_3 in the troposphere (ground level) is regionally important as a toxic air pollutant and greenhouse gas. Mixing with stratospheric air provides a natural global average background of around 10-20 parts per billion (ppb), though there is some debate about the concentration. Additional quantities of tropospheric O_3 are produced by photochemical reactions from nitrogen oxides (NO_x) and volatile organic compounds (VOCs), which include various hydrocarbons.

Formation and Sources

Ground-level ozone (O_3) is not emitted directly from anthropogenic sources. It is a "secondary" pollutant formed by a complicated series of chemical reactions in the presence of sunlight. Photochemical reactions of NOx and VOCs (originating from largely from combustion processes) govern the concentration of ground-level O_3 in the atmosphere. Under typical daytime conditions with a well-mixed atmosphere, three reactions reach equilibrium and no net chemistry occurs.

$NO + O_3$ -> $NO_2 + O_2$ (1)

NO_2 + hn -> $NO + O$ (2)

$O + O_2$ -> O_3 (+ M) (3) where hn = sunlight with wavelength 280-430 nm

M = any molecule eg N_2 or O_2

The chemical reactions do not take place instantaneously, but can take hours or days. Ozone levels at a particular location may have arisen from VOC and NO_x emissions many hundreds or even thousands of miles away. Maximum concentrations, therefore, generally occur downwind of the source areas of the precursor pollutant emissions.

Anthropogenic emissions of the ozone precursors (NOx/VOCs) can also cause large transient increases in ozone concentration, termed episodes or smog. These occur when high concentrations of precursors coincide with weather conditions favourable for ozone production such as when the air is warm and slow moving.These "ozone episodes" provide concentrations of O_3 (>40 ppb) which are toxic both to human health (with a long term threshold objective of 50 ppb daily 8-h mean, WHO) and vegetation.

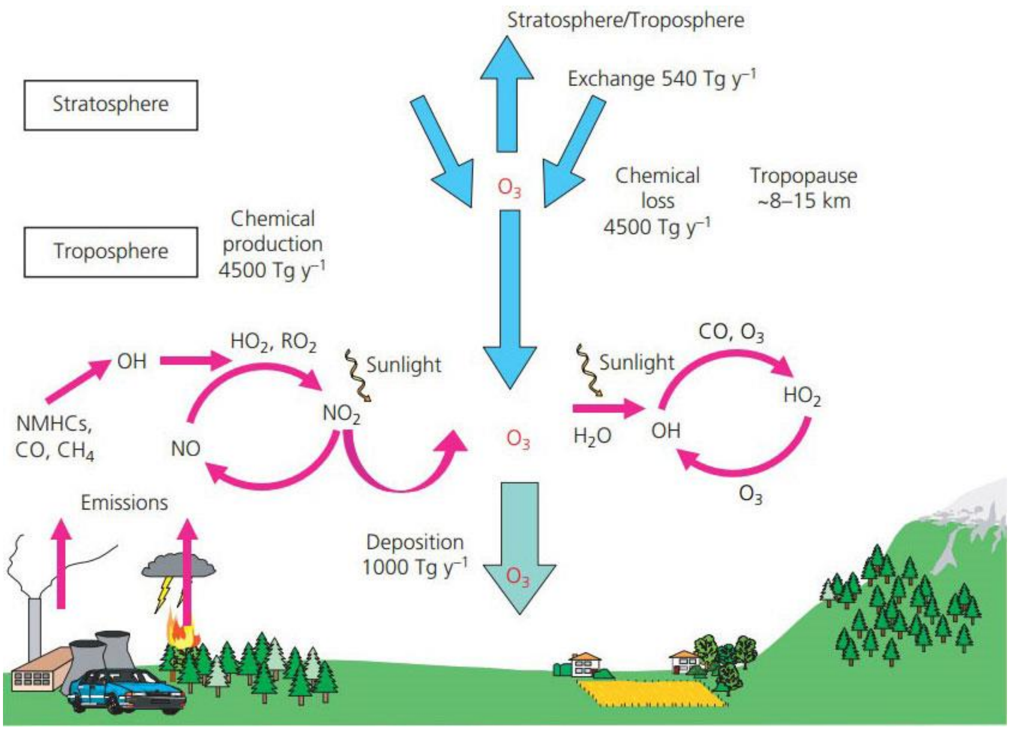


Figure 1: A schematic view of the sources and sinks of O3 in the troposphere. Annual global fluxes of O3 calculated using a global chemistry–transport model have been included to show the magnitudes of the individual terms. These fluxes include stratosphere to troposphere exchange, chemical production and loss in the troposphere and the deposition flux to terrestrial and marine surfaces. Data source: IPCC Fourth Assessment Report Working Group I Report "The Physical Science Basis" (Denman et al. 2007).

Prior to the industrial revolution natural sources of NO_x and VOCs (see table) would have generated O_3 in the troposphere, adding to that transported from the stratosphere. However, the large amounts of NO_x and VOCs released by human activities, such as the combustion of fossil fuels, has led to a large increase in the northern hemisphere background concentration. Evaluation of historical O_3 measurements indicate that since the 1950s, the background ozone concentration has roughly doubled (Volz and Kley, 1988, Vingarzan, 2004), although there has been some slowing down of this trend in the last decade (Derwent et al., 2013).

Table 1: Summary of Ozone formation from natural and anthropogenic NOX and VOC sources			
NO_x		**VOC**	
Natural	**Anthropogenic**	**Natural**	**Anthropogenic**
Soils, natural fires	transport (road, sea and rail), power stations, other industry and combustion processes	Vegetation, natural fires	Transport, combustion processes, solvents, oil production

Concentrations

Policy actions to date across Europe have reduced the emissions of ozone precursors NOx and VOCs. These emission controls have reduced peak ozone concentrations by typically 30 ppb in the UK, but over the last 20 years mean concentrations have been increasing in urban areas due to reductions in the local depletion of O_3 by NO. Background concentrations have increased in rural areas due to increases in the hemispheric background O_3 concentrations which have increased by approximately 0.2 ppb per year, or by about 5 ppb (Figure 2). The cause of the increases in background O_3 are increases in precursor emissions throughout the northern hemisphere, including shipping, aircraft, vehicle, and industrial emissions in developing economies.

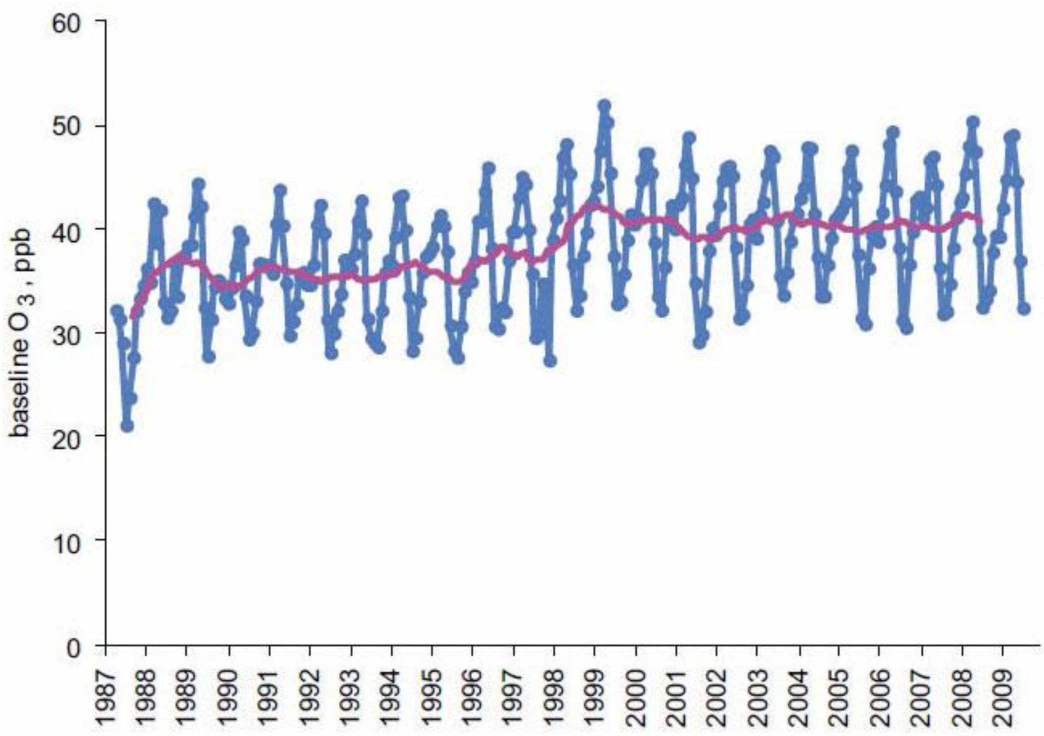


Figure 2: Trends in monthly mean baseline ozone levels at Mace Head on the west coast of Ireland between 1986 and 2006, including a running 12-month mean (pink line), showing the ~5 ppb increase in ozone mixing ratios over 20 years.

Ozone concentrations are highly variable, spatially and temporally. Figure 3 shows the high background concentration in March to May over the northern half of Scotland, upland Wales and northern England, and parts of southern and eastern England. In the summer months of May-July high concentrations cover parts of south-east England and some upland areas of Wales and northern Scotland. The summer time spatial pattern extends across Western Europe and is caused by regions where ozone production occurs more frequently in a combination of precursor emissions (NOx and VOCs), high solar radiation and temperatures. During the winter the highest concentrations occurring in the NW and the lowest in the SE as a consequence of ozone destruction in the NO polluted atmosphere of the dense urban/industrial regions of the southern UK and continental Western Europe.

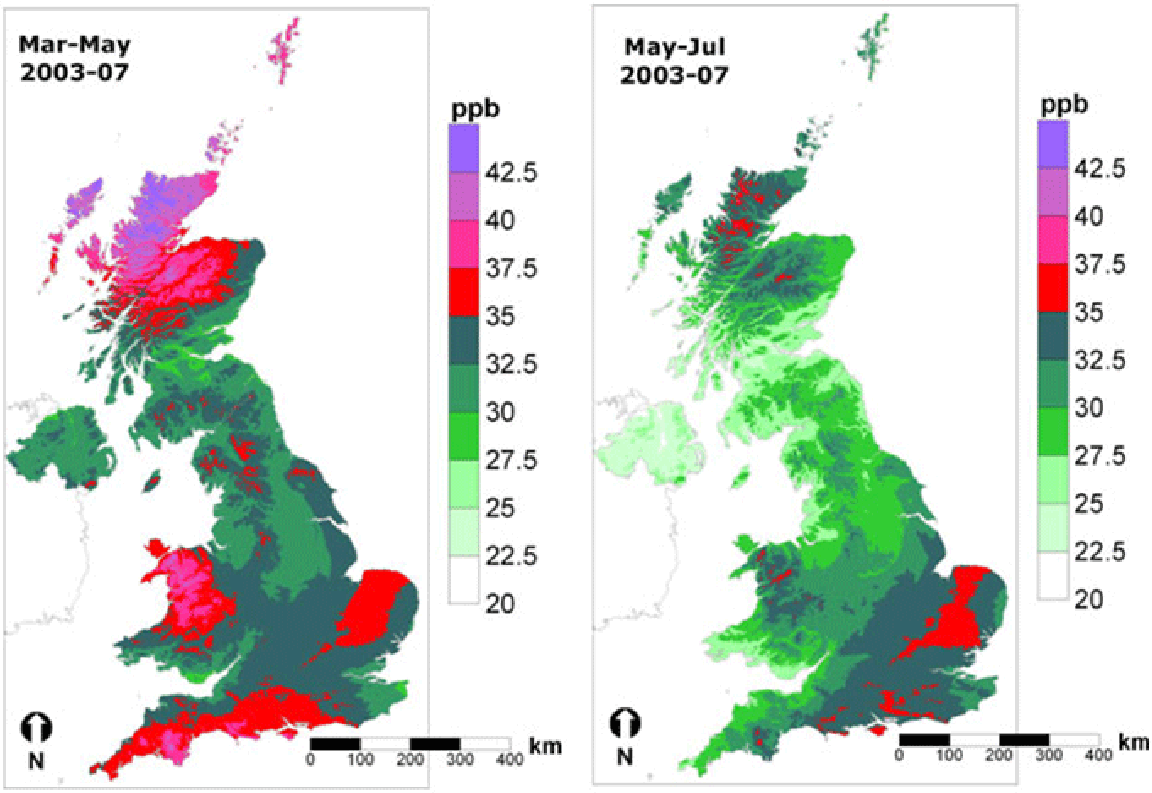


Figure 3: Distribution of mean ozone concentrations over the periods March-May and May-July, using data for the five year period 2003-07.

Effects

A large body of evidence has shown that ambient O_3 causes damage to O_3-sensitive vegetation. O_3 enters leaves via the stomatal pores on the leaf surface. Once inside the leaf, a series of chemical reactions occur leading to damage to cell membranes and other negative impacts on plant metabolism, including photosynthesis. These effects can be in response to short-term episodes or cumulative during the growing season, and can lead to:

- Visible leaf damage and premature aging of leaves;
- Reductions in above- and below-ground growth and biomass;
- Changes in the ratio between shoot and root biomass (including carbon allocation);
- Reductions in flower number, flower biomass and seed production;
- Reductions in crop yield quantity and quality, including cereal grains, potato tubers and tomato fruit;

- Changes in forage quantity and quality for pasture;
- Altered tolerance to abiotic stresses such as drought and frost and biotic stresses such as pest attacks and diseases

Critical Levels

Ozone pollution effects on vegetation are described by cumulative metrics that are based on either atmospheric concentration of O_3 over a threshold concentration (AOT40) or modelled uptake of O_3 through the stomatal pores on the leaf surface (Phytotoxic Ozone Dose above a threshold of Y, POD_Y). A recent analysis of field evidence for O_3 effects confirmed that maps generated using an O_3 flux metric more closely matched locations of damage than those based on O_3 concentration (Mills et al., 2011a). New O_3 flux-based critical levels have been derived for crops, tree species and for grassland species (Mills et al., 2011b). See the overview on Guide to Critical Loads and Levels.

Other Impacts

The role of ozone as a contributor to the direct radiative forcing of global climate has grown in importance (IPCC, 2007). In addition, the recognition that the effects of O_3 on carbon sequestration through its effects on primary productivity of vegetation is an additional reason for interest in ground level ozone (Sitch et al., 2007).

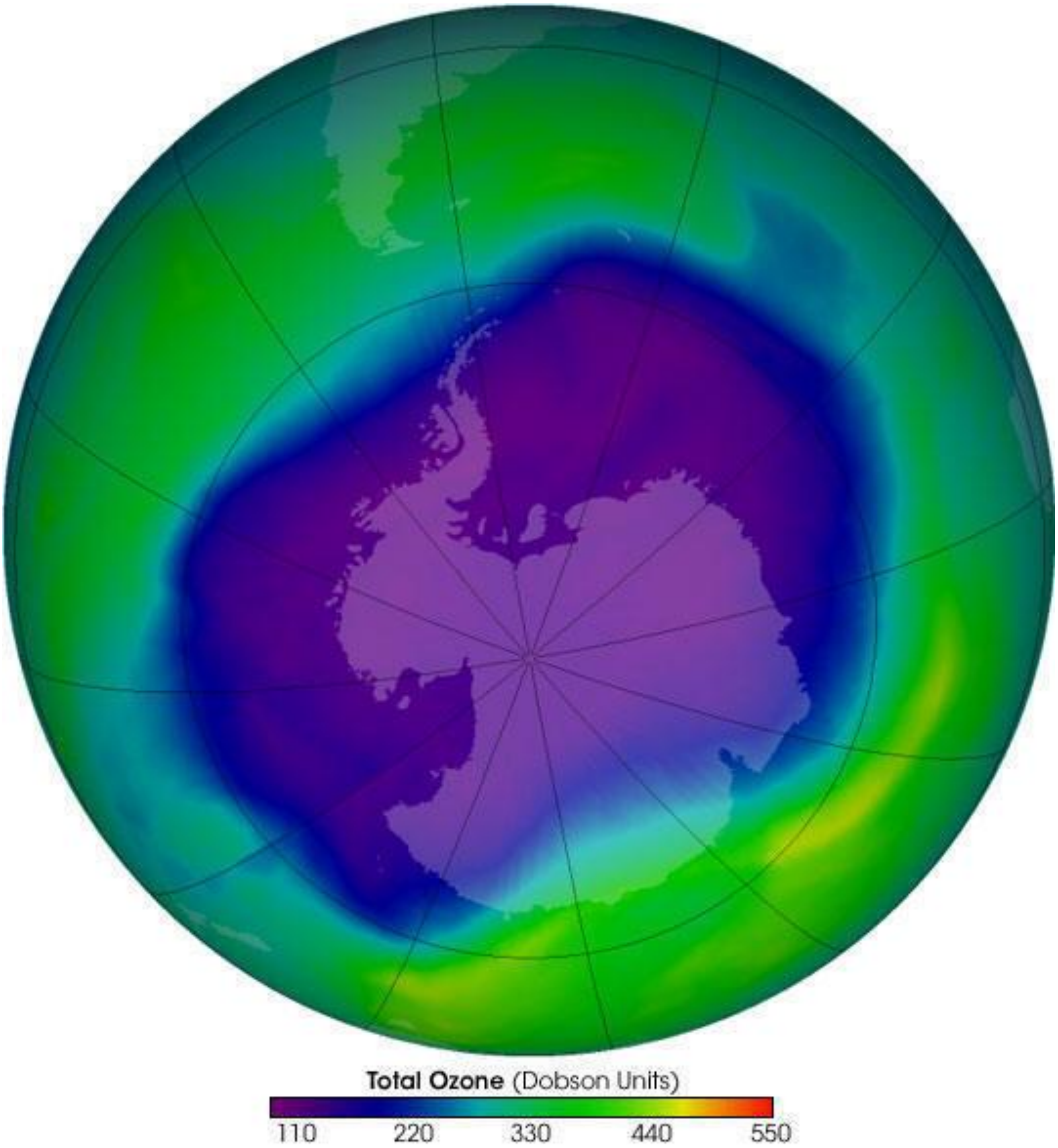
Further Reading (full references below)

The Royal Society (Royal Society, 2008) undertook at major review of ozone in 2008. They showed that action at a hemispheric scale is required effective control of ground level ozone.

The ICP Vegetation report 2017 on the *Mapping of critical levels of vegetation* provides an in-depth description of the methodology and approaches that has been researched for setting critical levels for ozone. Further reports and information are also available on the ICP Vegetation website.

The Ozone Hole in Antarctica

The seasonal thinning of the ozone layer of the earth's atmosphere above Antarctica, so allowing abnormal amounts of ultra-violet light to reach the earth's surface in those regions.



300 units (green) is normal, less than 220 (blue) is a "hole"

What is ozone?

Ozone is a gas* made of oxygen atoms. Usually oxygen atoms hang around in pairs - this is the sort of oxygen that we breathe and that helps things to burn. Oxygen sometimes however will form a molecule with three oxygen atoms, this is what we call ozone:

O_2 - two oxygen atoms - ordinary oxygen

O_3 - three oxygen atoms - Ozone

Ozone has a the particularly useful characteristic that it can absorb large quantities of ultra-violet (uv) light - more of that soon.

Where is the ozone?

Most of the ozone on earth isn't on earth at all, but in the layer of the earth's atmosphere called the stratosphere. This is the upper layer of the atmosphere and starts between 12.9 to 19.3 km (8 to 12 miles) above our heads and goes upwards to almost 50 km (30 miles). The stratosphere has virtually no clouds or other form of weather, it's thinnest at the equator and thickest at the poles.

Ozone is formed in the stratosphere by the action of sunlight on oxygen molecules. In particular it is the high energy ultra-violet light in sunlight that is effective, it causes an oxygen molecule to split into two oxygen atoms:

$$\mathbf{O_2} \text{ -------->} \ \mathbf{O + O}$$

oxygen molecule 2 oxygen atoms

These lone oxygen atoms are very reactive, each then joins with another oxygen molecule to form a molecule of ozone:

$$\mathbf{O + O_2 \text{--------->} O_3}$$

oxygen atom + oxygen molecule ozone molecule

Ozone may also be destroyed by joining with a lone oxygen atom to get back to oxygen again. **Ultra-violet light is required for ozone to form in the stratosphere, but then the ozone absorbs the ultra-violet light so stopping it reaching deeper into the earth's atmosphere. The result is that levels of ozone are greatest at around 20km up**. This is good news for us as it stops lots of ultra-violet light getting through to us and also keeps the ozone high up in the atmosphere out of the way.

Isn't ozone bad news? I'm sure I saw something on the weather forecast....

Ironically, at ground level ozone is very bad news. It is a major component of photochemical smog. Down here on the ground, It is caused by the effect of ultra-violet light on nitrogen oxide from vehicle exhausts and so particularly affects built up areas in regions of high sunshine.

Ozone affects lung function, it can aggravate asthma and other chronic respiratory tract and lung diseases and can reduce lung function in the short term or even permanently on repeated exposure. Ozone has an effect like sunburn on the lining of the respiratory tract damaging the cells.

Why is a ozone hole a problem?

Ozone in the stratosphere is nicely out of the way and has the wonderful benefit to life on earth that it specifically absorbs the harmful ultra-violet light from the sun while letting other light wavelengths through.

Though we talk about a "hole" in the ozone layer, it's not an actual hole, just a thin bit. Ozone is spread thinly throughout the stratosphere - and in low quantities too - **if all the ozone above your head was collected together in a continuous layer, it would only be about 3-5mm (1/8 of an inch) thick.**

It's better to think of the ozone as being like orange squash in a glass of water where the water is the rest of the atmosphere. When we talk about an "ozone hole" we actually mean a region where the squash is thinner and more diluted than we'd like it to be.

When a "hole" forms in the ozone layer then this means that more harmful ultra-violet rays get through than are good for us or many other life forms, plant or animal as there isn't enough ozone to stop it.

Too much ultra-violet light can result in:

- **Eye damage** such as cataracts.
- **Immune system damage.**
- **Reduction in phytoplankton in the oceans** that forms the basis of all marine food chains including those in Antarctica.
- **Damage to the DNA** in various life-forms So far this has been as observed in Antarctic ice-fish that lack pigments to shield them from the ultra-violet light (they've never needed them before).
- **Skin cancer.**
- Probably other things too that we don't know about at the moment.

Climate effects

The depletion of the ozone hole has also caused an overall cooling trend on the Antarctic continent, this has masked to some extent the effects of warming temperatures, particularly on the larger part of East Antarctica and areas away from the peninsula region.

The loss of ozone has also led to increased winds and storms, both in frequency and strength. Winds in the Southern Ocean have been estimated to have increased by 15-20%. It has caused a low pressure system to form in the Amundsen Sea again both with increased frequency and strength. This low pressure sucks cold air from the interior of Antarctic and across the Ross Sea leading to a great increase in the amount of sea-ice forming in this area in recent years.

Why does a ozone hole form over Antarctica?

The ozone hole is caused by the effect of pollutants in the atmosphere destroying stratospheric ozone. During the Antarctic winter something special happens to the Antarctic weather.

- **Firstly, strong circular winds form that blow around the whole continent**, this is known as the "polar vortex" - this isolates the air over

Antarctica from the rest of the world.

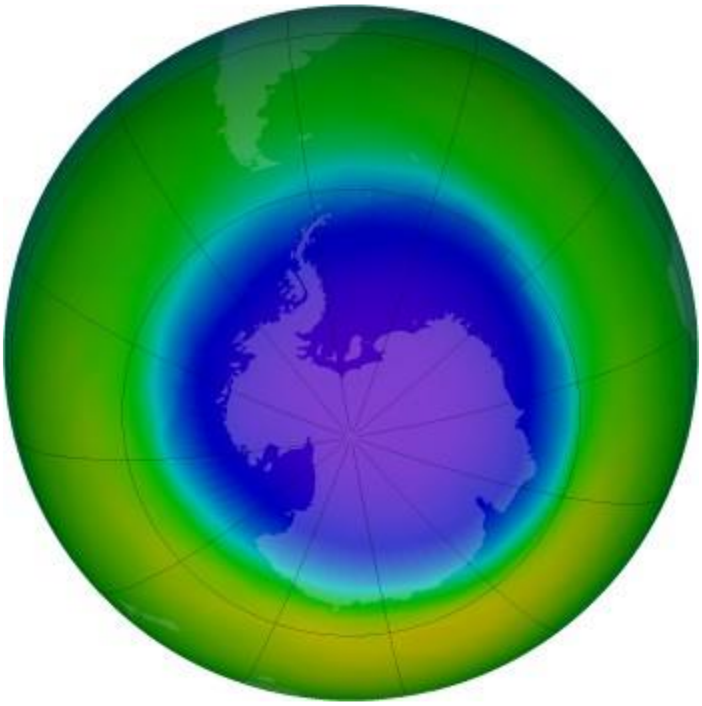
- **Secondly, special clouds form called Polar Stratospheric Clouds.** Clouds don't normally form in the stratosphere and these turn out to have the effect of concentrating the pollutants that break down the ozone, so speeding the process up.

Polar stratospheric clouds at about 25,000 m (80,000 ft) altitude. These are the highest flying of all clouds and only occur in polar regions where the temperature in the upper atmosphere dips below minus 70C (-100F). They are sometimes called "nacreous clouds" as they are coloured like the nacre of mother-of-pearl with coloured bands that move the position of cloud and observer.

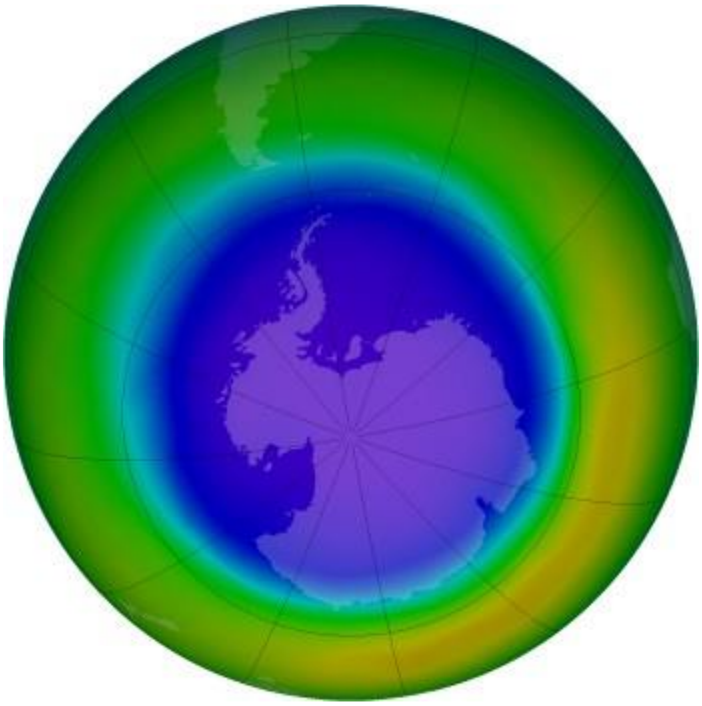
By the time spring arrives and the sun comes back after the long polar night, the ozone levels are severely depleted around the Antarctic continent causing the "ozone hole". Unfortunately, there then follows a particularly long period of high sunshine and long days, just to make the effect of the ozone hole worse with all that uv light around.

The concentration of ozone in the atmosphere is measured in "Dobson Units", the average concentration of ozone in the atmosphere is about 300 Dobson Units. The ozone hole is considered to be wherever the concentration drops below 220 Dobson Units.

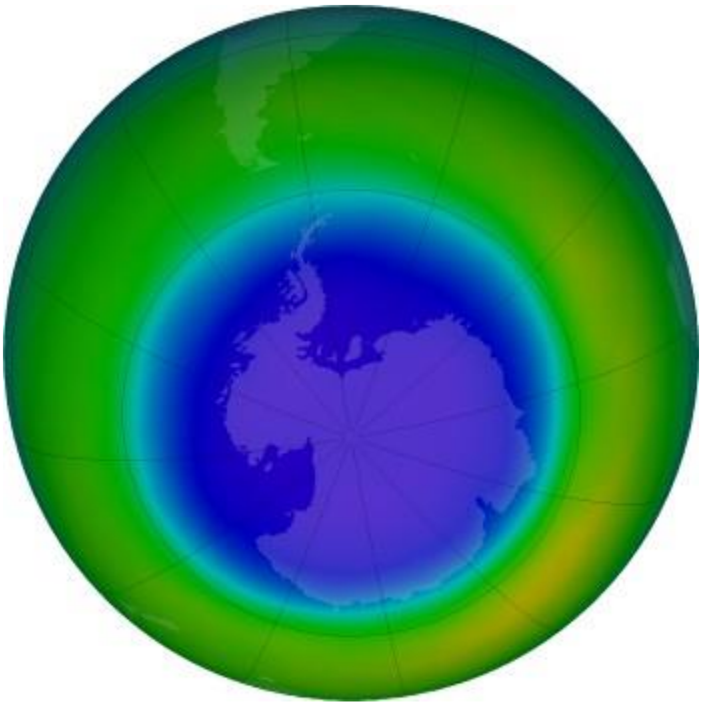
The following pictures show the extent of ozone thinning. Dark blue and purple colors correspond to the thinnest ozone, while light blue, green, and yellow pixels indicate progressively thicker ozone.



October 1999 (average)
Historically, the Antarctic ozone hole was largest during October. In recent years however, September has been the peak month.



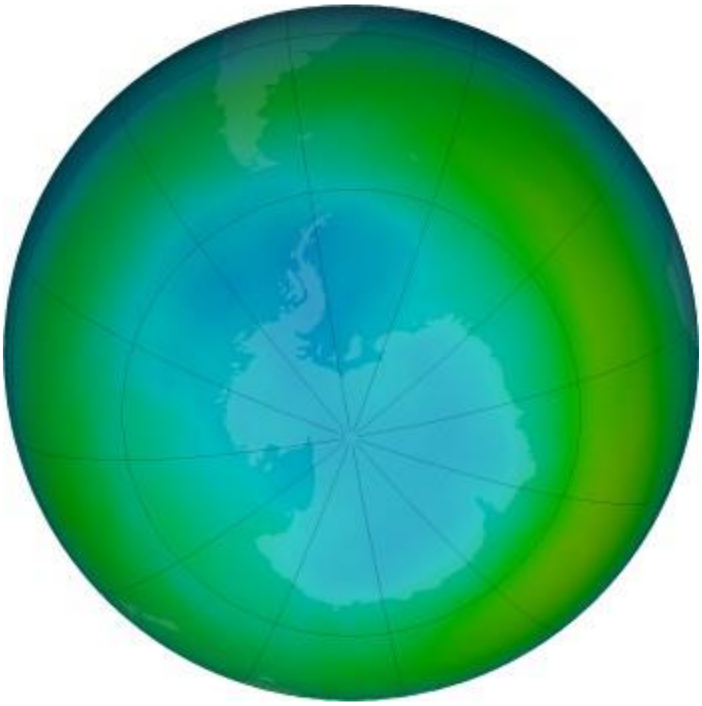
September 7th 2000
The ozone hole grew quicker than usual and exceptionally large. By the first week in September the hole was the largest ever at that time. For the first time it reached towards South America and to regions of high population.



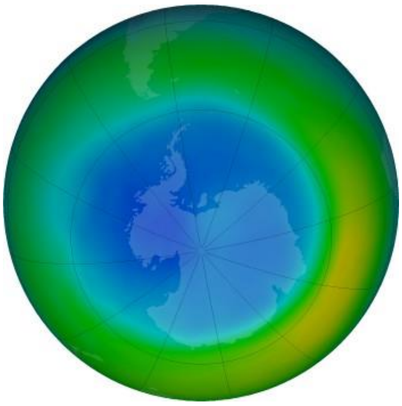
September 3rd 2020
In September 2020, the average area of the ozone hole was the largest ever seen, at 28.3 million square kilometres (11 million square miles). The previous record

was about 27.2 million square kilometres (5 million square miles) on September 19th, 1998.

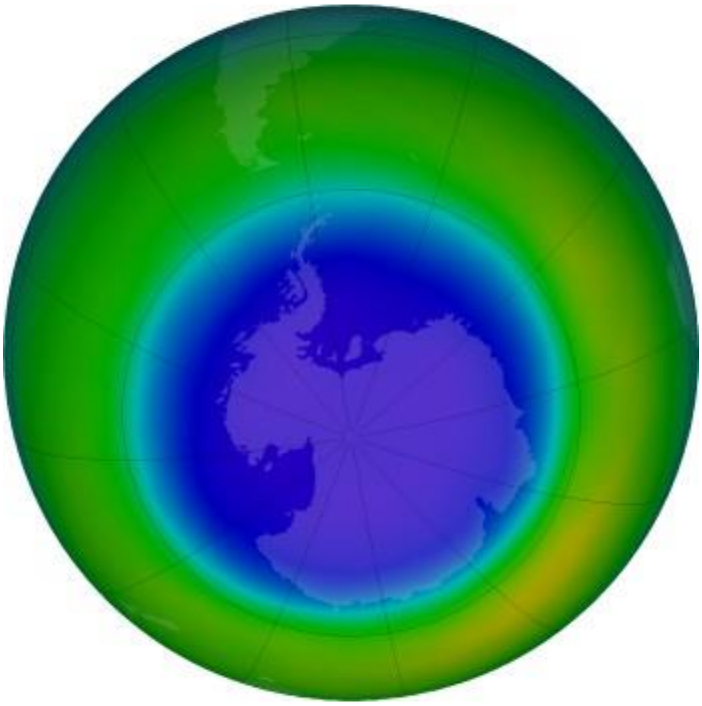
The ozone hole builds up over the winter months, peaking at around September and breaking up again by December, this data set from 2020.



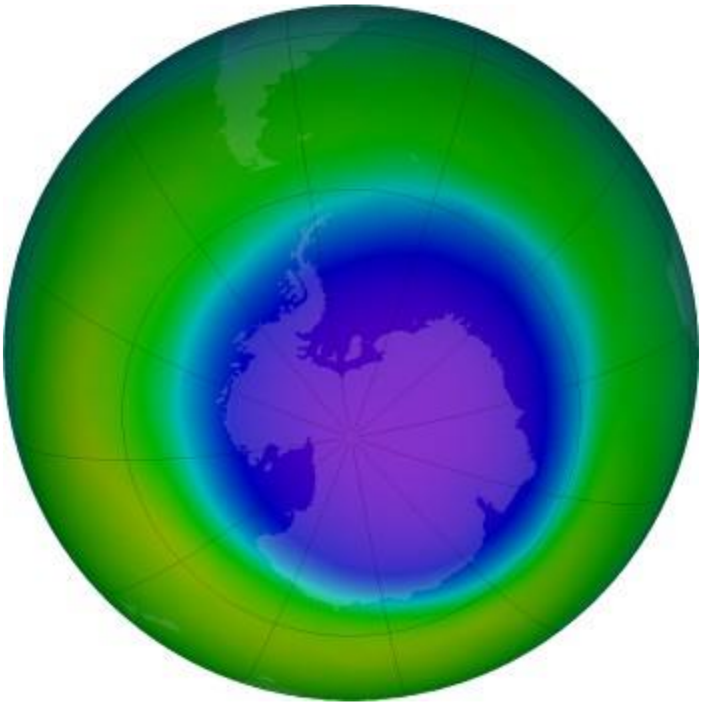
July



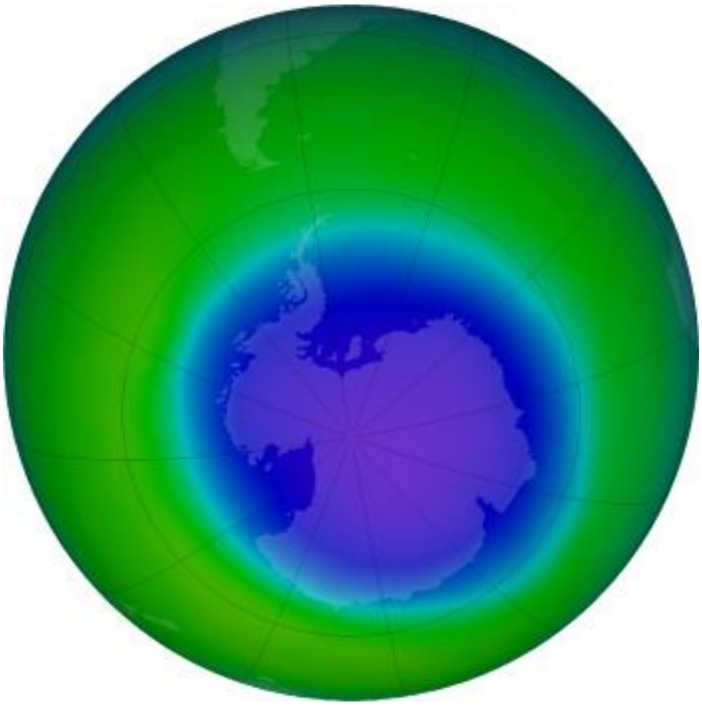
August



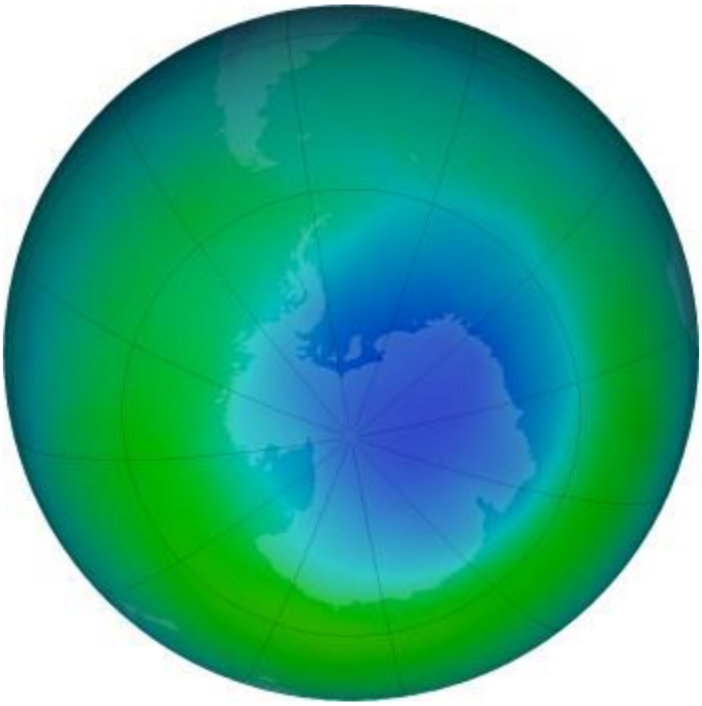
September



October



November



December

What causes the ozone depletion?

Ozone is mainly broken down by chemicals called ChloroFluoroCarbons CFC's and also by nitrogen oxides.

CFC's ironically were first used in large quantities because they were thought to be safe and inert (unreactive) chemicals. They are a group of chemically similar gases used in refrigeration systems, air conditioners, aerosols, solvents and in the production of some types of packaging. Nitrogen oxides are a by-product of fuel burning, they find their way to altitude where they can damage ozone come from jet aircraft exhausts.

CFC's don't occur naturally, they are man-made chemicals. They are very useful when they are where they are supposed to be, and doing what they are supposed to be doing. But once released into the atmosphere they are a serious pollutant. The problem was it took us many years to realise this during which time we thought they were perfectly harmless, but in fact were building up to levels that will take decades to disappear again even when we stopped producing them altogether.

The reactions that destroy ozone are very complicated. They take place on the surface of the ice particles of the Polar Stratospheric Clouds and **it takes only a small amount of CFC to destroy an awful lot of ozone.**

Is the ozone hole going to stay over Antarctica?

Since the annual thinning of the ozone layer over Antarctica was first discovered, measurements have been carried out in all regions. Ozone depletion has been measured everywhere in the world except in the tropics. Depletion is usually worse the further from the equator and recently an Ozone hole (as defined by a distinct area of very low ozone levels) has been detected above the North Pole in the arctic.

There is a lot to learn about the breakdown of ozone in the atmosphere and warmer regions, non polar depletion of ozone in particular is not properly understood.

So for the time being the "ozone hole" seems to be an Antarctic phenomena, but a less severe thinning of the ozone layer is pretty much a world-wide thing. How acute and important it will be in the future is not known.

Can the ozone hole recover?

Yes! - hurrah!

The way to stop the formation, growth and spread of ozone thinning is to reduce the production of those chemicals that cause the destruction of ozone, namely CFC's and nitrogen oxides.

In 1987, the **Montreal Protocol** was signed by many nations whereby those nations that signed agreed to reduce their emissions of CFC's to a half (of the 1987 levels) by 2000.

Potential problems come from nations that do not see the reduction of CFC's to be a priority, and also from the huge quantity of refrigeration and air conditioning systems in the world that still contain CFC's. If they are not disposed of correctly, then the CFC's will escape into the atmosphere and continue to destroy ozone.

The latest estimates are that as long as production and release of CFC's is regulated properly, global ozone levels should recover by 2050, this may seem a long way off, though the indications are that we are on target for it, the Montreal Protocol was 36 years ago, 2050 is 27 years away (I'm writing this in 2023), so it is doable. **We just have to make sure we don't start to use CFC's again.**

Ozone layer recovery is an environmental success story

Published

15 September 2021

The World Meteorological Organization joins the rest of the international community in marking World Ozone Day on 16 September. It highlights the importance of safeguarding the Earth's protective ozone layer and shows that collective action, guided by science, is the best way to solve major global challenges.

The ozone layer in the upper atmosphere blocks ultraviolet (UV) radiation that harms living tissue, including humans and plants. The ozone “hole,” which was discovered in 1985 is the result of human emited chlorofluorocarbons (CFCs), which are ozone-depleting chemicals and greenhouse gases used as coolants in refrigerators and in aerosol spray. Nearly 200 countries signed the Montreal Protocol in 1987, which phased out the production and consumption of CFCs.

A new study in Nature demonstrates that by protecting the ozone layer, which blocks harmful UV radiation, the Montreal Protocol also protects plants and their ability to pull carbon from the atmosphere.

“The Montreal Protocol began life as a mechanism to protect and heal the ozone layer. It has done its job well over the past three decades. The ozone layer is on the road to recovery. The cooperation we have seen under the Montreal Protocol is exactly what is needed now to take on climate change, an equally existential threat to our societies,” said UN Secretary-General Antonio Guterres in a message.

The most recent WMO /UN Environment Programme Scientific Assessment of Ozone Depletion, issued in 2018, concluded that the measures under the protocol will lead to the ozone layer on the path of recovery and to potential return of the ozone in the Arctic and Northern Hemisphere mid-latitude ozone before the middle of the century (~2035) followed by the Southern Hemisphere mid-latitude around mid-century, and Antarctic region by 2060 .

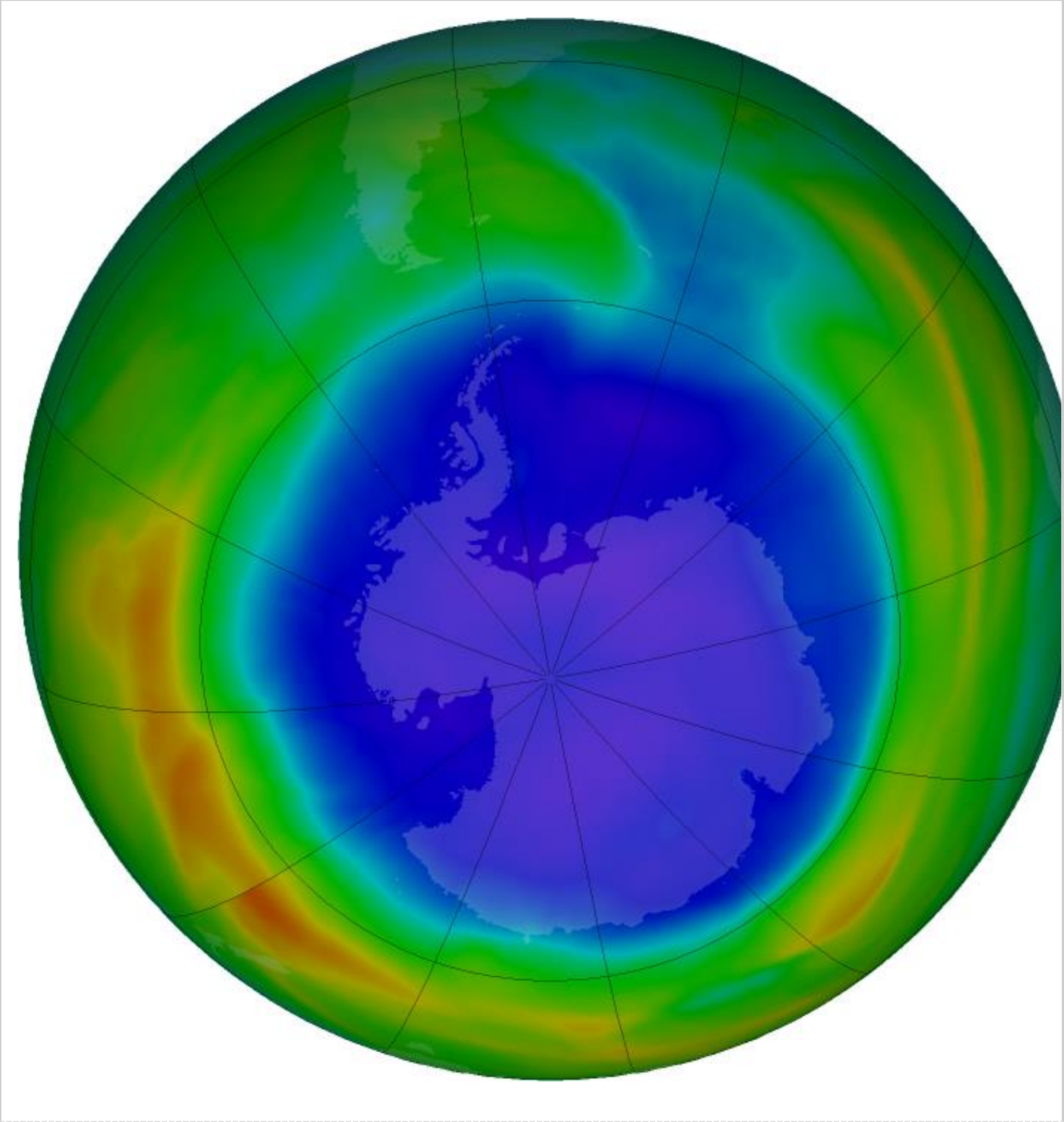
Although the use of halons and chlorofluorocarbons has been discontinued, they will remain in the atmosphere for many decades. Even if there were no new emissions, there is still more than enough chlorine and bromine present in the atmosphere to destroy ozone at certain altitudes over Antarctica from August to December. The formation of the ozone hole is still expected to be an annual spring event. Its size and depth are governed to a large degree by the meteorological conditions particular for the year.

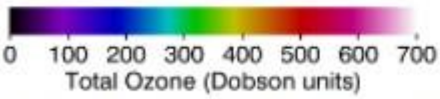
As of the first week of August 2021, the ozone hole reappeared and is rapidly growing and has extended to 23 million square kilometers on 13 September which is above the average since the mid 1980s. The lowest ozone value in the during this seasons was around 140 DU. The hole fluctuates in size annually and it usually reaches its largest area during the coldest months in the southern hemisphere, from late September to early October.

Its evolution is monitored by satellites and ground-based observing stations of WMO’s Global Atmosphere Watch Programme. Those observations are being combined with numerical modelling by different organizations and institutions (NASA, the Copernicus Atmospheric Monitoring Service implemented by

ECMWF, ECCC, KNMI and others) to provide near -real time information and analyses on the ozone levels at different parts of the stratosphere, the location and dimensions of the ozone depleted area.

In 2020, there were exceptionally large ozone holes over the Antarctic and Arctic, reflecting extreme meteorological conditions. Specific dynamic conditions in the stratosphere in 2019 led to the smallest Antarctic ozone hole since its discovery. This shows the need for continued vigilance and observations.





False-color view of total ozone over the Antarctic pole.
The purple and blue colors are where there is the least ozone,
and the yellows and reds are where there is more ozone (NASA).

Ozone and climate

The theme for this year is ***Montreal Protocol – keeping us, our food and vaccines cool.***

Ozone depleting substances (ODS) are also greenhouse gases (GHG) and their abundance in atmosphere over the years has made an important contribution to the radiative forcing of climate.

While ODS concentrations are expected to keep decreasing, the concentrations of long-lived greenhouse gases have been increasing.

The distribution and amount of stratospheric ozone depends on temperature and circulation, so that changes in climate will affect the distribution of ozone. Long-lived greenhouse gases warm the troposphere, but cool the stratosphere, leading to changes of the global circulation, affecting the stability of the polar winter vortices, and changing weather patterns.

Therefore, the future evolution of the ozone layer will be influenced by the concentrations of these long-lived greenhouse gases, and by climate change.

The Montreal Protocol has led to very significant avoided warming and the Kigali amendment which regulates the hydrofluorocarbons (HFCs), CFCs and hydrochlorofluorocarbons (HCFCs) replacement gases, adds a further layer of important climate protection. The avoided ultraviolet radiation and climate change also have co-benefits for plants and their capacity to store carbon through photosynthesis.

Some recent scientific findings point that the ozone depletion in the Arctic polar vortex could intensify by the end of the century unless global greenhouse gases are rapidly and systematically reduced. In the future, this could also mean more UV radiation exposure in Europe, North America and Asia when parts of the polar vortex drift south.

Scientists are monitoring the extent to which climate change is leading to stratospheric cooling, which enhances the possibilities for observing temperatures under -78°C especially in the Arctic where there is evidence that the coldest stratospheric winters are becoming colder. Those temperatures are needed for the polar stratospheric cloud formation where the destruction of the ozone takes place.

UV Radiation

Several feedbacks to climate change occur from the effects of UV radiation on the biosphere. For example, the breakdown or photodegradation of dead plant material releases carbon to the atmosphere, increasing the amount of carbon dioxide and other greenhouse gases.

Increased thawing or melting of snow, ice and permafrost in the Arctic also releases GHGs and has a negative effect on the exposed ecosystems.

Studies are showing that temperature, UV radiation and frequency of rainfall are key factors that determine the availability or range of suitable habitats for certain plant species to survive. UV-B radiation and factors associated with climate change affect plant growth, pathogen and pest defence, and food crop quality.

For human health, UV radiation can have significant negative effects, for example, in causing skin cancers and certain eye diseases, such as cataract. However, the Montreal Protocol has played a major role in avoiding large numbers of cases and deaths.

With regard to pollution, UV radiation can have a substantial impact on the composition and quality of the atmosphere; on human, terrestrial and aquatic environment health. It drives the breakdown of plastic pollutants with implications for human health and the environment.

UV radiation is major factor in urban and continental-scale air pollution because many organic compounds emitted by human activities and natural processes are transformed by solar UV radiation into toxic products, decreasing air quality and damaging human health. UV radiation is also a key driver of contaminant breakdown in aquatic environments generating toxic and carcinogenic products.

Air quality is dependent on solar UV radiation in the troposphere and thus on the thickness of stratospheric ozone layer and is influenced by transport of ozone from the stratosphere to the troposphere.

Heavy metal ions in aquatic systems are concentrated by microplastics after surface oxidation by UV radiation, promoting metal binding to the polymers, thereby increasing toxicity.

WMO Global Atmosphere Watch Programme observing network

The atmospheric measurements and analysis allowed to detect the renewed emissions of some of the controlled substances and led to their timely reduction. WMO's Global Atmosphere Watch Network has stations in the

Arctic and these are performing high-quality measurements of ozone, ODS, GHG and UV radiation.

Global Wind Patterns and Wind Belts

There are several kinds of wind blowing across the surface of the Earth, depending on the origin, destination, and distance traveled, among other factors. These are called prevailing or planetary winds. Global winds are one such type of wind.

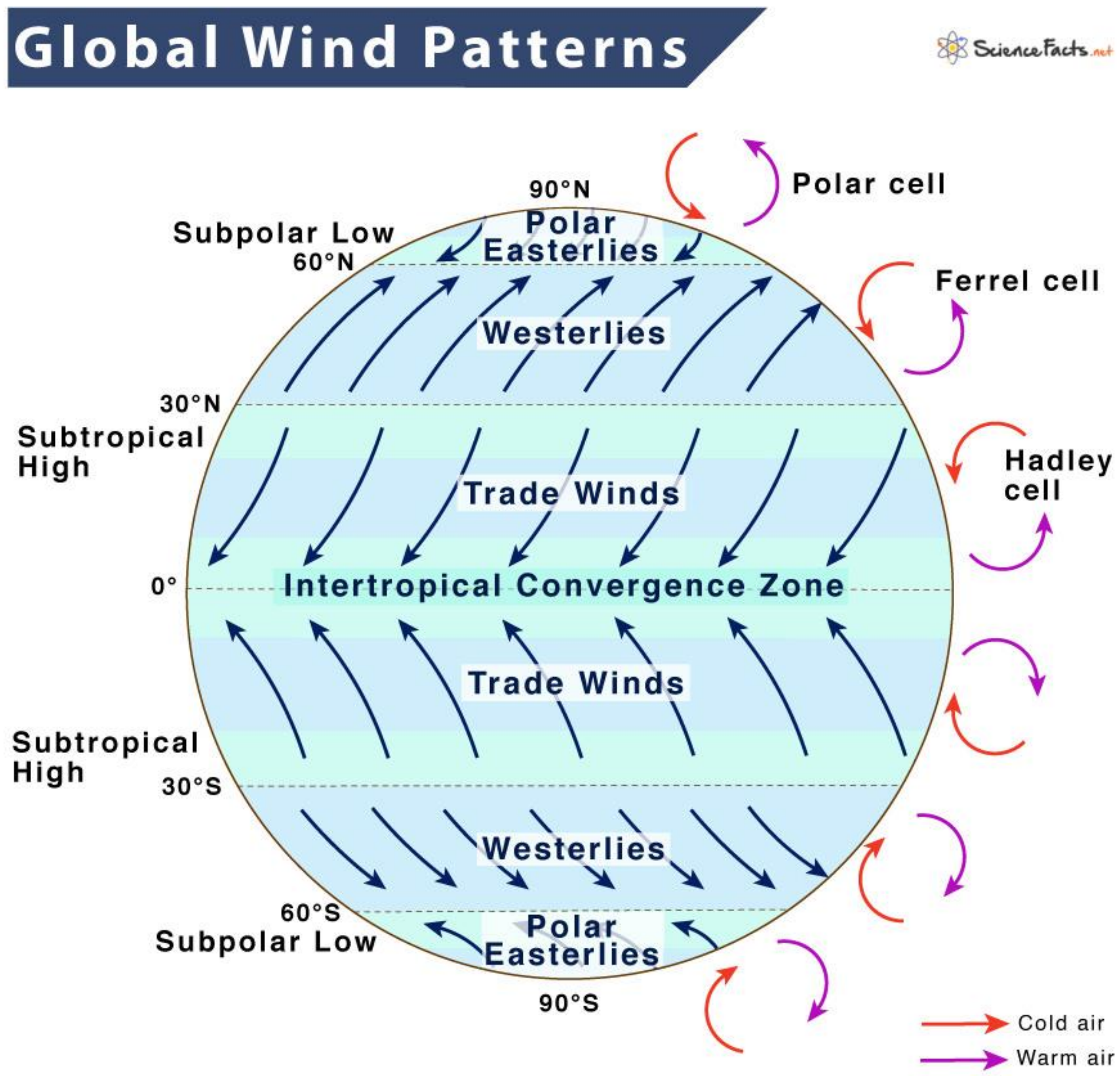
Global winds are winds that develop in belts distributed all around the world. Like local winds, the leading cause of global winds is unequal heating of the atmosphere, causing a difference in air pressure.

What Causes Global Wind Patterns

Earth is hottest at the Equator and cooler toward the poles. In the area near the Equator, the sun is overhead for most of the year. As a result, the warm air rises and moves towards the poles. This phenomenon creates a low-pressure zone near the equatorial area called the Equatorial Low-Pressure Zone, also known as the Intertropical Convergence Zone (ITCZ). At the polar front, the cooler air sinks and moves back towards the Equator.

Almost 30° north and 30° south of the Equator, this air cools and descends to create high-pressure zones called Subtropical High-Pressure Zones. Cold air at the poles descends and creates Polar High-Pressure Zones. The cold polar air moves at about 60° north and south towards the Equator and starts rising.

This unique, permanent movement of winds across the surface of the Earth due to permanent air pressure zones creates a distinct pattern known as global wind patterns or global wind systems.



Global Winds Patterns Belts

However, global winds do not move directly from north to south or south to north due to the rotation of the Earth. All winds in the Northern Hemisphere curves to the right as they move. Contrastingly, winds appear to curve to the left in the southern hemisphere. This apparent shift in the path of the wind moving about the Earth's surface due to the Earth's rotation is called the Coriolis Effect.

The global movement of air on the Earth in the three atmospheric convection cells is summarized below:

Convection Cell	Location	Air Movement	Direction of Air	Wind Name
Hadley Cell	0° to 30° latitude north and south	Rising at the Equator, sinking at 0° to 30° north and south	Towards the Equator	Tropical Easterlies (Trade Winds)
Ferrel Cell	30° to 60° latitude north and south	Rising at 60° north and south, sinking at 30° north and south	Towards the poles	Prevailing Westerlies
Polar Cell	60° to 90° latitude north and south	Rising at 60° north and south, sinking at 90° north and south	Towards the Equator	Polar Easterlies

What Factors Contribute to Global Winds

Thus the pattern of global winds is affected by:

- Unequal heating of the Earth
- High and low-pressure areas (Pressure Gradients)
- Coriolis Effect
- Convection Cells

What are Global Wind Belts

As we now know, three types of prevailing winds on the Earth form three major global wind belts created due to the Coriolis Effect. They are named based on their origin and the area where they blow. We will now describe them in more detail.

1. Tropical Easterlies (Trade Winds)

Location – 0° to 30° Latitude

They are persistent winds that blow westward and toward the Equator from the subtropical high-pressure belt. Trade Winds finally converge at an area near the ITCZ, producing a narrow band of clouds, and thunderstorms encircle portions of the globe. It is more robust and consistent over the oceans than the land, often producing partly cloudy sky conditions characterized by shallow cumulus clouds or clear skies.

Most tropical storms, including hurricanes, cyclones, and typhoons, develop as trade winds.

2. Westerlies

Location – 30° to 60° Latitude

They are moist prevailing winds that blow from the subtropical high-pressure belts towards sub-polar low-pressure belts. Westerlies come from the southwest in the Northern Hemisphere and the northwest in the Southern Hemisphere.

Westerlies are essential in carrying the warm, equatorial winds to the western coasts of continents.

3. Polar Easterlies

Location – 60° to 90° Latitude

They are dry, cold prevailing winds that blow from the high-pressure areas of the polar highs in both hemispheres. They flow towards low-pressure areas within the Westerlies at high latitudes. Cold air subsides at the pole creating high pressure, creating a northward flow of wind towards the Equator in the southern hemisphere. The flow is then deflected westward by the Coriolis Effect. Thus, these prevailing winds blow from the east to the west.

Since the winds originate in the east, they are then known as easterlies.

How Do Global Winds Affect Climate

The global wind belts are associated with typical weather conditions, and the prevailing winds lead to seasonal climatic changes in many places on Earth. All low-pressure areas are associated with high precipitation (tropical rainforests), and all high-pressure areas are associated with arid climates (deserts).

Jet Streams and Global Wind Belts

The polar jet stream or the jet stream blows high up in the atmosphere where the two cells meet. They are fast-moving air currents at the boundary between the troposphere and the stratosphere.

They also develop due to the unequal heating of the Earth's surface, creating a vast temperature difference between two air masses. Jet streams circle the planet, mainly from west to east, but the flow often shifts to the north and south. The polar jets are the strongest jet streams that blow between about 30°N and 50° to 75°N. They blow towards the south in the winter and north in the summer.

El Niño and La Niña: Their Impact on the Environment

We know that there are many anthropogenic forcings on the climate, particularly the volume of carbon and greenhouse gases pumped into the atmosphere as a part of our everyday lives. Yet there are a number of natural processes that affect local weather, regional climate and global conditions. Some effects on our climate are a result of fluctuations and anomalies in the complex water conveyor belts of the ocean currents of the world. These fluctuations are known as "oscillations" and the two best-known oscillations are El Niño and La Niña .

The latter is the opposite of the former and make up an oscillation known as ENSO. Understanding them requires knowledge of a broad range of data from multiple disciplines. Typically, researchers who understand the processes and study their causes and effects have post-graduate degrees in such disciplines as oceanography, geography, climatology and meteorology. The phenomena and the data extrapolated from them, have applications for palaeoclimatology (the study of climate in the past), anthropology, palaeobotany and archaeology, particularly in what we can extrapolate from the changes to tree ring data (dendrochronology) .

Oscillations occur naturally in oceans all across the world; some have a limited impact on the regional weather and wider climate, and some have a much greater impact . El Niño and La Niña are examples of oscillations that have a greater impact on our climate with effects that are perhaps surprisingly felt all over the globe . In economies that are dependent on certain weather conditions occurring regularly and on time (annual summer rainfall, spring ice melt etc), erratic oscillations can cause problems in these areas leading to drought. Knock on effects can lead to fish migrations and economic hardship for areas that rely on fish stocks. Marginal areas suffer or thrive depending on the effects of El Niño and La Niña leading to further knock on effects elsewhere

El Niño and La Niña: What Are They?

Both El Niño and La Niña are opposite effects of the same phenomenon: the ENSO (El Niño Southern Oscillation). Both are an oscillation in the temperatures between the atmosphere and the ocean of the eastern equatorial Pacific region, roughly between the International Dateline and 120 degrees west . El Niño - the conditions for which build up between June and December - is caused by a change in the wind patterns . Here, the Pacific Trade Winds fail to replenish following the summer monsoons of Asia . This warmer air leads to an oscillation between the cooler and warmer waters, leading to warmer ocean temperatures than normal.

It was Peruvian fishermen roughly around the start of the 20th century who first noticed the correlation between temperature changes and anchovy stocks that led to the development of study in this area , though they had noticed variations in fish stock for centuries. Every three to seven years and between December and January, there is a massive tailing off of stocks of the fish that the local economy relies on . Why does this happen? Up-swellings from the sea bed occur in normal years that bring nutrients up to the plankton to feed on and in turn abundance of plankton is beneficial to marine life up the food chain. In an El Niño year, that swelling does not occur so the plankton is reduced and in turn, so are the fish stocks, mostly through failure to reproduce .

La Niña is effectively the opposite of El Niño, indicated by prolonged periods of sea temperatures in the same region , and the effects stated above are generally reversed. During non El Niño years, atmospheric pressure is lower than normal over the western Pacific area and higher over the colder waters of the western Pacific, as already discussed. With La Niña, the Trade Winds are particularly strong in carrying warmer water westwards across the Pacific leading to colder than average temperatures in the east and warmer than average temperatures in the west . The result is that plankton increases in the areas where the temperature is cooler, leading to a positive effect on the marine life that depends on plankton or depends on those creatures that depend on plankton .

It is commonly expected that La Niña will follow immediately on from an El Niño event, but this is not always the case . Typically, both occur every three to five years but they have varied anything between two and seven years. Both phenomena last anything between nine and twelve months.

How Does El Niño Affect Conditions?

Typically, it comes around every five years and what usually happens is that warming in the oceans caused by the winds leads to diffusion of this warming all over the globe. It changes atmospheric pressures with consequences for rainfall, wind patterns, sea surface temperatures and can sometimes have a positive, and sometimes a negative effect on those systems . In Europe for example, El Niño reduces the instances of hurricanes in the Atlantic . The beginning of the El Niño system will be seen over North America in the preceding winter ; typically they include:

- Mild winter temperatures over western Canada and north western USA
- Above average precipitation in the Gulf Coast, including Florida
- A drier than average period in Ohio and pacific northwest

The effects of El Niño can sometime be erratic and are not always be predictable. For example, conditions at the start of 2014 were remarkably similar to the 1997/8 ENSO event and so therefore it was expected to be an El Niño year. Yet, as late as August, the initial warning signs were not appearing in the atmosphere to precede warming in the oceans meaning that the likelihood of El Niño occurring was dropping off but not entirely eliminated . Whether this is another effect of climate change is yet to be seen. However, some oceanographic institutes still predict that El Niño will take place in the autumn months, pointing to warming throughout August and the sometimes late nature of the development of El Niño .

How Does La Niña Affect Conditions?

Like El Niño, it too affects atmospheric pressure and temperature, rainfall and ocean temperature. In Europe, El Niño reduces the number of autumnal hurricanes. La Niña has less of an effect in Europe but it does tend to lead to milder winters in Northern Europe (the United Kingdom especially) and colder winters in southern/western Europe leading to snow in the Mediterranean region. Elsewhere in the world, areas that are affected by La Niña experience the opposite of the effects they experience with El Niño . It is continental North America where most of these conditions are felt. The wider effects include :

• Stronger winds along the equatorial region, especially in the Pacific
• Decreased convection in the Pacific leading to a weaker jet stream
• temperatures are above average in the southeast and below average in the northwest
• Conditions are more favourable for hurricanes in the Caribbean and central Atlantic area
• Greater instances of tornados in those states of the US already vulnerable to them

In the western Pacific, the formation of cyclones shifts westwards which increases the potential for landfall in those areas most vulnerable to their affects, and especially into continental Asia and China . There is greater rainfall in the west too, especially in Australia, Indonesia and Malaysia and further westwards toward the southern countries in the African continent. Consequently, over the US and Canada there will be lower than average precipitation and this pattern follows the coast southwards where the western portion of South America will also experience lower than average rainfall.

How El Niño and La Niña will change or affect climate change in the future is now of tremendous importance thanks to the known effects over the last century or more - and the conditions are still not very well understood, though the phenomenon has been known since the early 1600s . For climate scientists, this is a grey area as to whether it will have an impact on the climate, or whether they will be affected by climate change . Some recent research has suggested that the effects of the ENSO will worsen as the climate changes .

It is important for lives and livelihood, for economies and insurers to understand the potential for damage caused by extreme weather such as El Niño and La Niña so their continued study is of vital importance. Historical records have built up to the point that researchers have a clear idea of what the likely effects of each ENSO is likely to be in any given year. A paper in 2013 compiled the effects from records going back 700 years and how they have impacted global conditions in this time . The question is not settled, though many scientific institutes have devoted much time to the study of the ENSO and conflicting data says that ENSO may become the increasing norm.

IMPORTANT READ THIS

What are El Niño and La Niña?

El Niño is a term for the warming phase of the El Niño Southern Oscillation (ENSO), a cyclical weather pattern that influences temperature and rainfall across the global. It is a warming of the central to eastern tropical Pacific Ocean. During an El Niño event, sea surface temperatures across the Pacific can warm by 1–3°F or more for anything between a few months to two years. El Niño impacts weather systems around the globe, triggering predictable disruptions in temperature, rainfall and winds.

La Niña is the opposite – a cooling phase of ENSO that tends to have global climate impacts opposite to those of El Niño.

Though ENSO is a single climate phenomenon, it has three states, or phases, it can be in “El Niño”, “La Niña” (the two opposite phases) or “neutral” (neither El Niño nor La Niña). To qualify as an El Niño event, according to the US National Oceanic and Atmospheric Administration (NOAA), sea surface temperatures (SSTs) must remain at or above 0.5°C (about 1°F) for at least three months. For La Niña, the SSTs are below average rather than above.

Does a La Niña always follow an El Niño?

A La Niña episode may, but does not always, follow El Niño.

How often does it occur?

The pendulum between an El Niño and La Niña phase swings back and forth, on average, every three to seven years.

Wait, didn’t we just have an El Niño episode?

Yes, the last episode began just two years ago, in 2015. The 2015/2016 El Niño was one of the three strongest episodes on record. There were super El Niño events in 1972-73, 1982-83 and in 1997-98.

How long do El Niño and La Niña typically last and when do they develop?

El Niño and La Niña episodes typically last 9-12 months. It is somewhat easier for a La Niña event to last longer (up to 2–3 years) than an El Niño, which rarely persists for more than a year at a time. They both tend to develop during the spring (March-June), reach peak intensity during the late autumn or winter (November-February), and then weaken during the spring or early summer (March-June).

<u>How do El Niño and La Niña occur?</u>

El Niño and La Niña events are natural occurrences in the global climate system resulting from variations in ocean temperatures in the Equatorial Pacific. In turn, changes in the atmosphere impact the ocean temperatures and currents. The system oscillates between warm (El Niño) to neutral or cold (La Niña) conditions.

<u>Is El Niño caused by climate change?</u>

No. El Niño events are not caused by climate change – they are a natural reoccurring phenomenon that have been occurring for thousands of years. Some scientists believe they may be becoming more intense and/or more frequent as a result of climate change, although exactly how El Niño interacts with climate change is not 100 percent clear. Climate change is likely to affect the impacts related to El Niño and La Niña, in terms of extreme weather events. Further research will help separate the natural climate variability from any trends due to human activities.

<u>Can we prevent El Niño and La Niña from occurring?</u>

No, El Niño and La Niña are naturally occurring climate patterns and humans have no direct ability to influence their onset, intensity or duration.

<u>What is the humanitarian impact of El Niño?</u>

El Niño and La Niña can make extreme weather events more likely in certain regions, including droughts, floods and storms. Over 60 million people were impacted by the 2015/2016 El Niño although an exact number is hard to pinpoint. East Africa, Southern Africa, the Pacific Islands, South East Asia and Central America were most affected by extreme weather, including below-normal rains and flooding. The humanitarian fallout in certain areas included increased food insecurity due to low crop yields and rising prices; higher malnutrition rates; devastated livelihoods; and forced displacement. Excessive rainfall also triggered and exacerbated outbreaks of waterborne diseases such as cholera and typhoid as well as vector-borne diseases such as malaria. Twenty-three countries [1] issued humanitarian appeals totaling more than US$5 billion.

What are the global impacts of La Niña?

While El Niño and La Niña do impact global climate patterns; however, they neither affect all regions nor do are their impacts in a given region the same. In many locations, especially in the tropics, La Niña (or cold episodes) produces roughly the opposite climate variations from El Niño. For instance, parts of Australia and Indonesia are prone to drought during El Niño but are typically wetter than normal during La Niña. For the most accurate information at national or local level, it is important to consult National Meteorological and Hydrological Services. WMO Regional Climate Centres may also provide more particular information at national and regional levels.

The impacts of each La Niña event are never exactly the same. They depend on the intensity of the event, the time of year when it develops and the interaction with other climate patterns. La Niña is often associated with wet conditions in eastern Australia, and with heavy rainfall in Indonesia, the Philippines and Thailand. La Niña usually leads to increased rainfall in North Eastern Brazil, Colombia and other northern parts of South America and is associated with rainfall deficiency in Uruguay and parts of Argentina. Drier-than-normal conditions are generally observed along coastal Ecuador and North Western Peru. La Niña episodes feature a very wave-like jet stream flow over the United States and Canada in the northern winter, with colder and stormier than average conditions across the North, and warmer and less stormy conditions across the South. La Niña events are generally associated with increased rainfall in southern Africa, although they are not the only contributing factors. La Niña is associated with rainfall deficiency in equatorial eastern Africa – for instance Somalia and eastern Kenya.

It is important to stress that such factors as the Indian Ocean Dipole, the North Atlantic Oscillation/Arctic Oscillation can also have an important influence on seasonal climate.

Can we predict El Niño and La Niña episodes before they occur?

Yes, scientists can often predict the onset of El Niño and La Niña several months to a year in advance, thanks to modern climate models and observations data (which includes sensors on satellites and ocean buoys), which constantly monitors changing conditions in the ocean and atmosphere.

How do we predict El Niño/La Niña?

The first model-based ENSO predictions started in the late 1980s. Today, a number of computer models around the world use current ocean temperatures and atmospheric conditions to project the state of ENSO, looking a year or more into the future. Forecasters examine multi-model ensembles, scrutinizing where these models agree or disagree in order to issue El Niño and La Niña forecasts. Among the leading sources of regular ENSO forecasts are NOAA's Climate Prediction Center, the International Research Institute for Climate and Society at Columbia University (working with NOAA), and the Australian Bureau of Meteorology. (Note that the Australian BOM uses a higher threshold for El Niño development than the US definition; see above).

The World Meteorological Organization (WMO) facilitates development of a consensus-based El Niño/La Niña Update that is issued on a quasi-regular basis (approximately once every three months) through a collaborative effort with the International Research Institute for Climate and Society (IRI) and based on contributions from all the leading centres around the world.

Generally, the strongest ENSO events are predicted more accurately than weaker ones. A 2012 analysis in the Bulletin of the American Meteorological Society evaluated ENSO forecasts from 20 different prediction models for the period 2002 to 2011. The study found that the predictive skill of these models actually declined in the 2000s as compared to the 1980s and 1990s—not because of any loss in model quality (the models themselves had actually improved), but because the weaker, more variable ENSO events during this period made forecasting a far greater challenge.

Sources: UNOCHA, WMO, US National Oceanic and Atmospheric Administration (NOAA).

Global Dimming

Introduction

Global dimming is the reduction in the amount of sunshine reaching Earth's surface. Both global dimming and global brightening, the opposite effect, have been observed. Dimming is caused by an increased blockage in the atmosphere of light from the sun. Clouds and aerosols—small particles emitted by burning fuels—can both contribute to dimming. Because dimming reduces the amount of solar energy reaching Earth, it tends to cool the planet, masking or offsetting global warming.

Global dimming has been confirmed for the period of 1960 to 1990, during which time the amount of sunlight reaching Earth's surface decreased about 4%. However, the dimming trend has been reversed from about 1990 to at least 2007 over most of the world. There has been much confusion in the scientific literature over the measurement of global dimming, as well as some sensational media coverage of the subject. Dimming has been hailed both as an imminent threat to human life and as salvation from global warming (and therefore a reason not to curtail the burning of fossil fuels). However, both claims are inaccurate. Although global dimming has reversed, as of 2007 the brightening had not proceeded far enough in all areas to undo the earlier decades of dimming.

Impacts and Issues

Wide agreement that global dimming was indeed a global issue was not reached until about 2004, and even since then a large number of scientists have maintained that the phenomenon is not truly global but regional, being especially confined to urban areas, where there are more aerosols in the air. This dispute is partly about words, because if many regional dimming effects occur, then they will produce a dimming of the world as a whole. In any case, most scientists agree that global dimming has reversed itself since about 1990.

What is global dimming?

Fossil fuel use, as well as producing greenhouse gases, creates other by-products. These by-products are also pollutants, such as sulphur dioxide, soot, and ash. These pollutants however, also change the properties of clouds.

Clouds are formed when water droplets are seeded by air-borne particles, such as pollen. Polluted air results in clouds with larger number of droplets than unpolluted clouds. This then makes those clouds more reflective. More of the sun’s heat and energy is therefore reflected back into space.

This reduction of heat reaching the earth is known as Global Dimming.

Impacts of global dimming: millions already killed by it?

Global warming results from the greenhouse effect caused by, amongst other things, excessive amounts of greenhouse gases in the earth’s atmosphere from fossil fuel burning. It would seem then, that the other by-products which cause global dimming may be an ironic savior.

A deeper look at this, however, shows that unfortunately this is not the case.

Health and environmental effects

The pollutants that lead to global dimming also lead to various human and environmental problems, such as smog, respiratory problems, and acid rain.

The impacts of global dimming itself, however, can be devastating.

Millions from Famines in the Sahel in the 70s and 80s

The death toll that global dimming may have already caused is thought to be massive.

Climatologists studying this phenomenon believe that the reflection of heat have made waters in the northern hemisphere cooler. As a result, less rain has formed in key areas and crucial rainfall has failed to arrive over the Sahel in Northern Africa.

In the 1970s and 1980s, massive famines were caused by failed rains which climatologists had never quite understood *why* they had failed.

The answers that global dimming models seemed to provide, the documentary noted, has led to a chilling conclusion: **what came out of our exhaust pipes and power stations [from Europe and North America] contributed to the deaths of a million people in Africa, and afflicted 50 million more** with hunger and starvation.

Billions are likely to be affected in Asia from similar effects

Scientists said that the impact of global dimming might not be in the millions, but billions. The Asian monsoons bring rainfall to half the world's population. If this air pollution and global dimming has a detrimental impact on the Asian monsoons some 3 billion people could be affected.

As well as fossil fuel burning, contrails is another source

Contrails (the vapor from planes flying high in the sky) were seen as another significant cause of heat reflection.

During the aftermath of the September 11, 2001 terrorist attacks in the United States, all commercial flights were grounded for the next three days.

This allowed climate scientists to look at the effect on the climate when there were no contrails and no heat reflection.

What scientists found was that the temperature rose by some 1 degree centigrade in that period of 3 days.

Global Dimming is hiding the true power of Global Warming

The above impacts of global dimming have led to fears that **global dimming has been hiding the true power of global warming**.

Currently, most climate change models predict a 5 degrees increase in temperature over the next century, which is already considered extremely grave. However, global dimming has led to an underestimation of the power of global warming.

Addressing global dimming only will lead to massive global warming

Global dimming can be dealt with by cleaning up emissions.

However, if global dimming problems are only addressed, then the effects of global warming will increase even more. This may be what happened to Europe in 2003.

In Europe, various measures have been taken in recent years to clean up the emissions to reduce pollutants that create smog and other problems, but without reducing the greenhouse gas emissions in parallel. This seems to have had a few effects:

- This may have already lessened the severity of droughts and failed rains in the Sahel.

- However, it seems that it may have caused, or contributed to, the European heat wave in 2003 that killed thousands in France, saw forest fires in Portugal, and caused many other problems throughout the continent.

The documentary noted that the impacts of addressing global dimming *only* would increase global warming more rapidly. Irreversible damage would be only about 30 years away. Global level impacts would include:

- The melting of ice in Greenland, which would lead to more rising sea levels. This in turn would impact many of our major world cities
- Drying tropical rain forests would increase the risk of burning. This would release even more carbon dioxide into the atmosphere, further increasing global warming effects. (Some countries have pushed for using carbon sinks to count as part of their emission targets. This has already been controversial because these store carbon dioxide that can be released into the atmosphere when burnt. Global dimming worries increase these concerns even more.)

These and other effects could combine to lead to an increase of 10 degrees centigrade in temperature over the next 100 years, not the standard 5 degrees which most models currently predict.

This would be a more rapid warming than any other time in history, the documentary noted. With such an increase,

- Vegetation will die off even more quickly
- Soil erosion will increase and food production will fail
- A Sahara type of climate could be possible in places such as England, while other parts of the world would fare even worse.
- Such an increase in temperature would also release one of the biggest stores of greenhouse gases on earth, methane hydrate, currently contained at the bottom of the earth's oceans and known to destabilize with warming. This gas is eight times stronger than carbon dioxide in its greenhouse effect. As the documentary also added, due to the sheer amounts that would be released, by this time, whatever we would try to curb emissions, it would be too late.

This is not a prediction, the documentary said, it is a warning of what will happen if we clean up the pollution while doing nothing about greenhouse gases.

Root causes of global warming also must be addressed

If we were to use global dimming pollutants to stave off the effects of global warming, we would still face many problems, such as:

- Human health problems from the soot/smog
- Environmental problems such as acid rain
- Ecological problems such as changes in rainfall patterns (as the Ethiopian famine example above reminds us) which can kill millions, if not billions.

Climatologists are stressing that **the roots of both global dimming causing pollutants and global warming causing greenhouse gases have to be dealt with together and soon**.

We may have to change our way of life, the documentary warned. While this has been a message for over 20 years, as part of the climate change concerns, little has actually been done. Rapidly, the documentary concluded, we are running out of time.